

CharXiv — Rubric-Demand → Reasoning-Failure Random Forest

How to read this document

What this PDF contains

Every page after this one is ONE Random Forest model. Each RF tries to predict, from a chart item's 12 DeLeAn rubric-demand scores alone, whether a given vision-language model gets that item's reasoning question WRONG. Page order: COMBINATION_ALL (all 39 models pooled into one dataset) first, then one page per model, ordered by reasoning accuracy (strongest first).

All models are trained and cross-validated on the 800-item TRAIN split only; the 200-item test set is untouched.

Setup (same for every page)

- Target: $y=1$ = INCORRECT (failure is the positive class), $y=0$ = correct.
- Features: the 12 model-agnostic demand dims — VL, AS, MCr, MCu, MA, VO, AT, GS, QLI, KNf, QLq, CL. ♦ / purple = emergent figure-grounding dims (VL, GS, MA); the rest are standard DeLeAn.
- Validation: paper-grouped StratifiedGroupKFold, 5 folds × 3 repeats = 15 folds (no paper straddles train/test within a fold). RandomForest, 200 trees. Importance = permutation, 5 shuffles per fold, scored by ROC-AUC.

The two panels on each page

LEFT — AUC distribution. Box-and-whisker of ROC-AUC across the 15 folds (mean marked); dashed line at 0.5 = chance. Panel title shows mean ± SD. y-axis = ROC-AUC (0.3-1.0).

RIGHT — Permutation importance. One bar per demand dim, sorted high→low. y-axis = Δ AUC: how much held-out AUC DROPS when that dim's values are randomly shuffled (bigger = more predictive). Error bars = SD ACROSS THE 15 FOLDS (fold-to-fold wobble)— not SEM, and not the within-shuffle spread. Bars at/below 0 = the dim carries little or no predictive signal.

What you can conclude

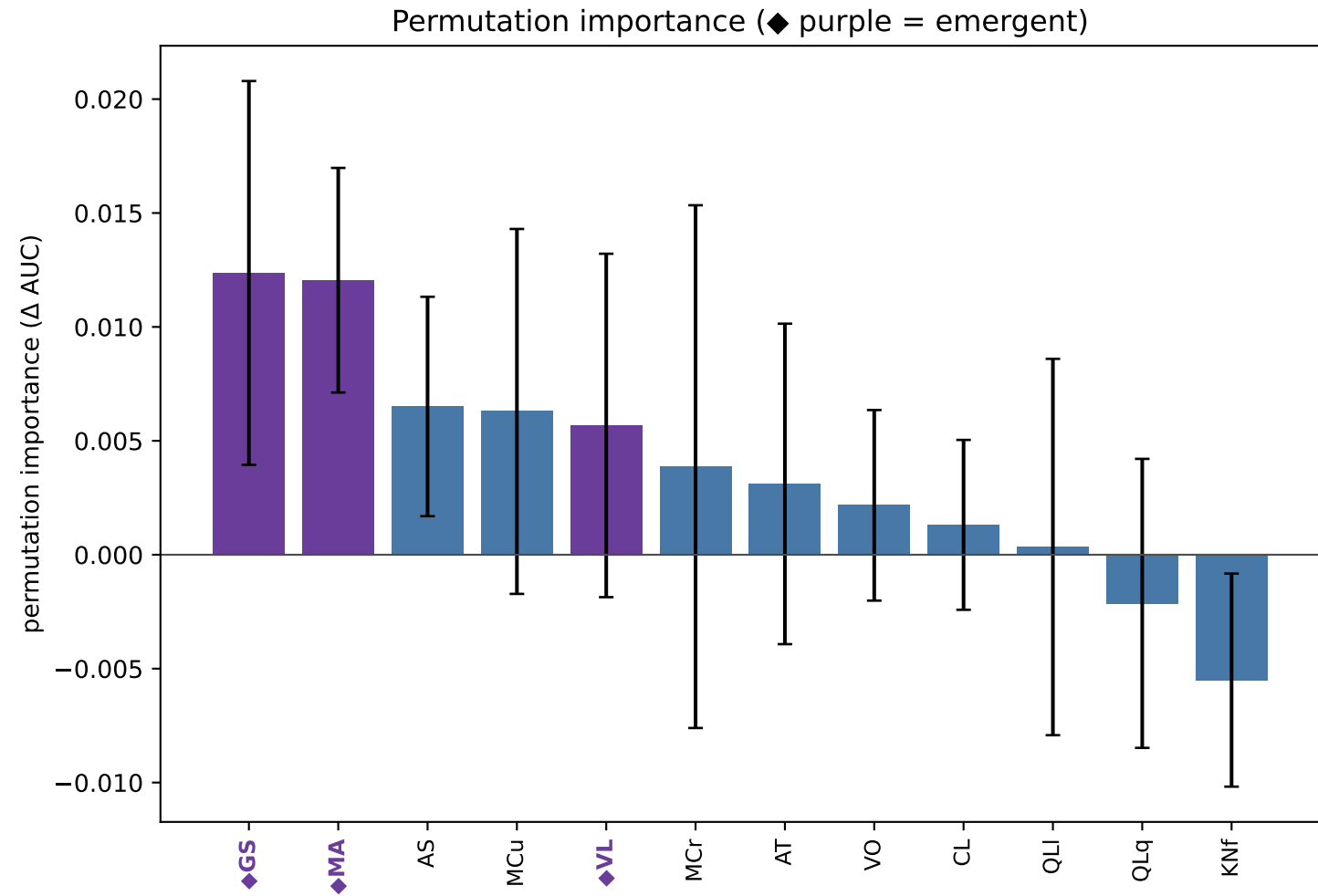
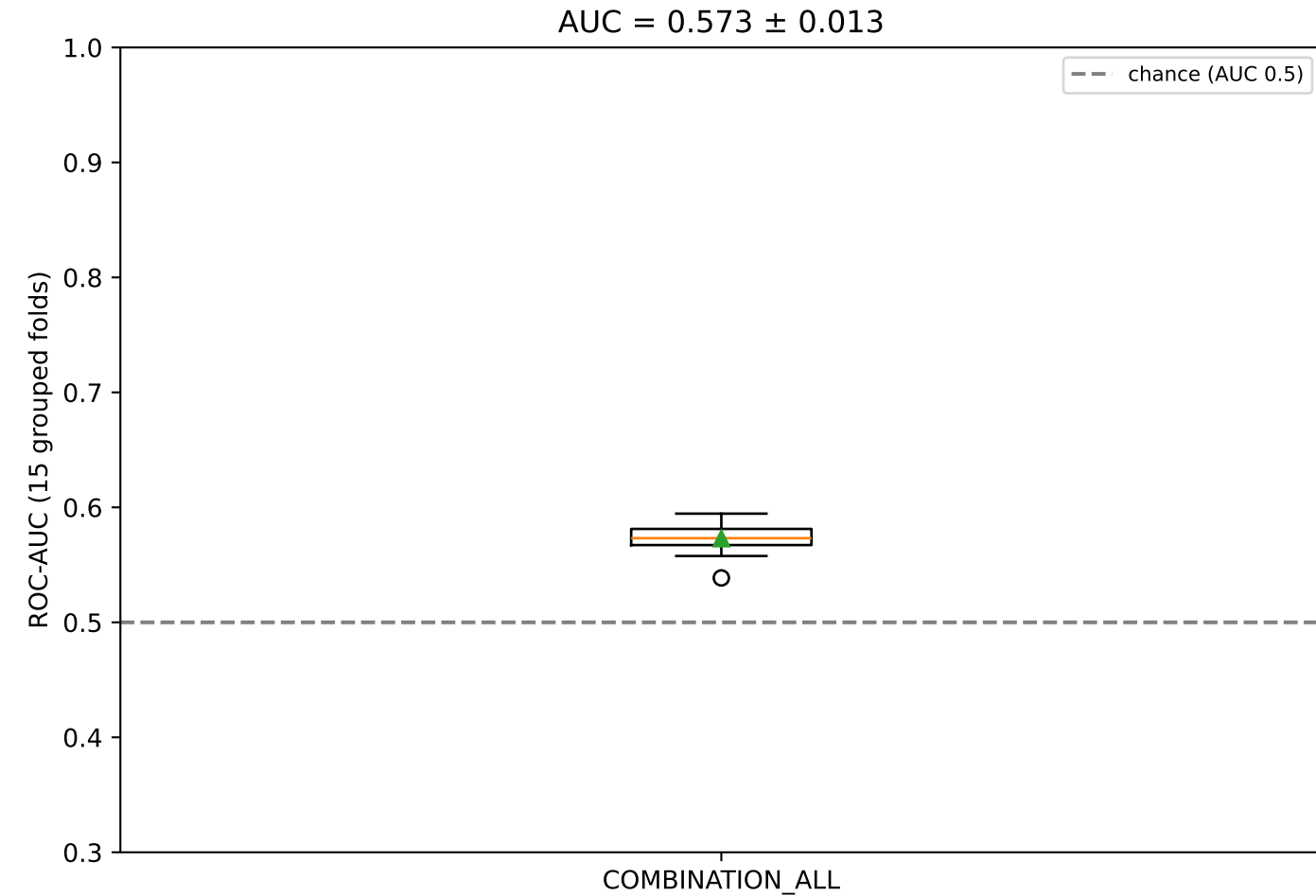
- Is demand predictive of failure? If the AUC box sits above 0.5 by more than ~1 SD, that model's failures are predictable from item demand better than chance (true for 35/39 models here, though the effect is modest— $AUC \approx 0.59$).
- Which demands are most PREDICTIVE (direction-blind — see caveats)? Tallest bars: GS (Gestalt/Shape), MA (Multi-Element Aggregation), then MCu, VL, AS. But the logistic cross-check shows GS is PROTECTIVE and MA RF-specific, so read bars as 'informative', not 'raises failure odds'.
- Do models share a signature? Compare the right panel across pages — the same few visual-grounding dims recur, so the (weak) signal is shared across models, not model-specific quirks.
- COMBINATION_ALL is the aggregate signal; with ~31k rows its AUC has the tightest SD and is the single most reliable estimate.

Caveats

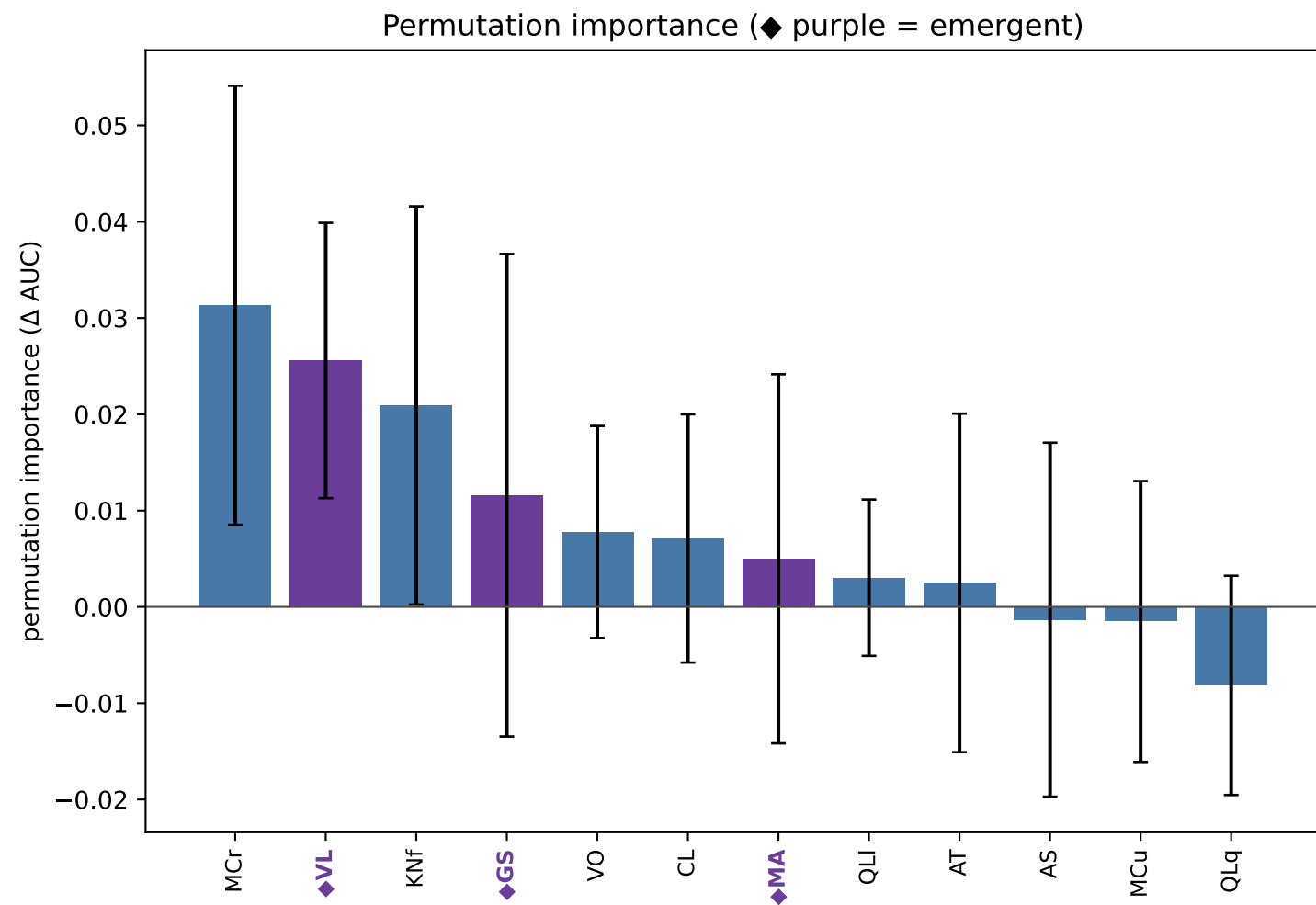
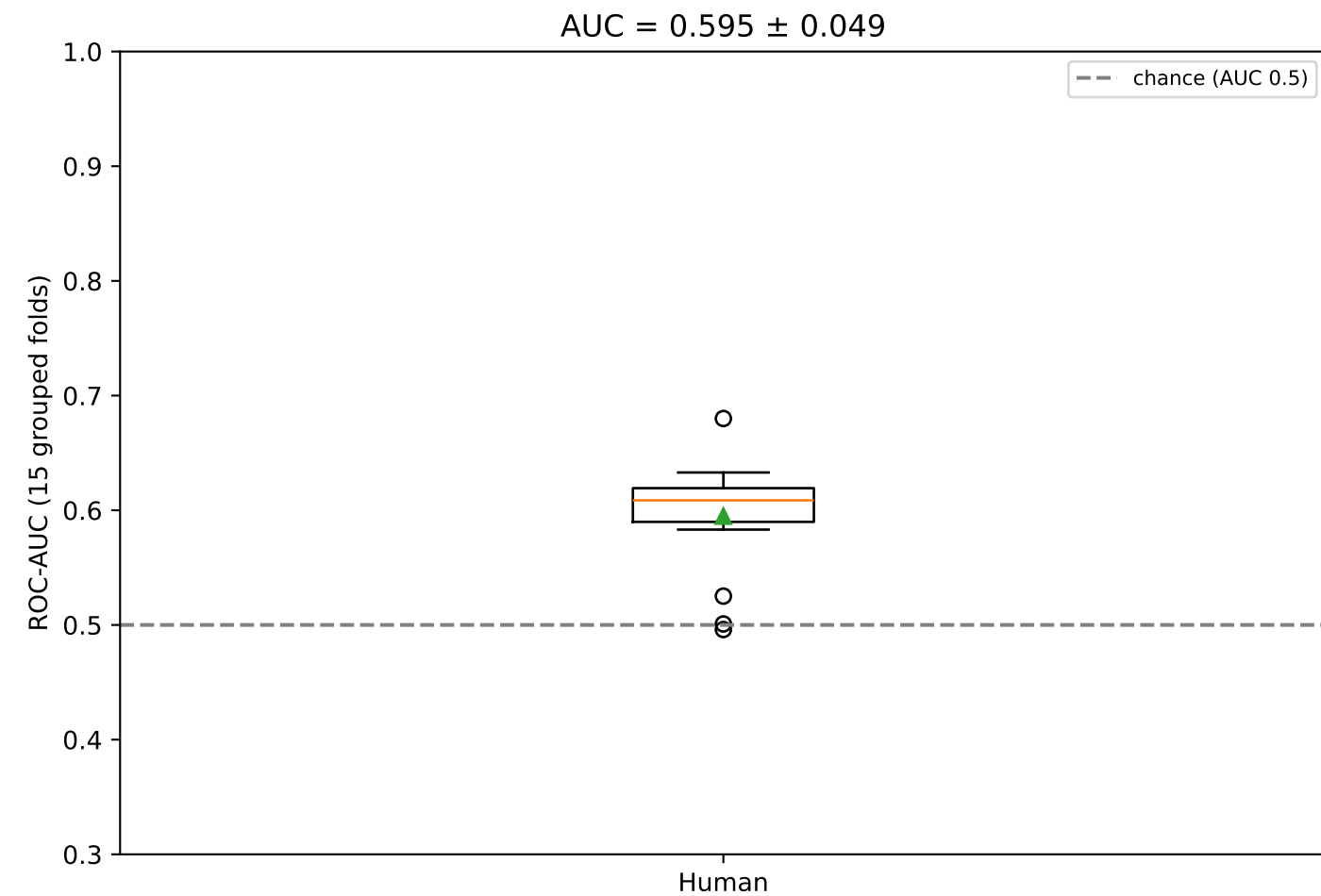
- Weak predictor ($AUC \approx 0.59$): use these for EXPLANATION (which dims matter), not deployment.
- Signal is essentially LINEAR: a logistic on the same dims matches/beats this RF (0.603 vs 0.573; both ≈ 0.60 even after nested tuning), so the RF's nonlinearity adds nothing. See logistic_cross_check.py / tuned_model_comparison.py.
- Direction $\neq$ magnitude. The logistic cross-check resolves sign: GS is PROTECTIVE (coef -0.18 , higher GS demand → FEWER failures, matching Mann-Whitney); MA is RF-SPECIFIC (rank 2 here, 10 under a linear model). Don't read tall bars as 'failure-driving'.
- Permutation $\approx$ TreeSHAP (Spearman ≈ 0.67) on GS/MA; VO is a perm-only blip, but VIFs ≤ 1.8 rule out collinearity (just method sensitivity). See perm_shap_agreement.py.

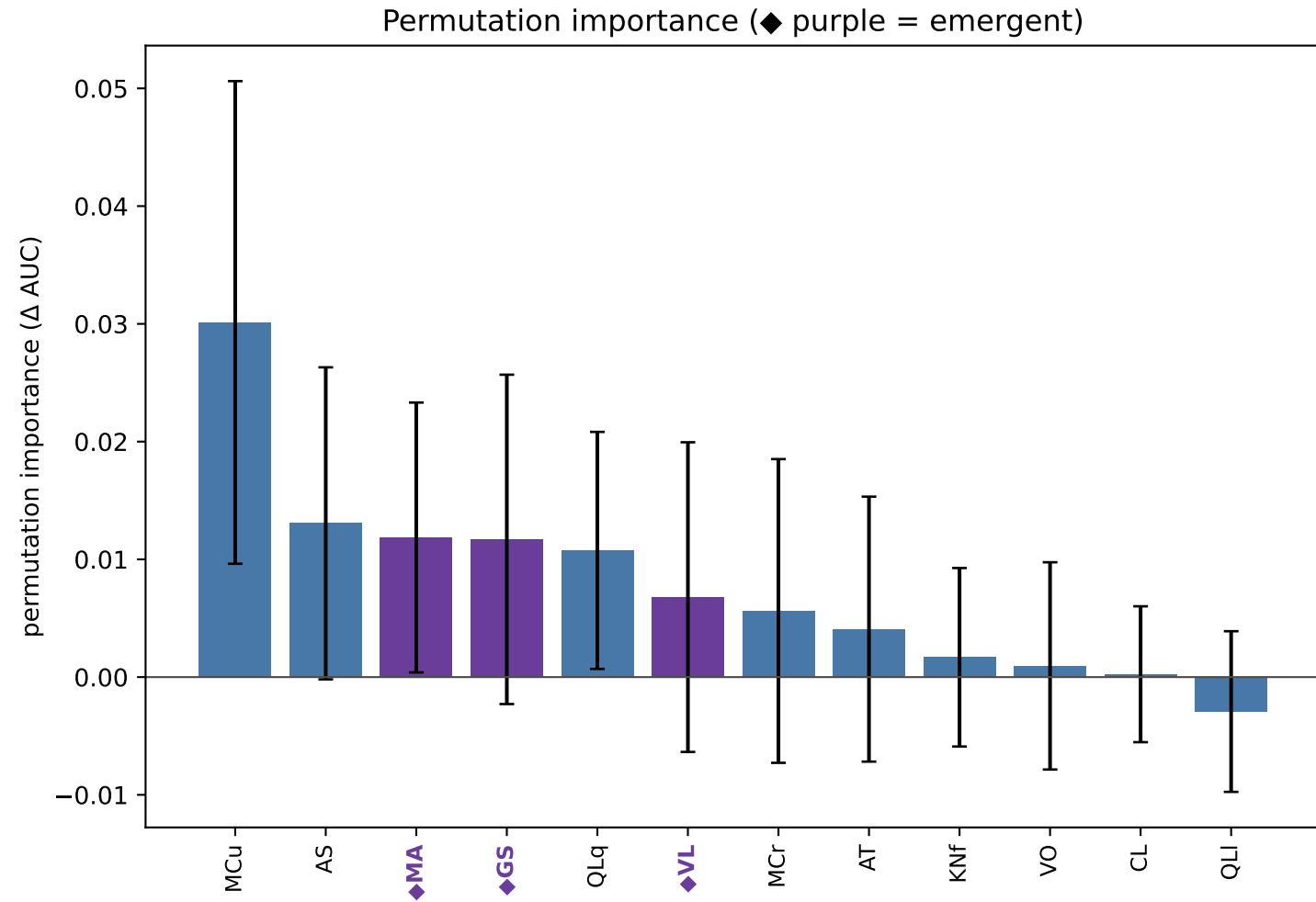
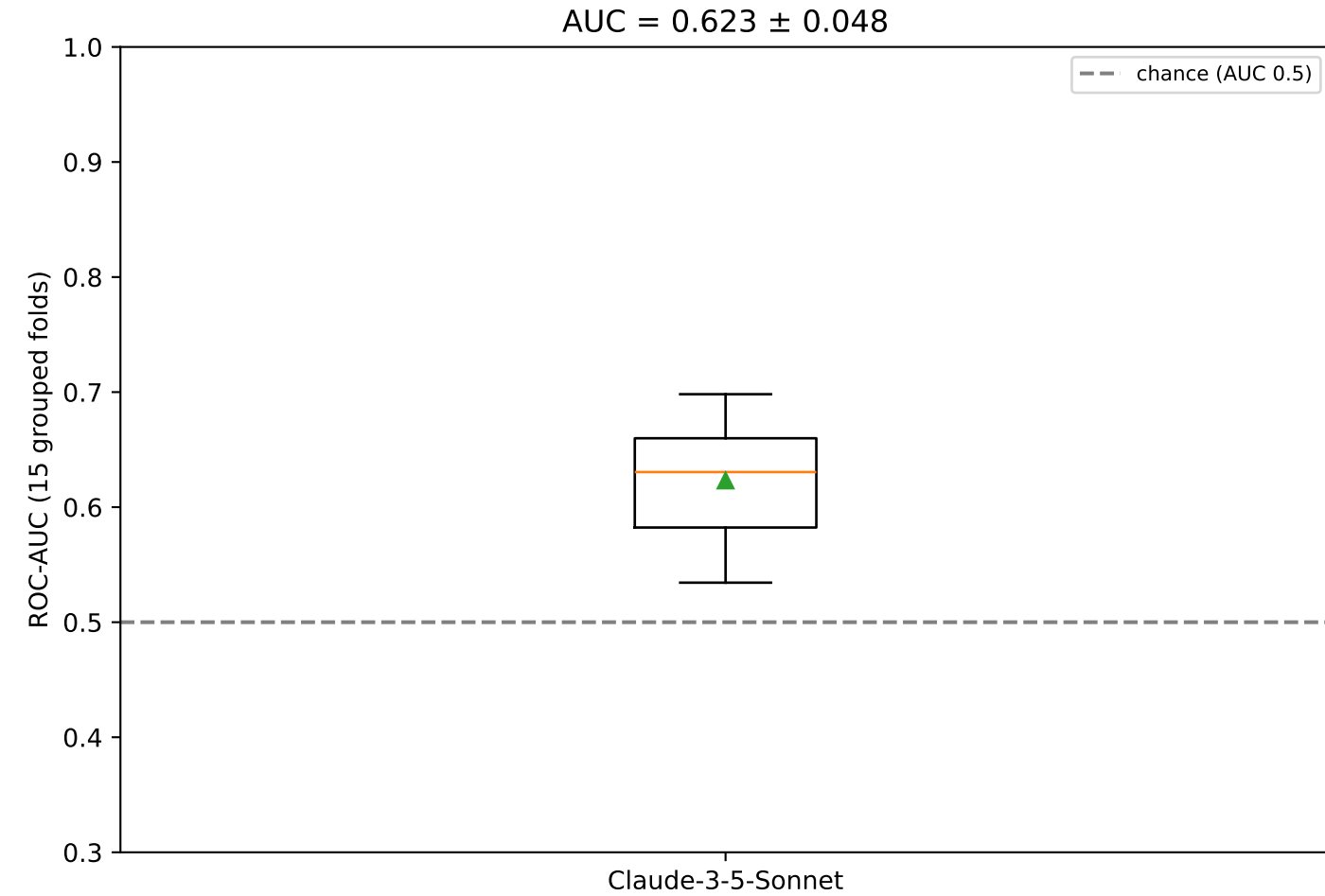
• AUC $\pm$ and error bars are population SD (ddof=0) over only 15 folds — itself a noisy SD estimate

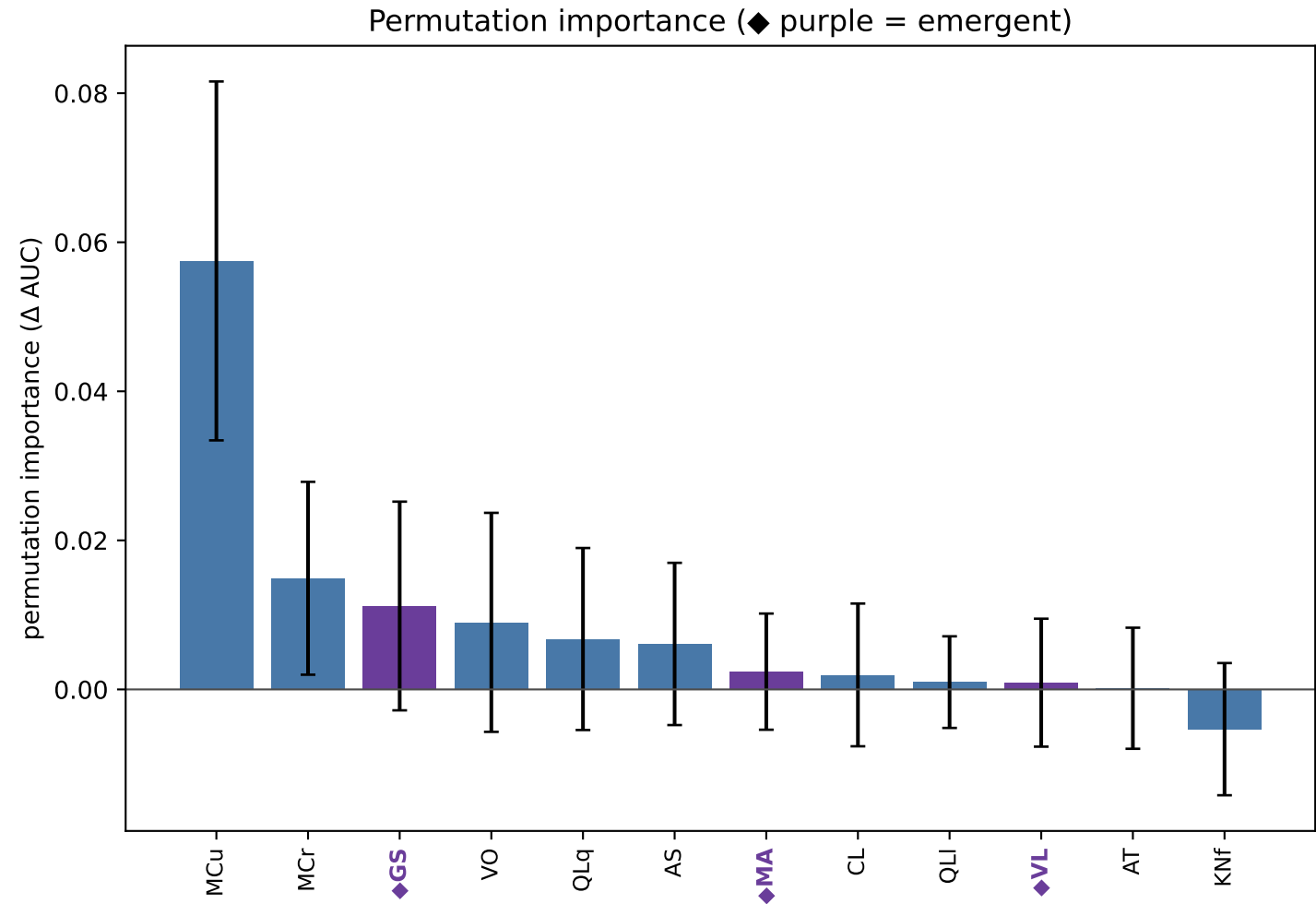
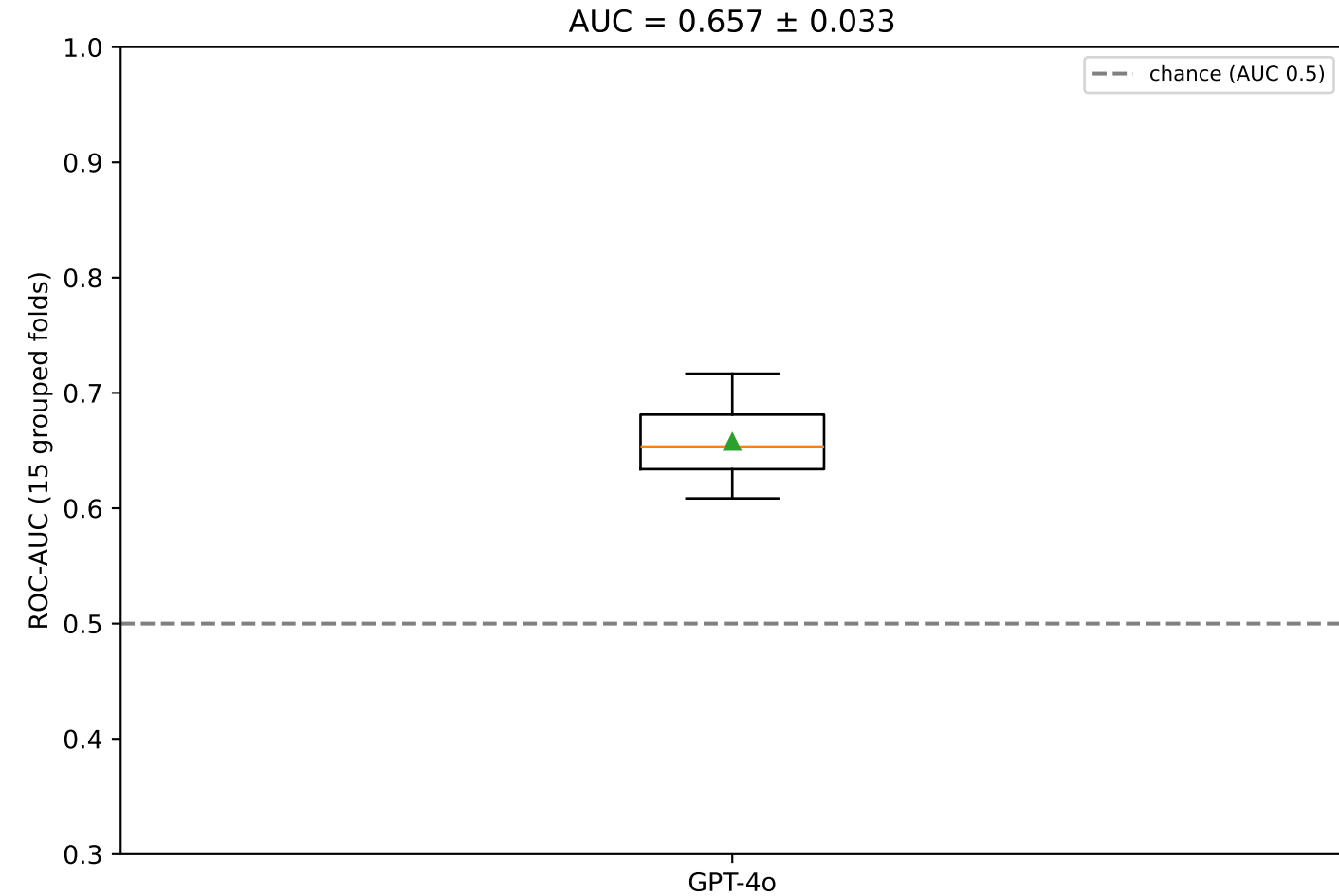
Outputs: charxiv_analyze/random_forest/ — this PDF, rf_1_P05, rf_1v_results.csv, rf_importance_long.csv, plus charxiv_random_forest_shap.pdf (out-of-fold TreeSHAP beeswarms), *_shap.png, rf_shap_importance_long.csv.

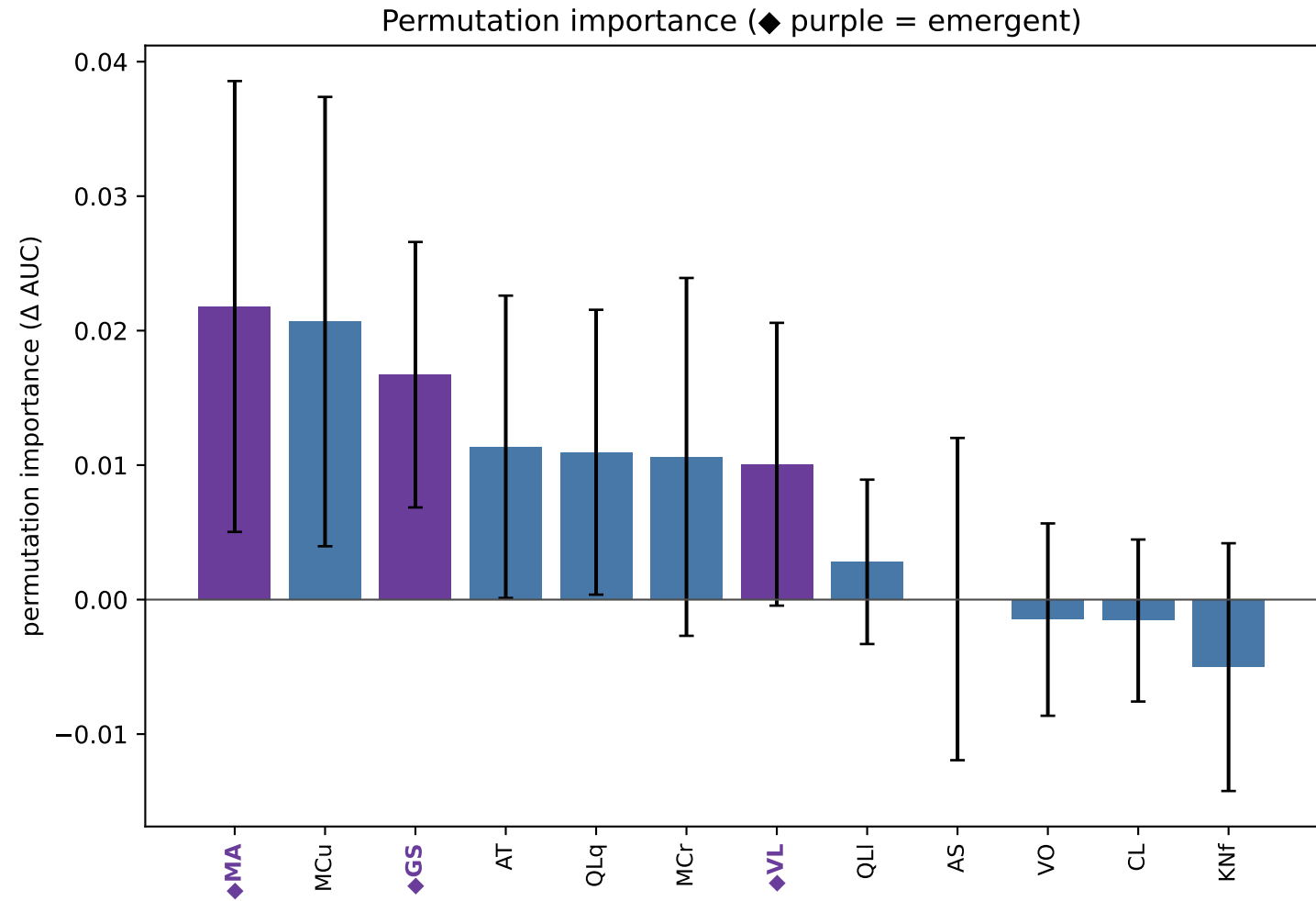
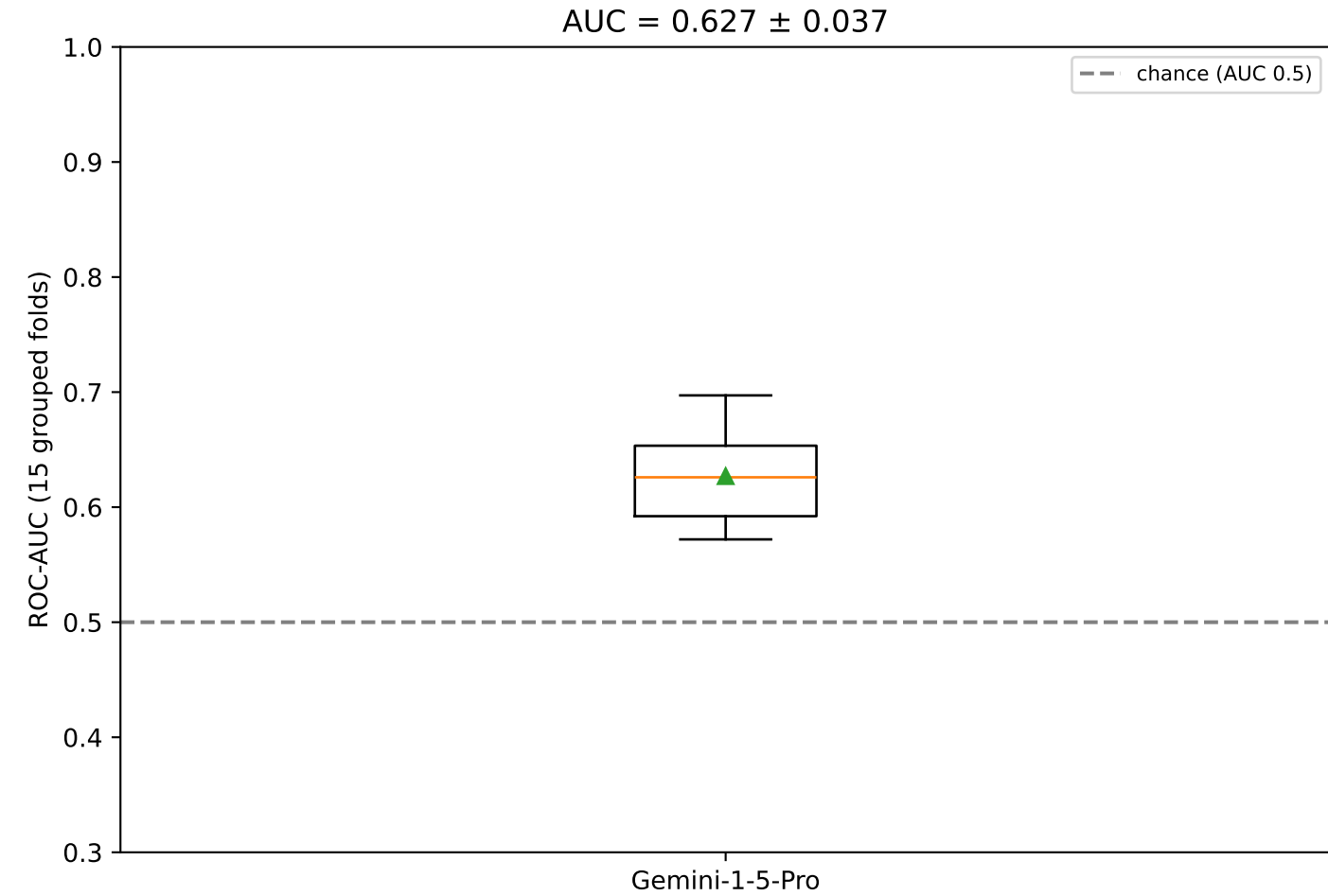


Human — rubric→failure RF (y=1 INCORRECT; base rate 0.193, n=800, paper-grouped 5-fold ×3)

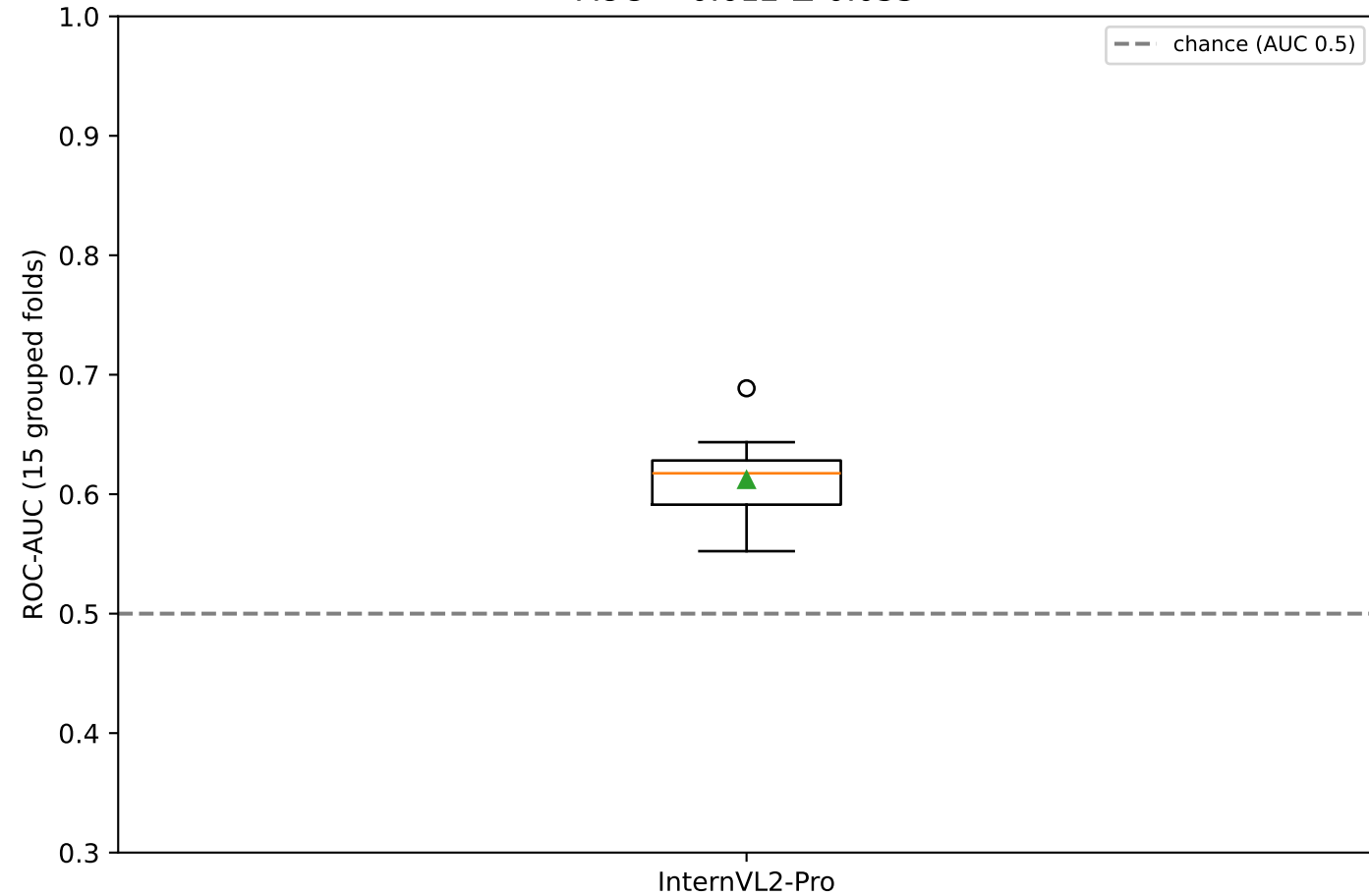




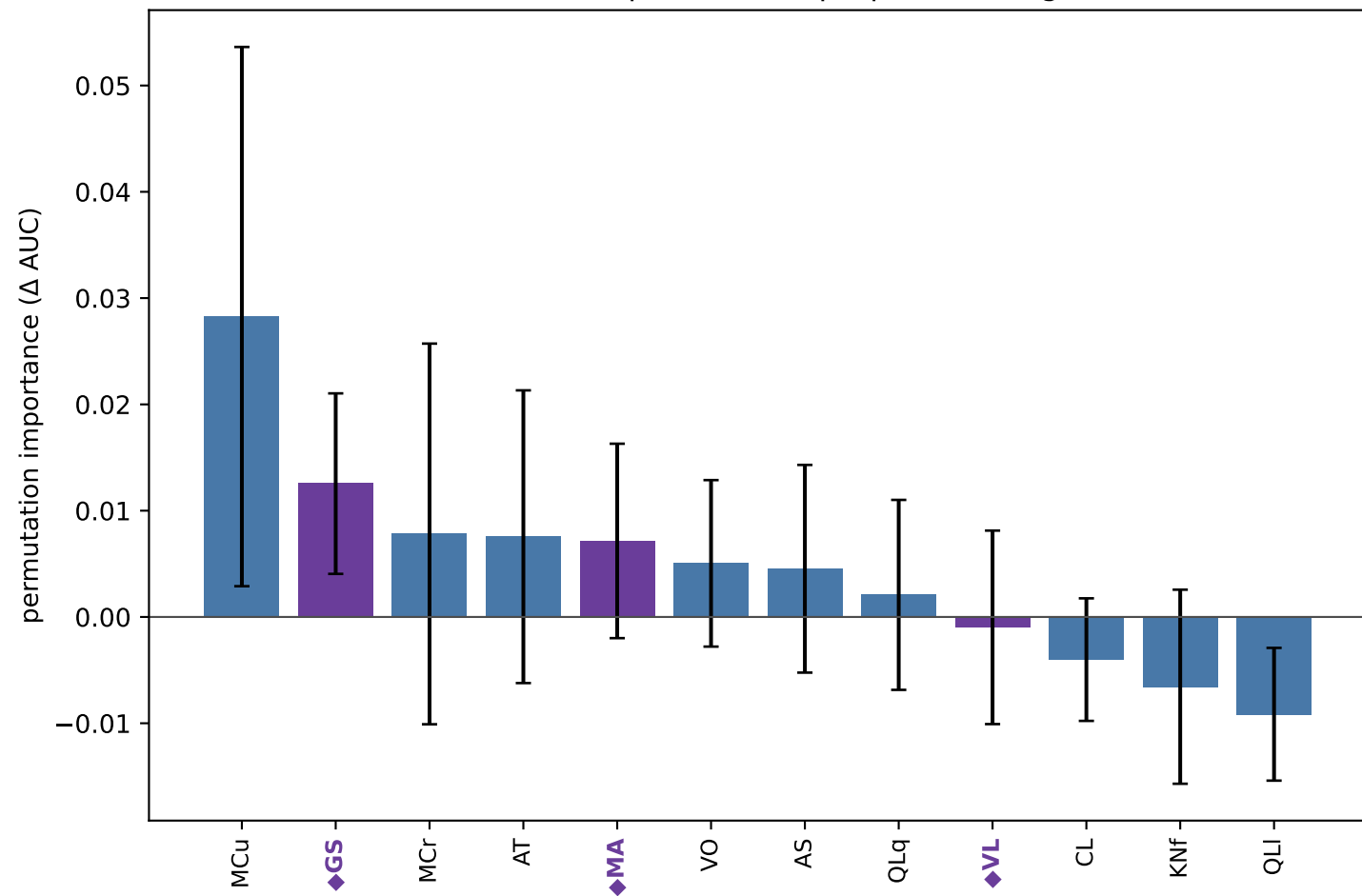




AUC = 0.612 ± 0.033

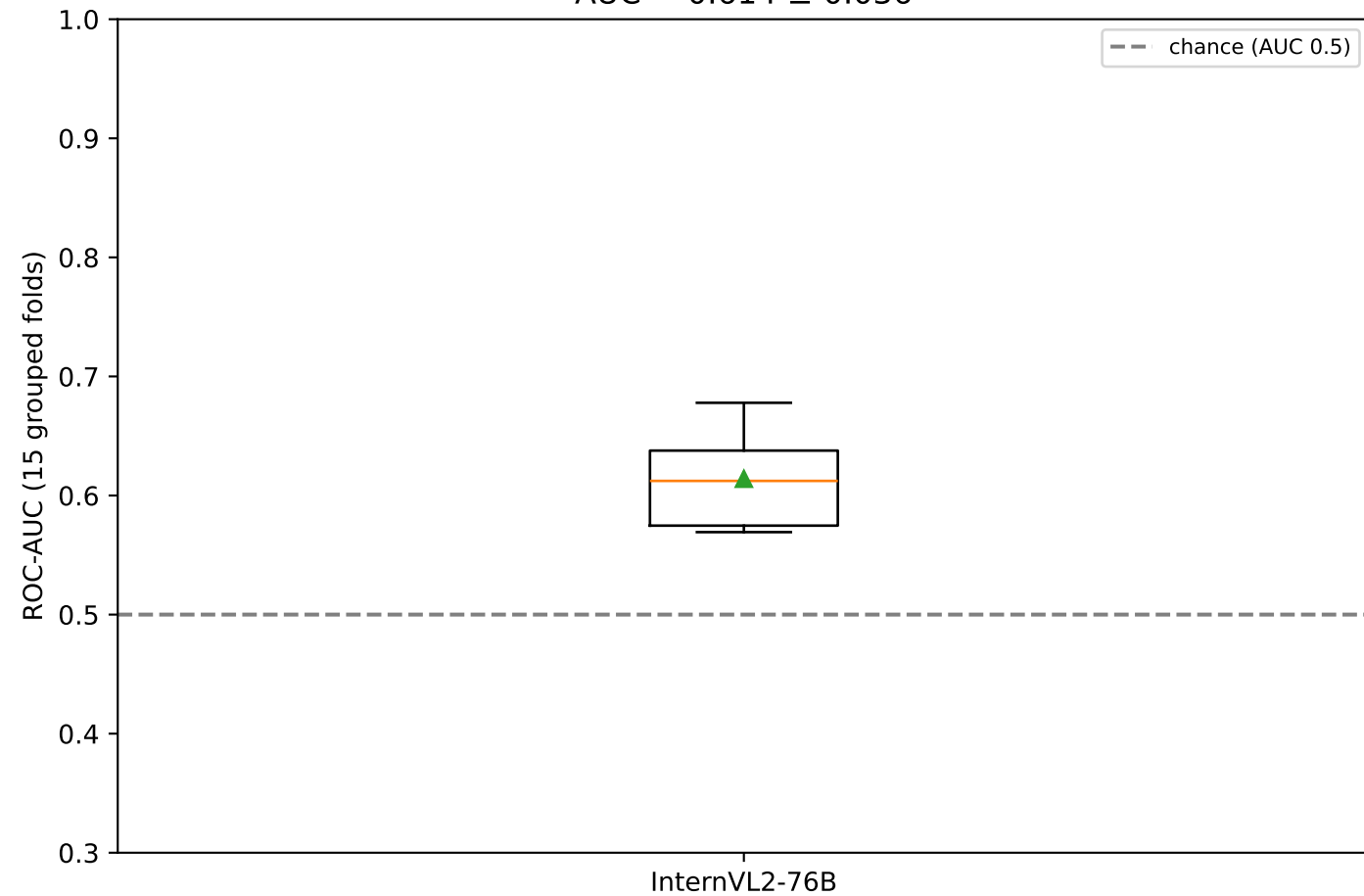


Permutation importance (◆ purple = emergent)

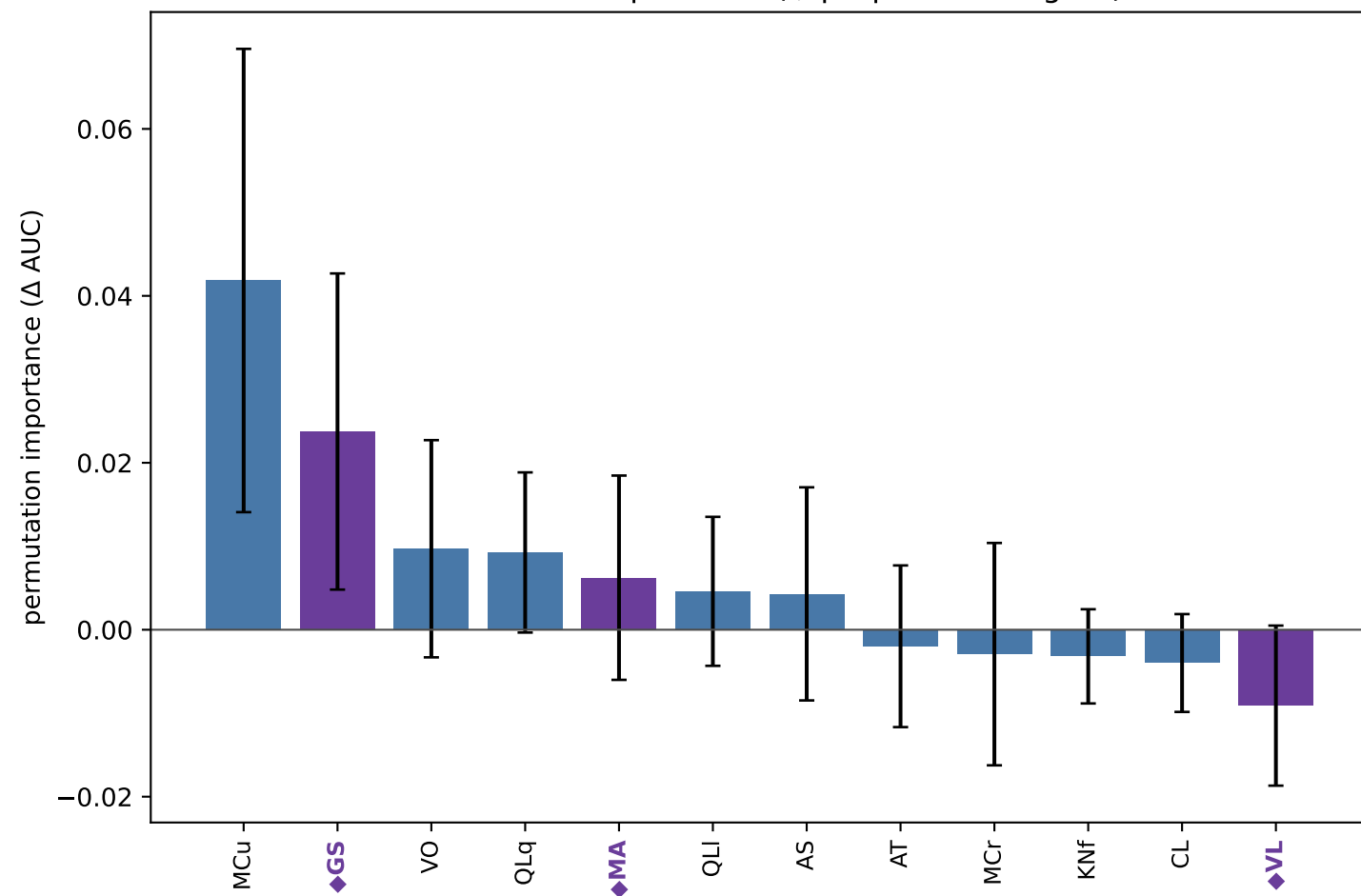


InternVL2-76B — rubric→failure RF (y=1 INCORRECT; base rate 0.598, n=800, paper-grouped 5-fold ×3)

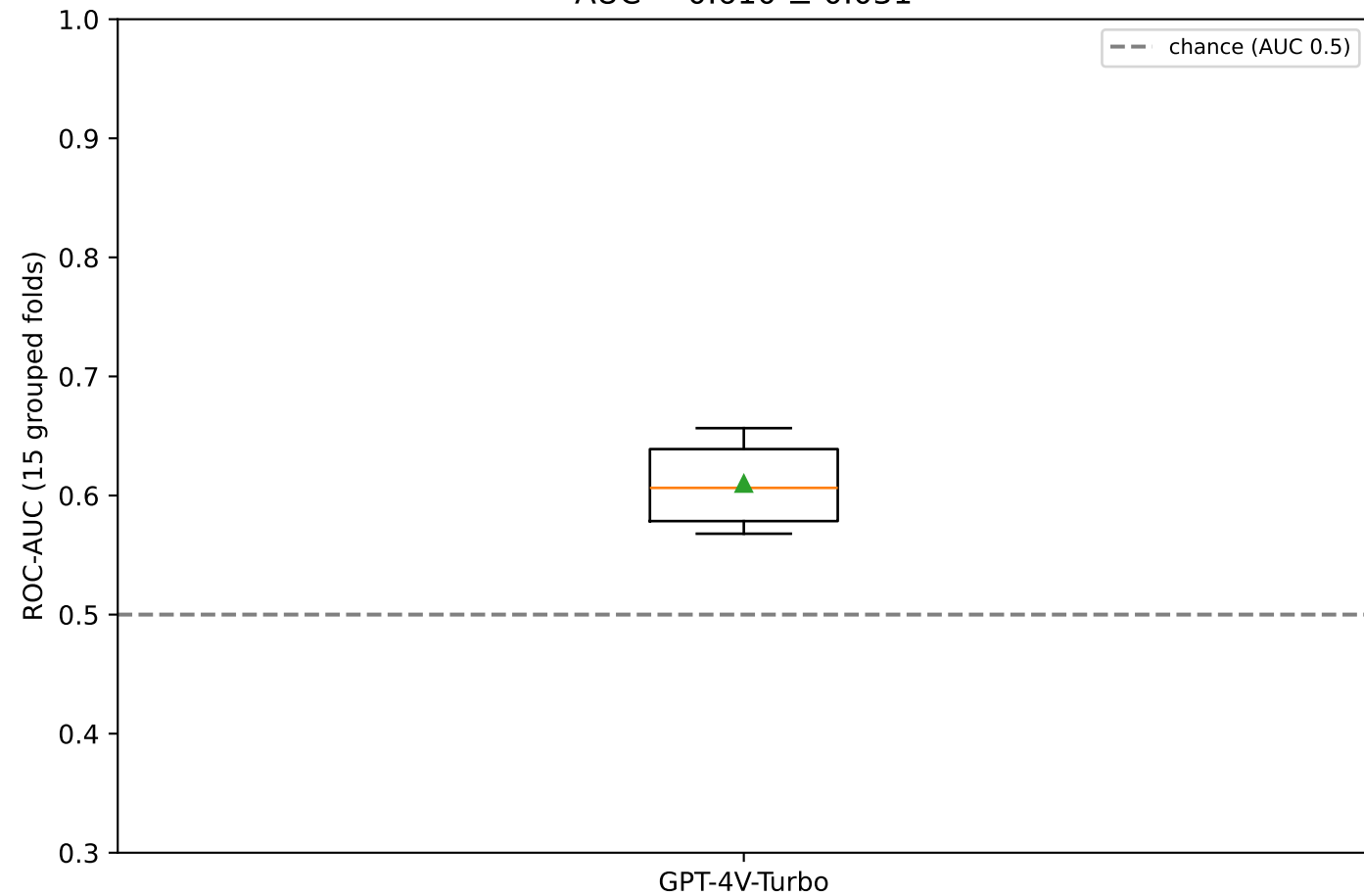
AUC = 0.614 ± 0.036



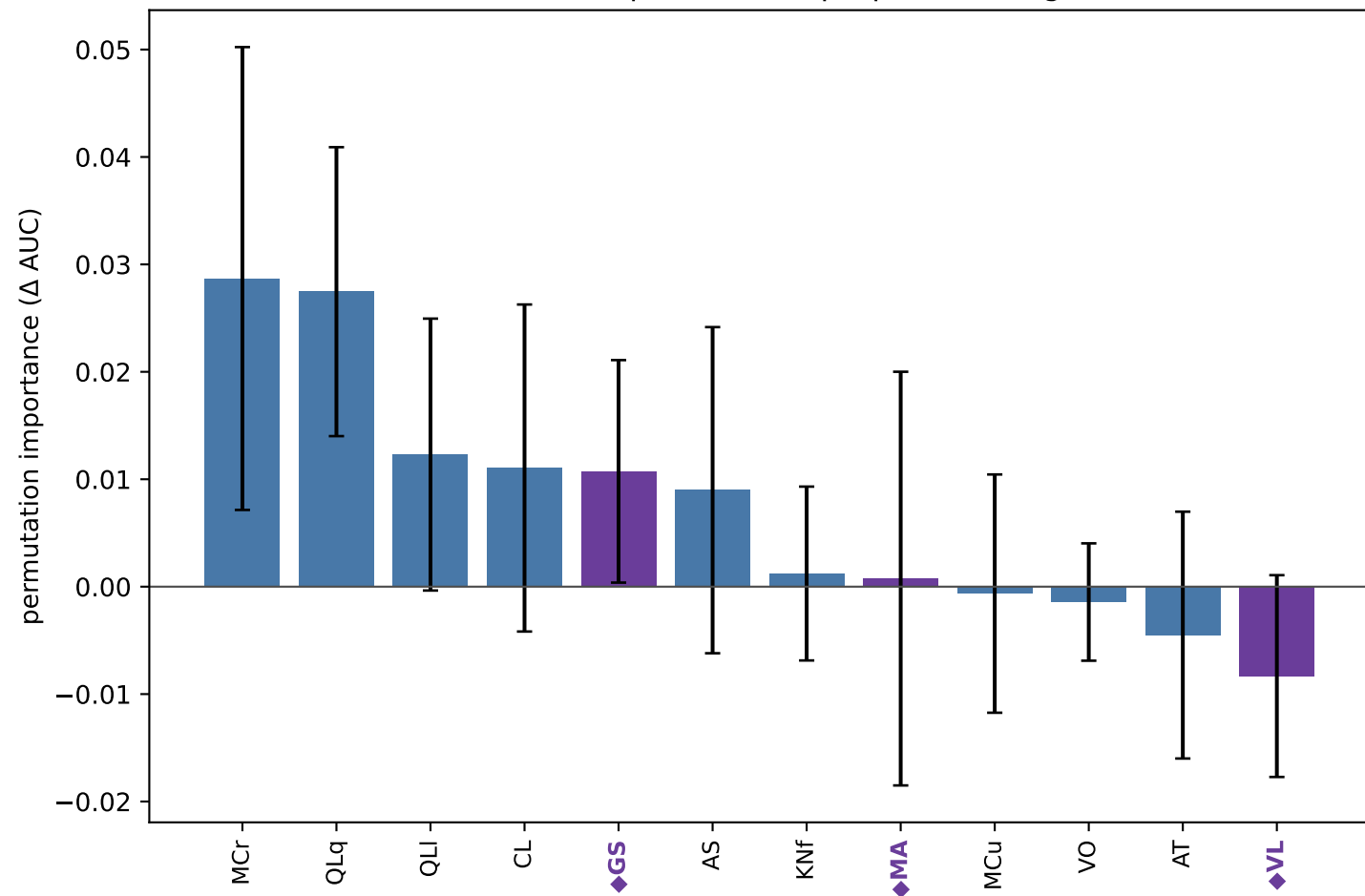
Permutation importance (◆ purple = emergent)



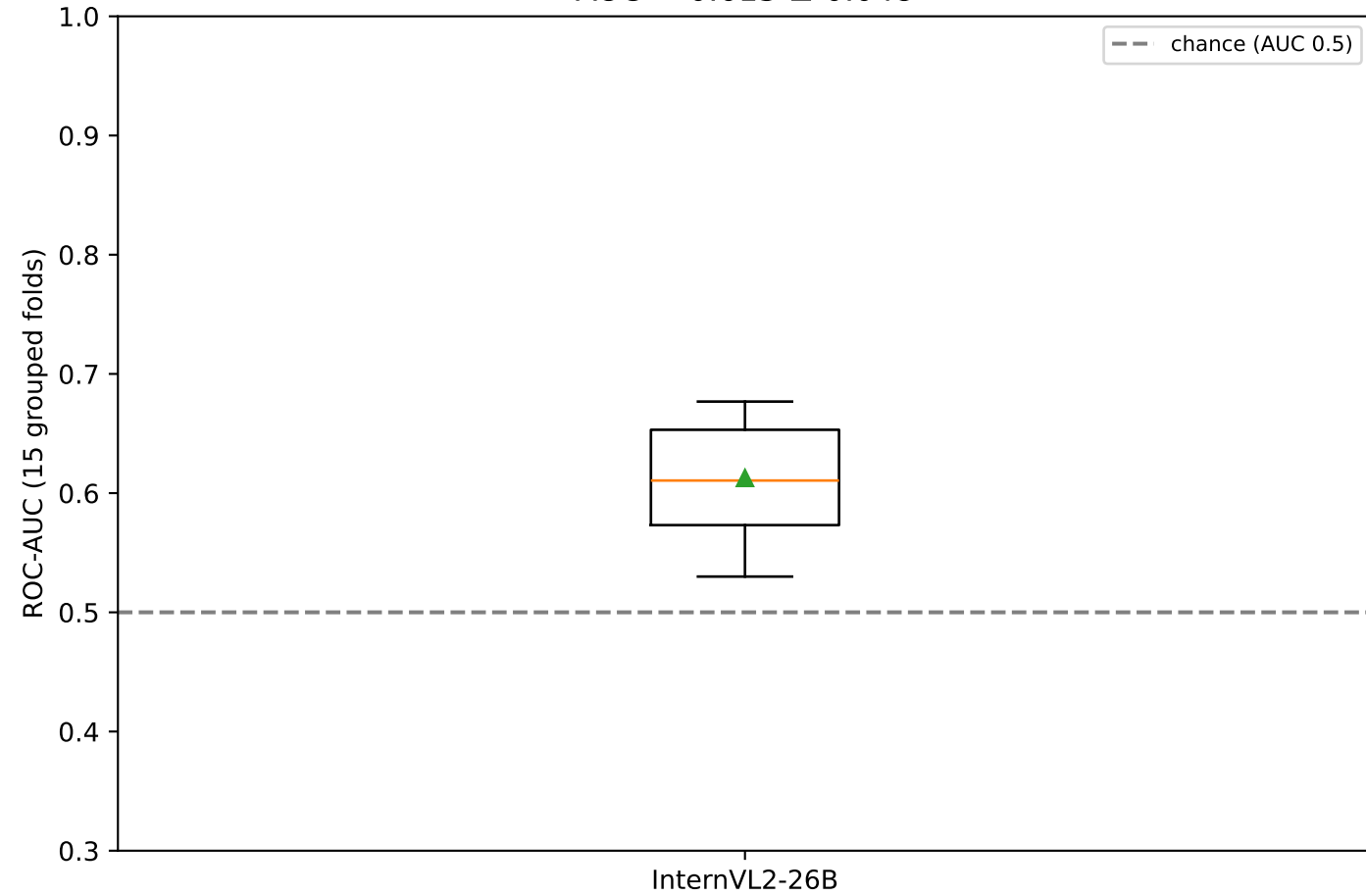
AUC = 0.610 ± 0.031



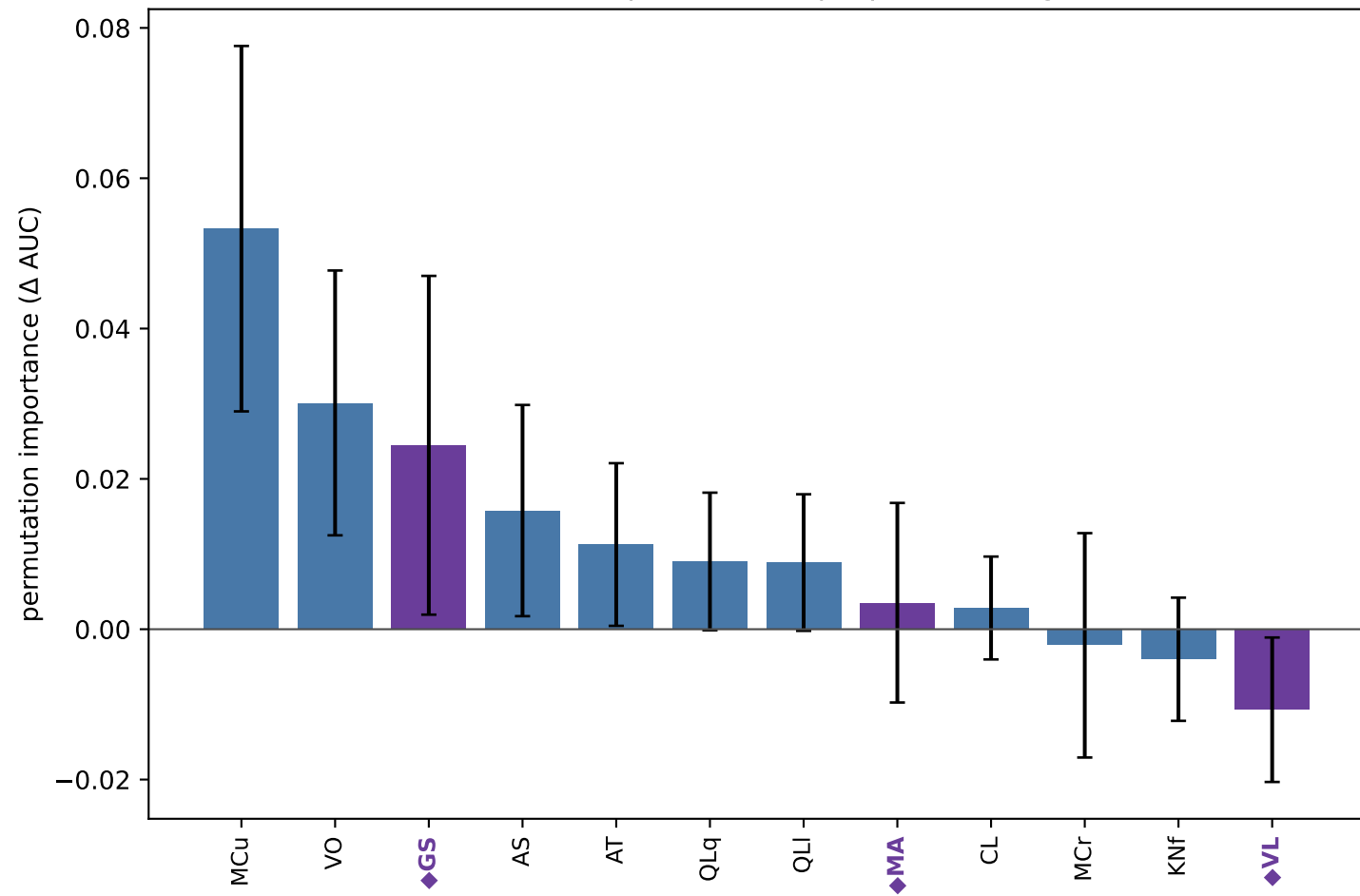
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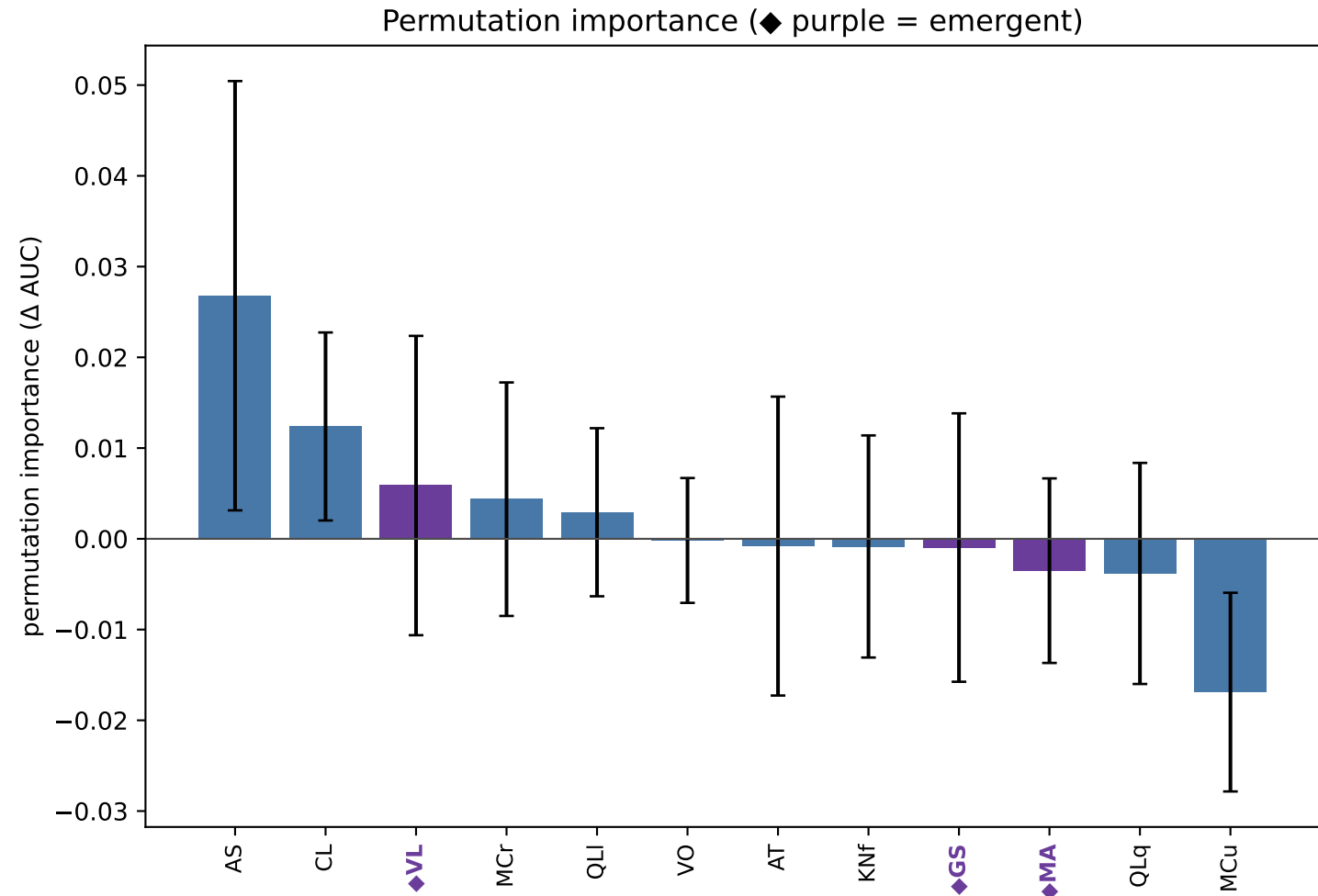
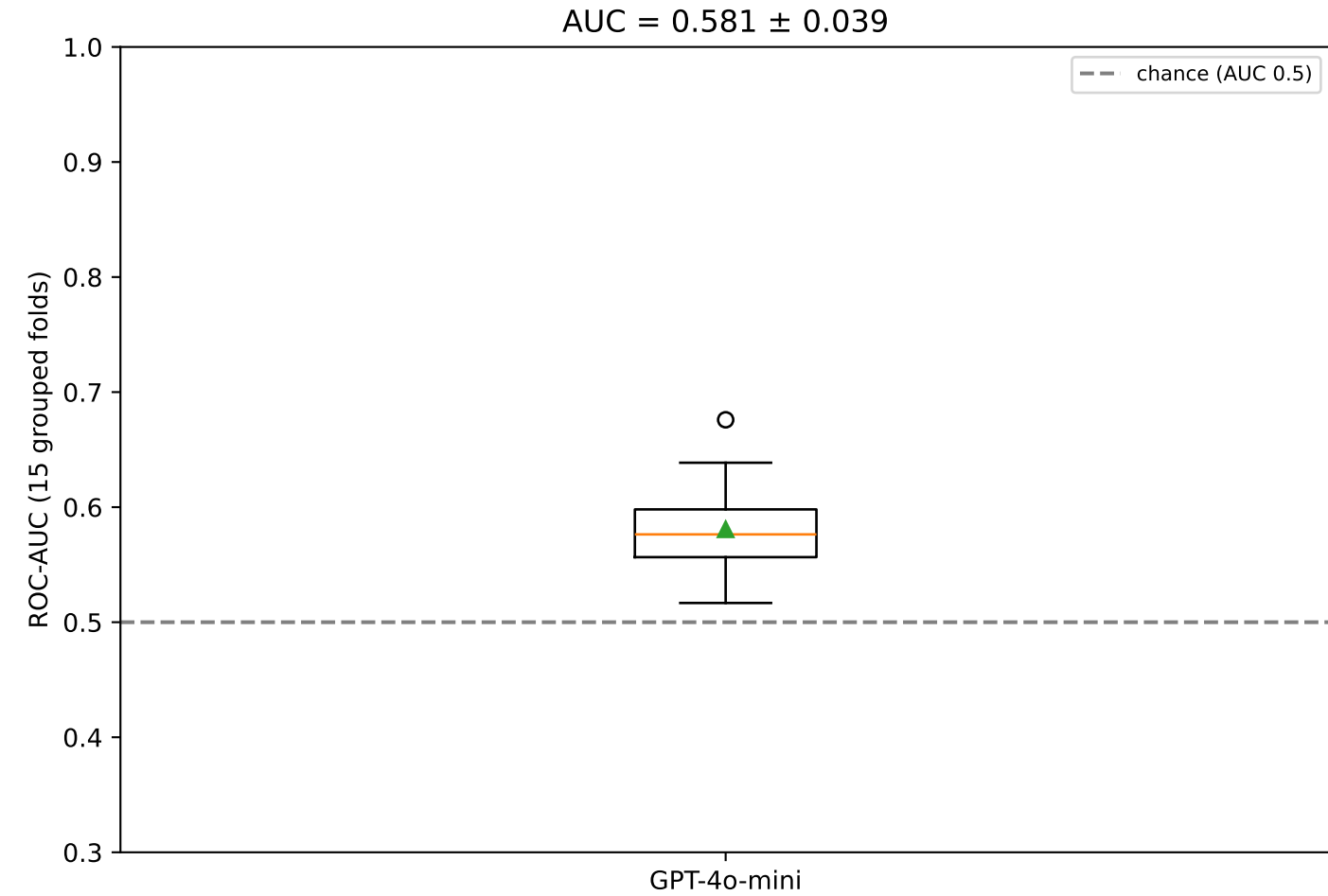


AUC = 0.613 ± 0.048

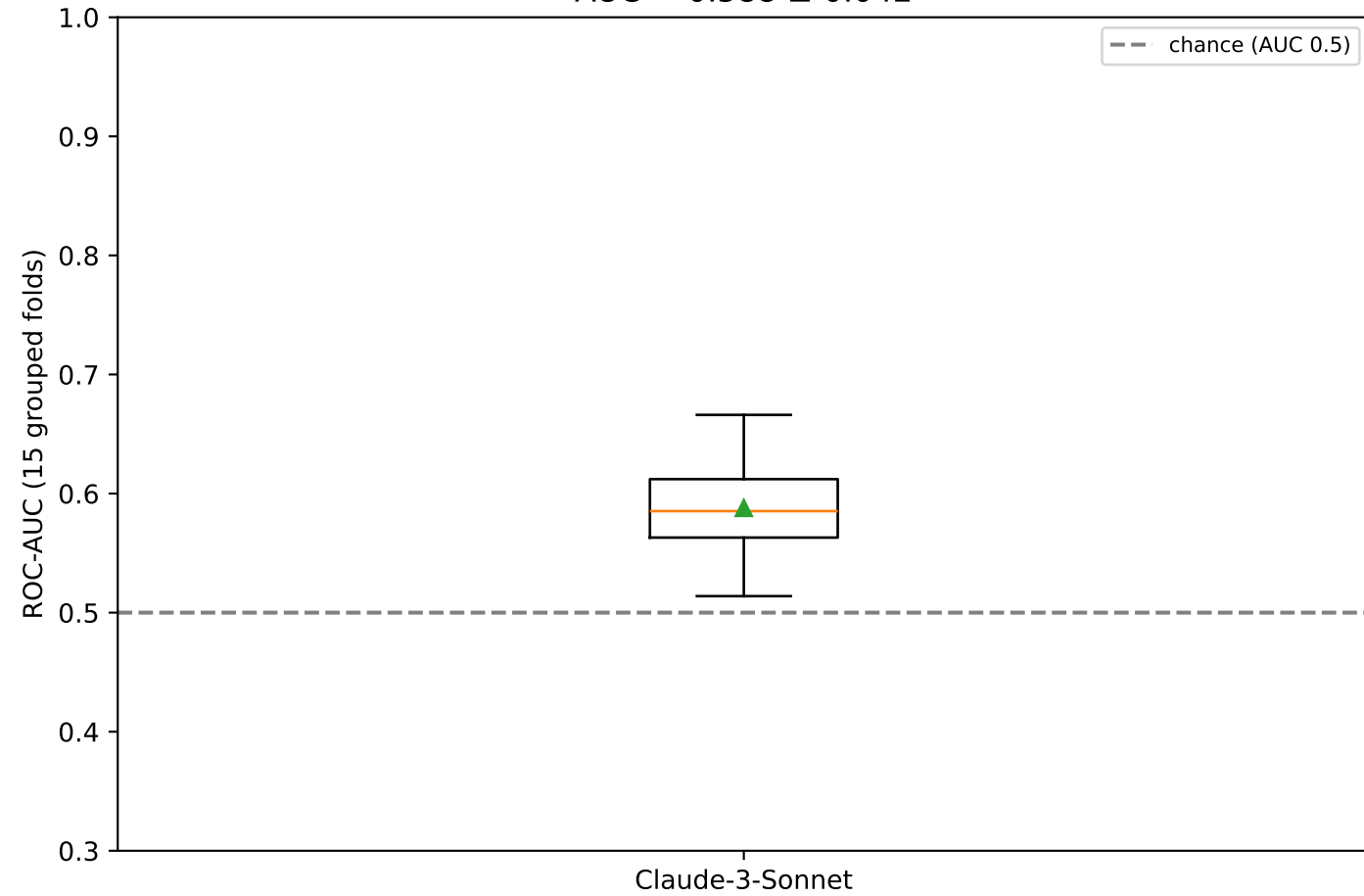


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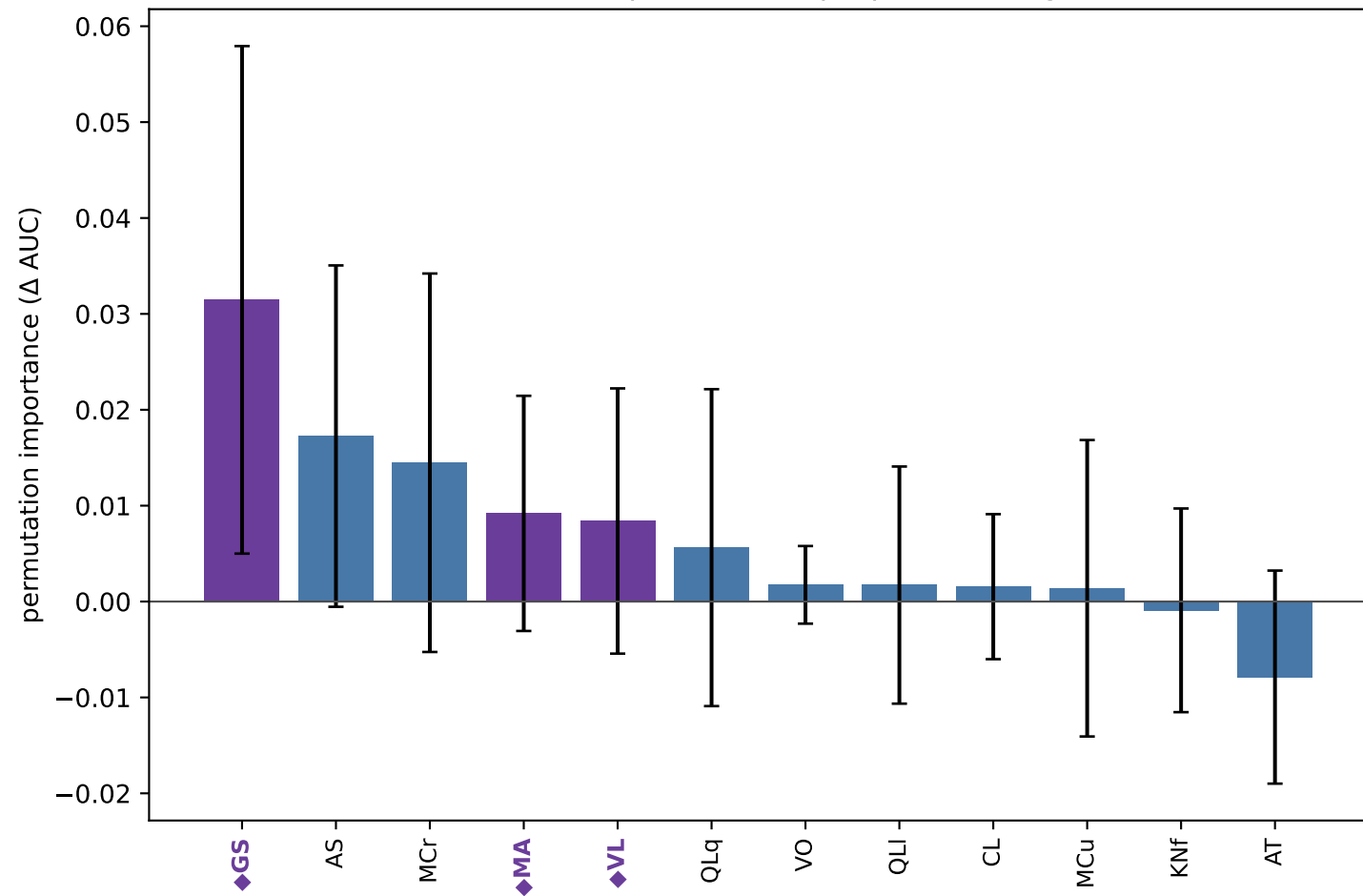




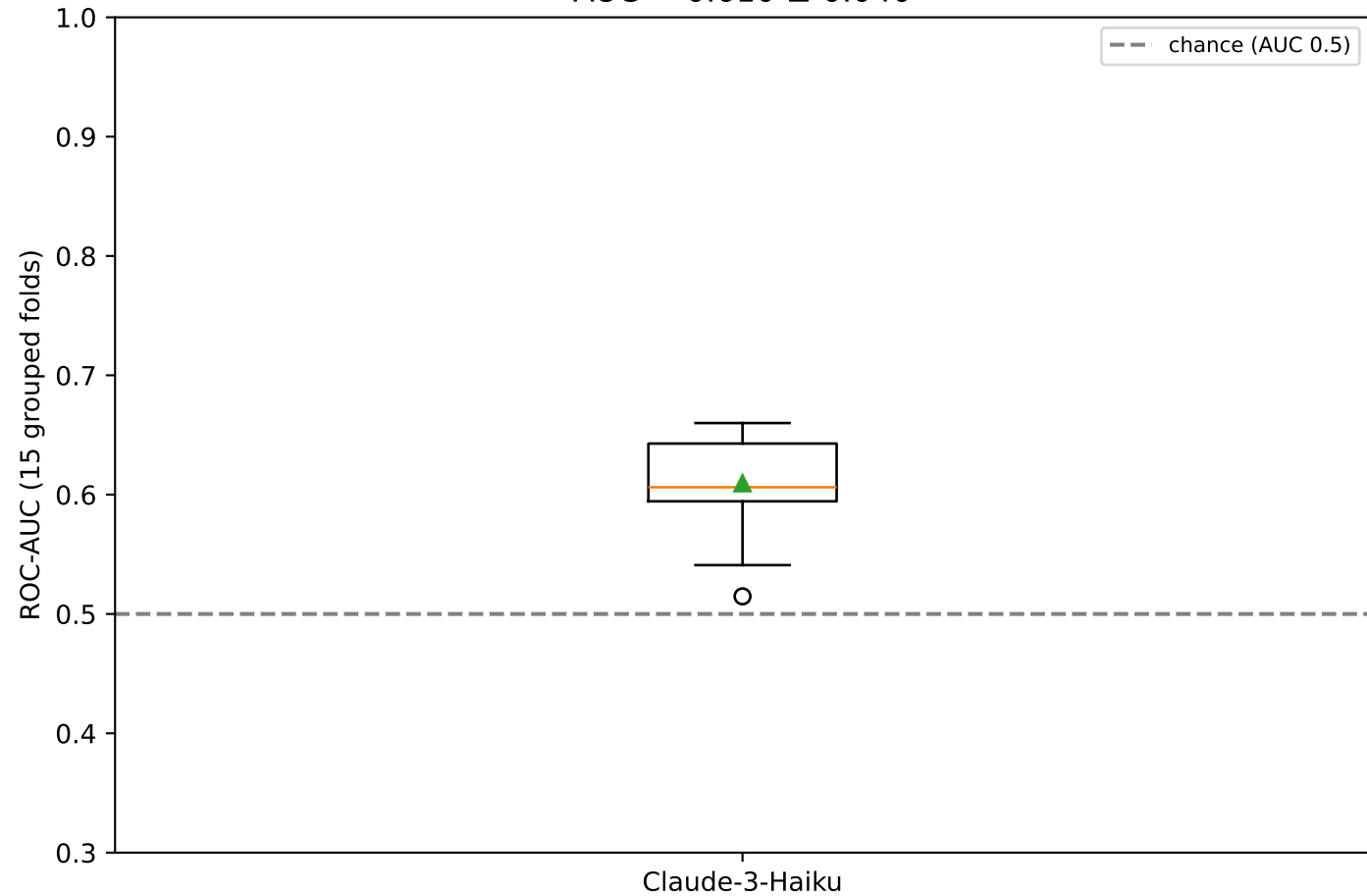
AUC = 0.588 ± 0.042



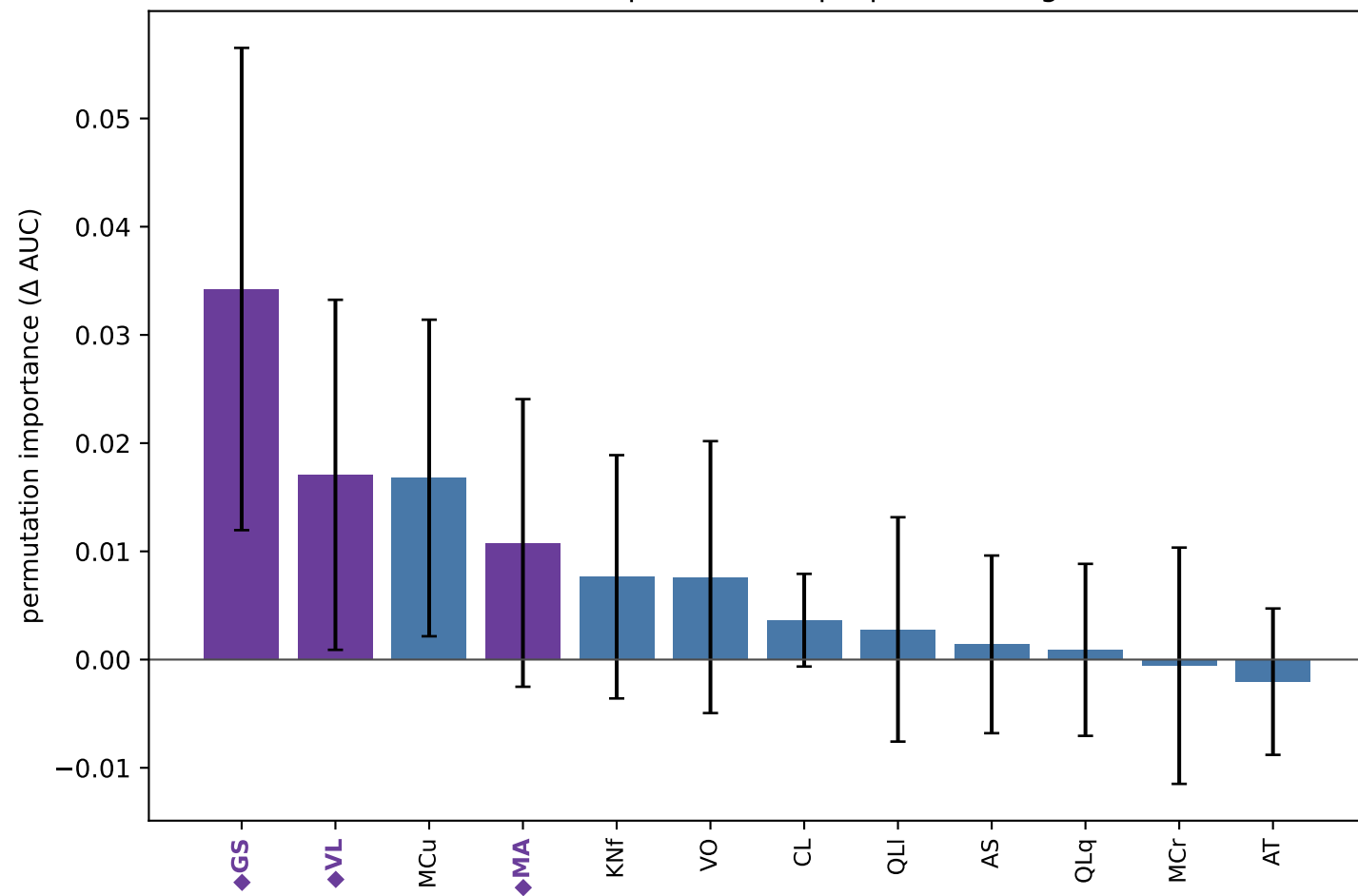
Permutation importance (◆ purple = emergent)



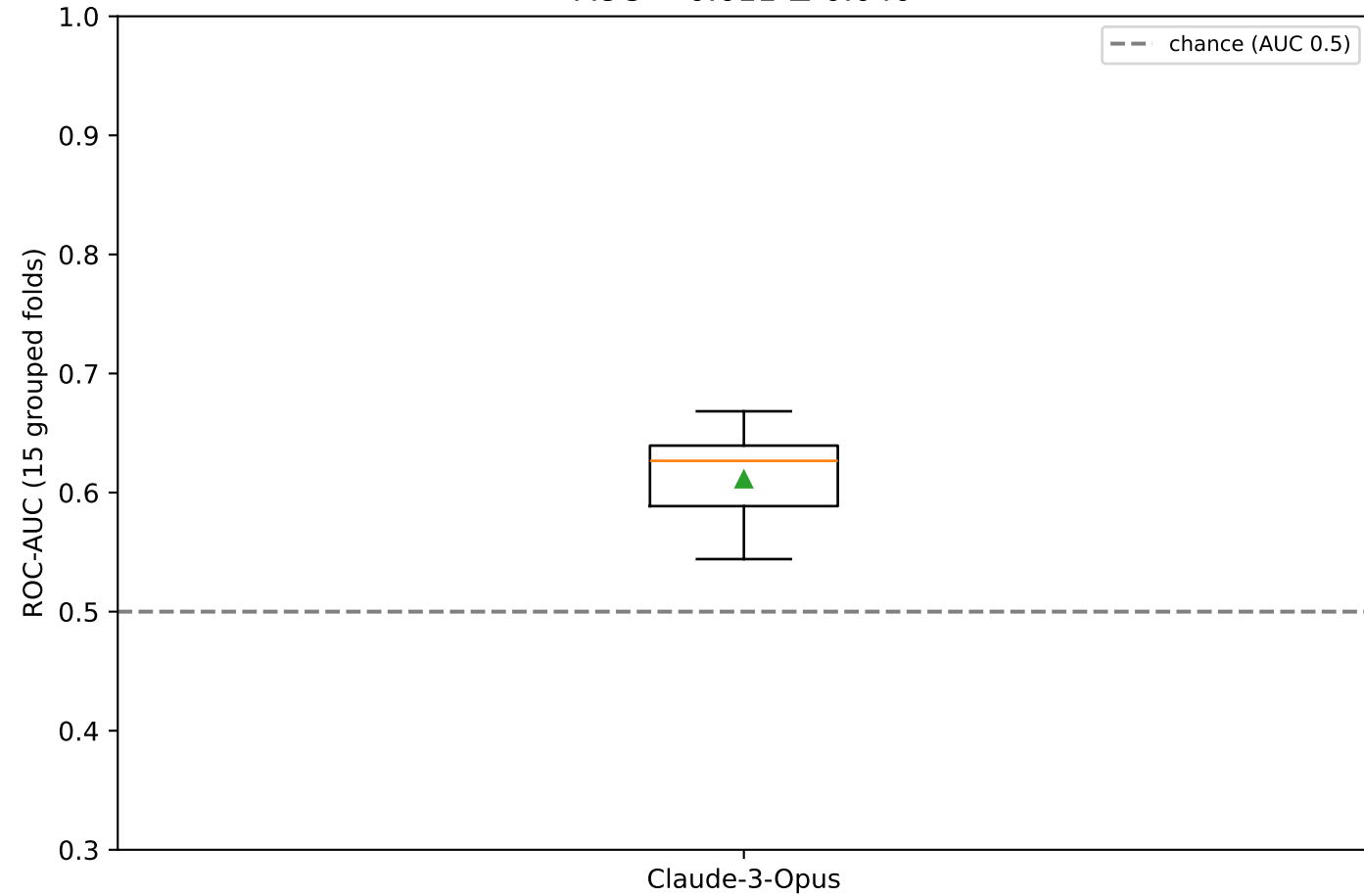
AUC = 0.610 ± 0.040



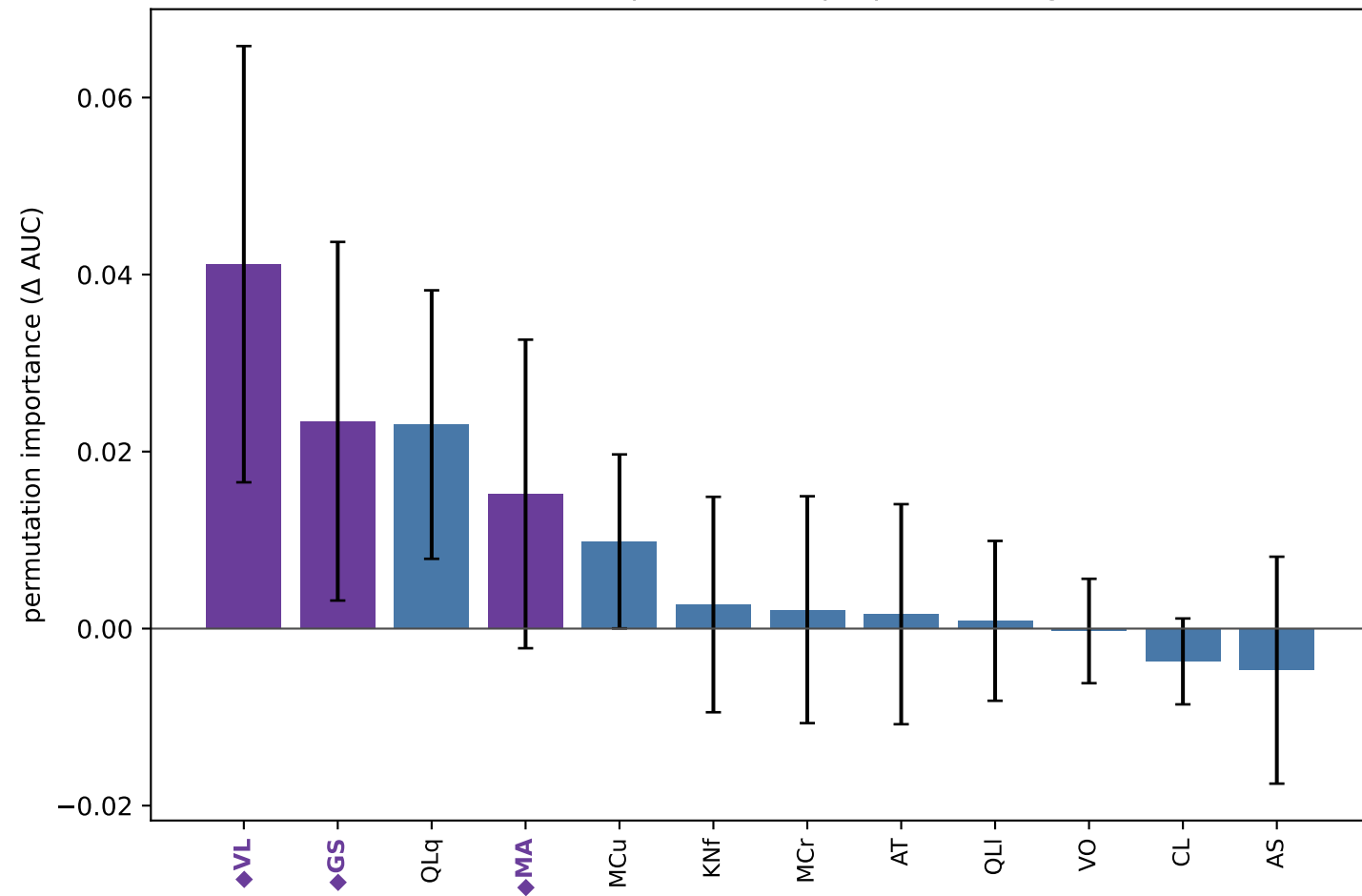
Permutation importance (◆ purple = emergent)



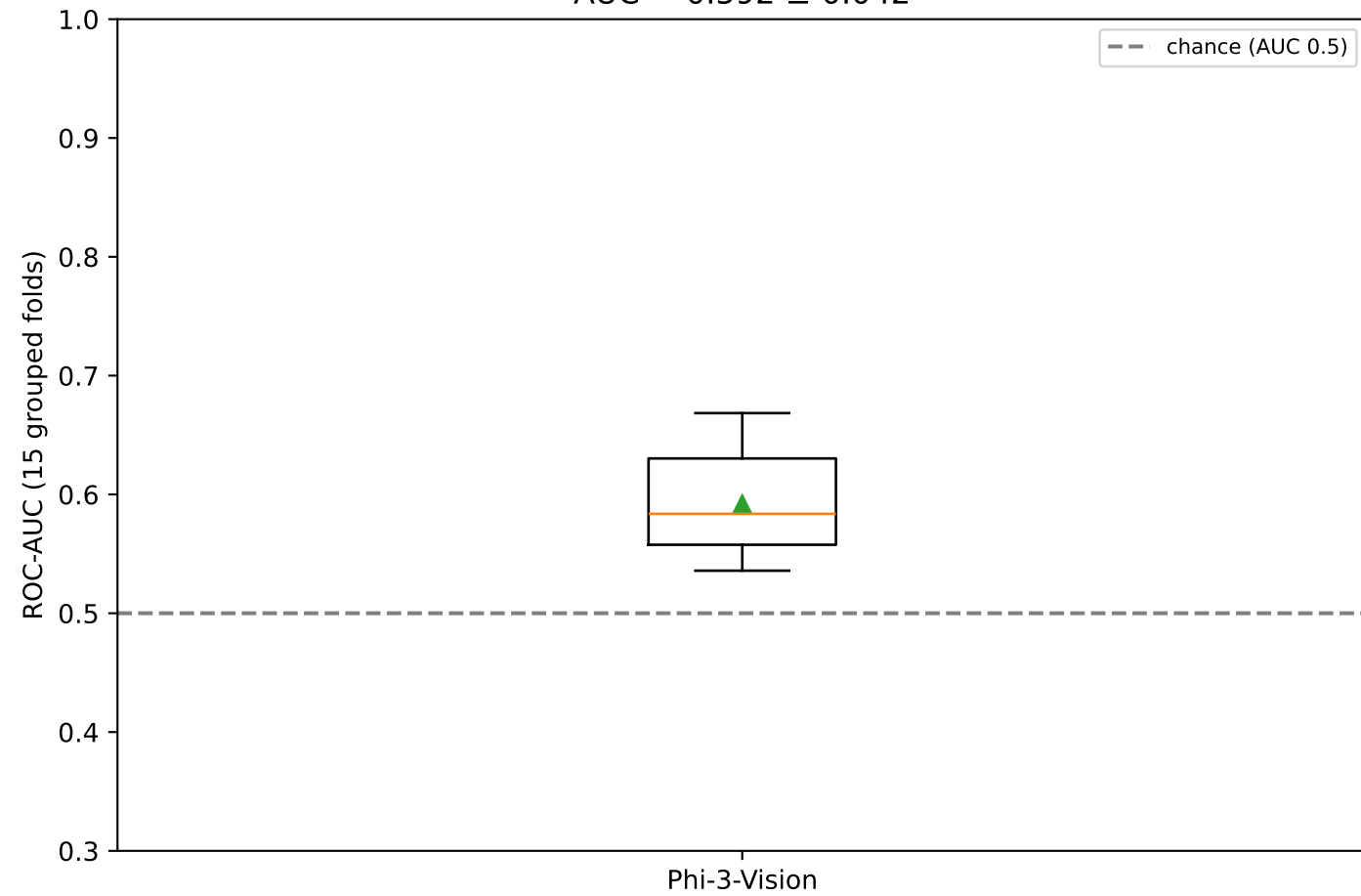
AUC = 0.611 ± 0.040



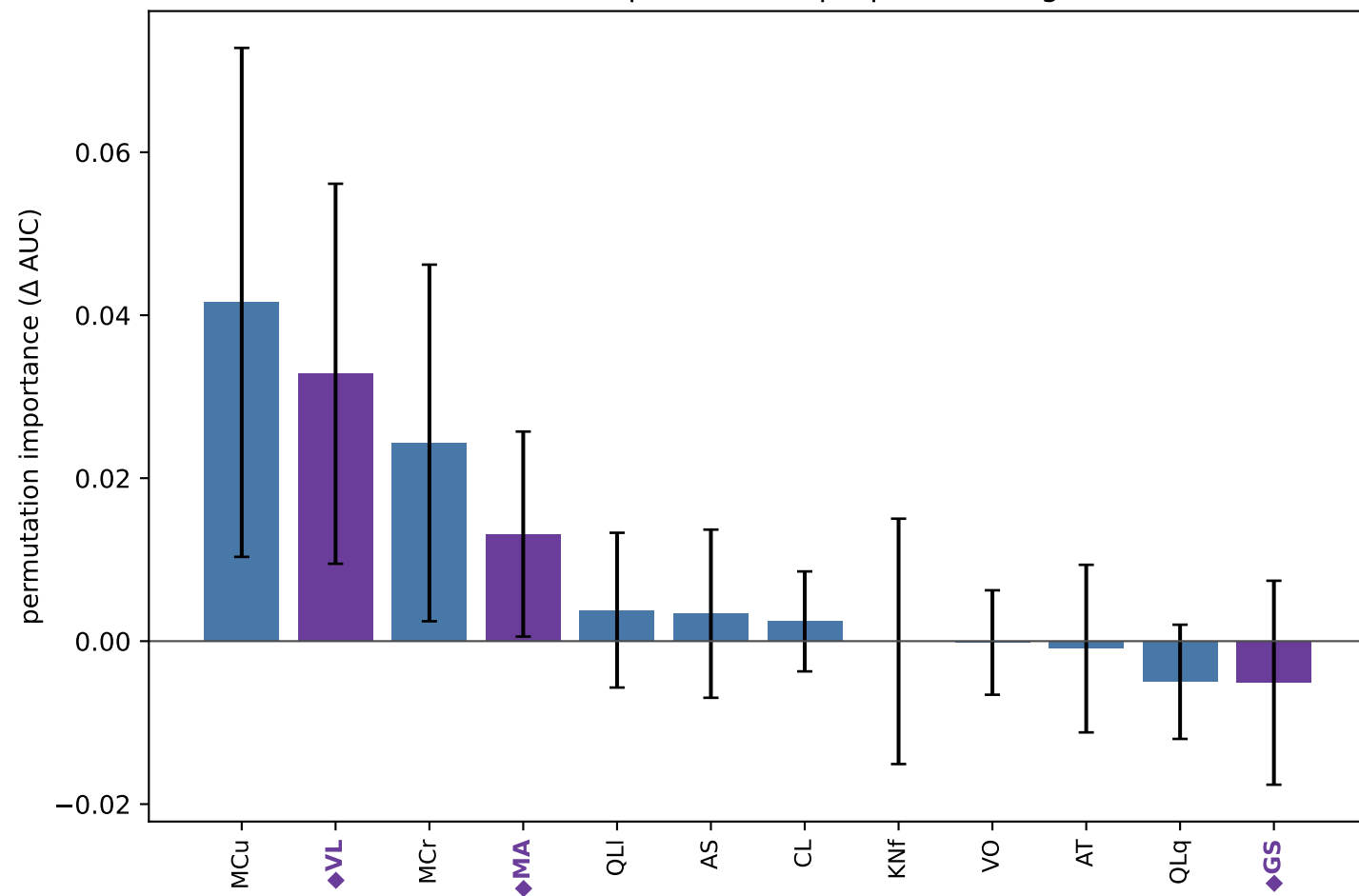
Permutation importance (◆ purple = emergent)

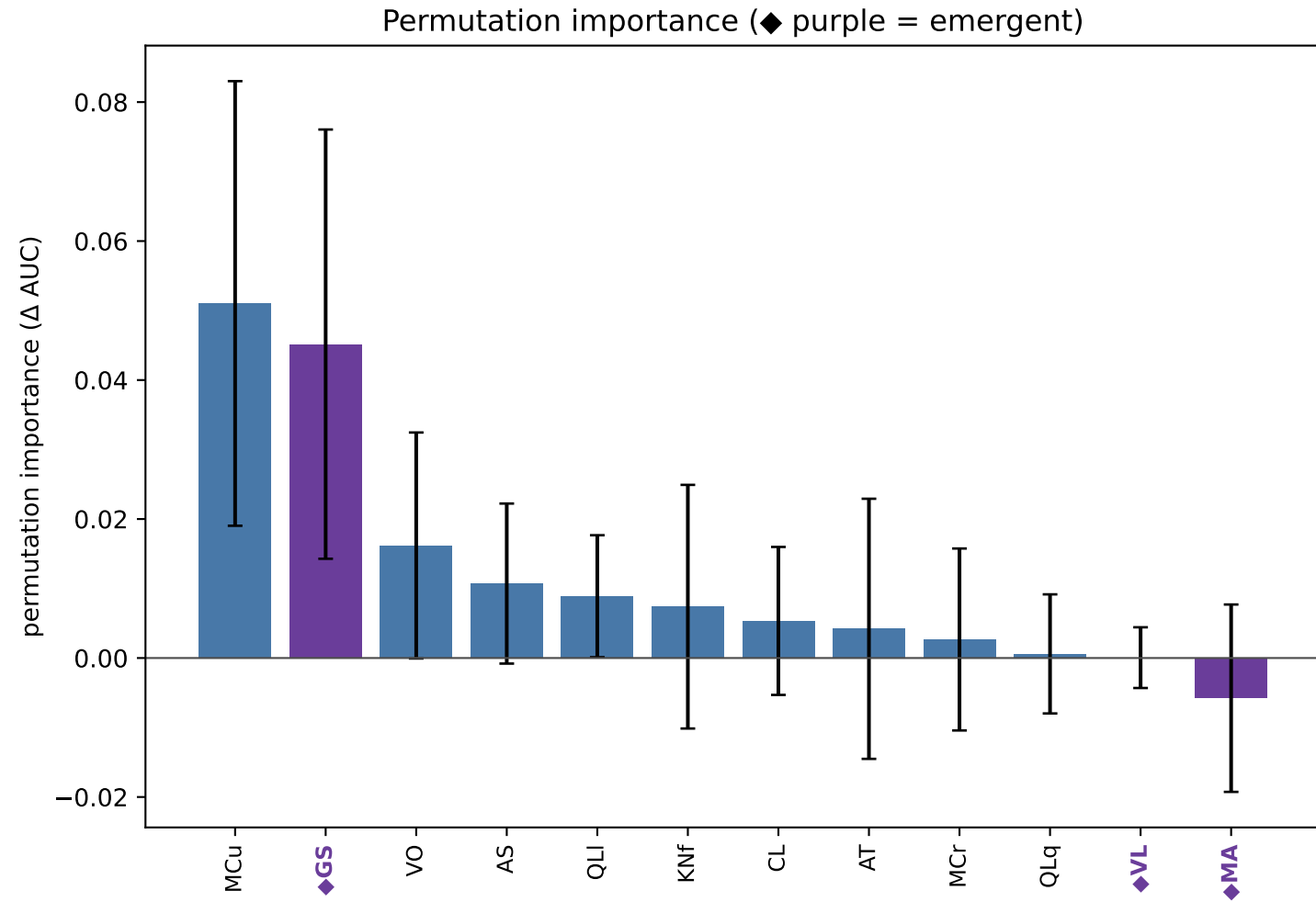
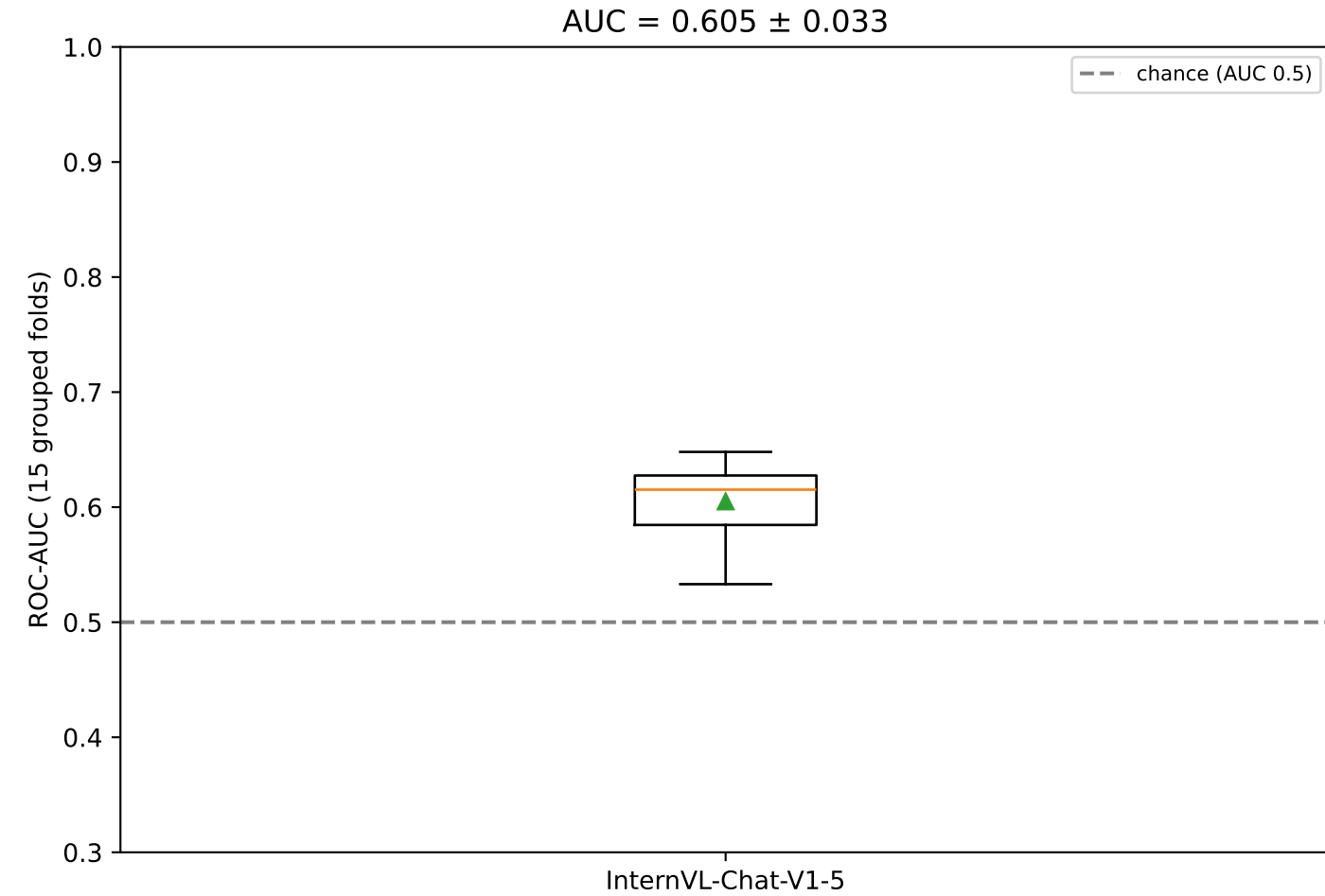


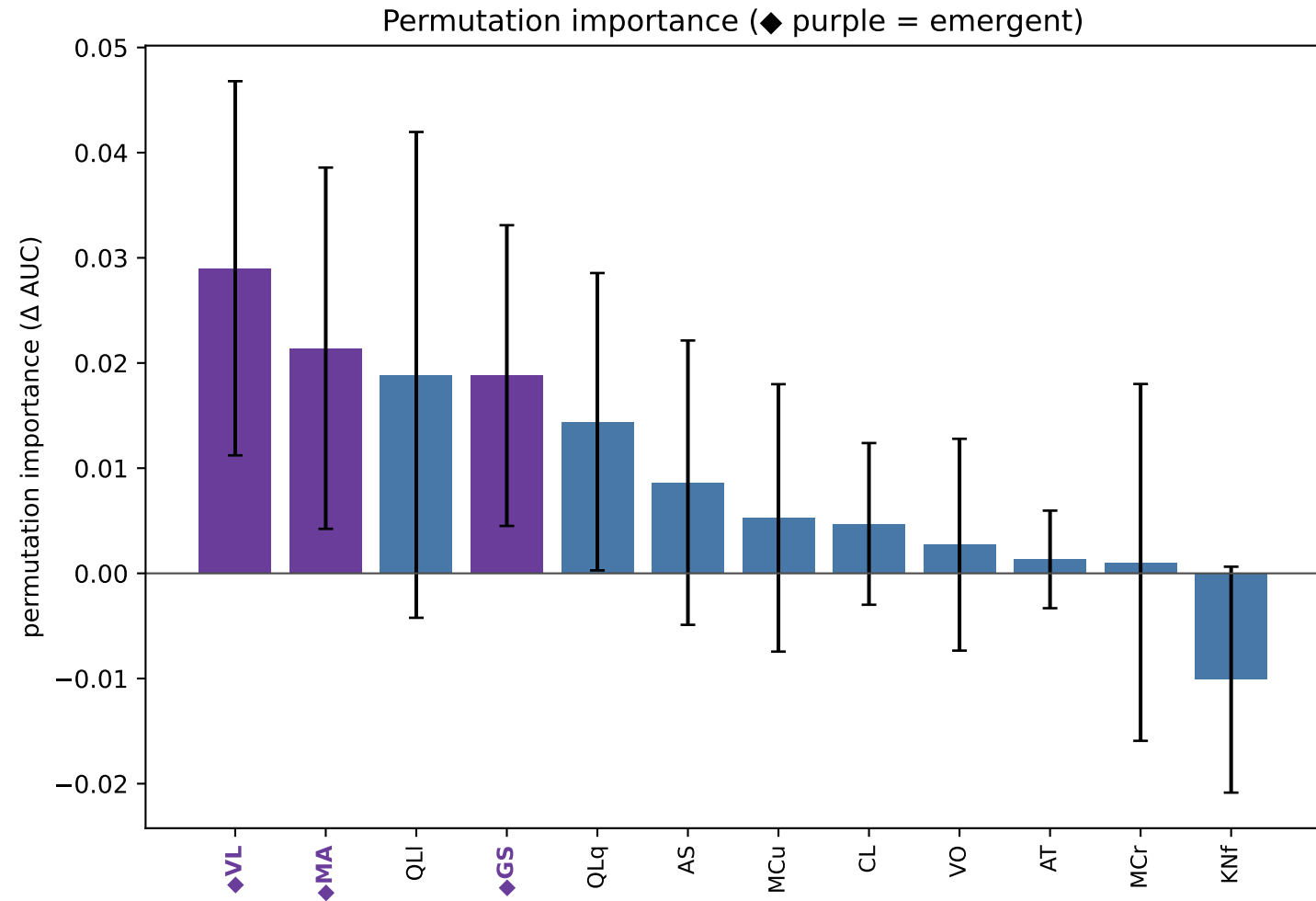
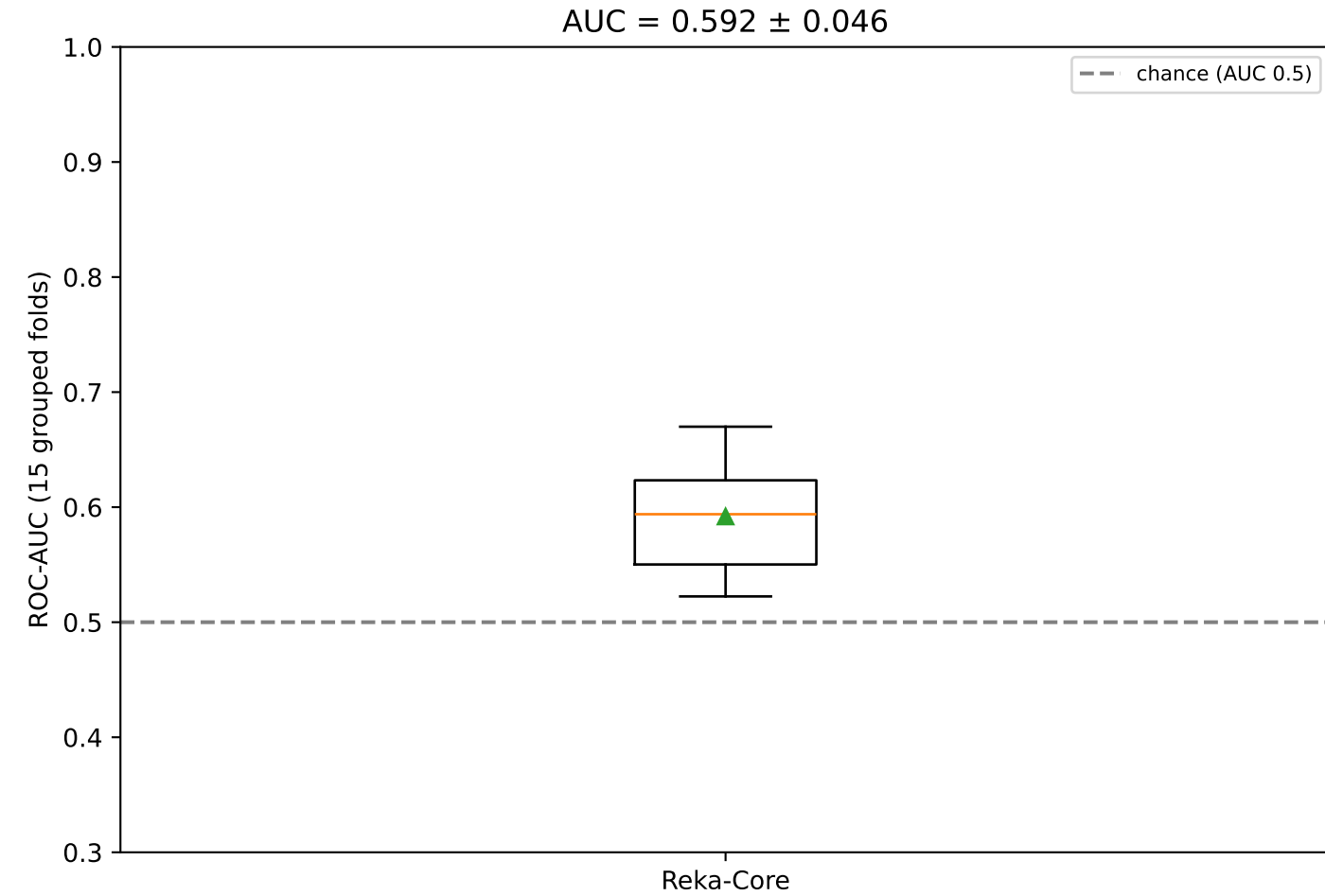
AUC = 0.592 ± 0.042

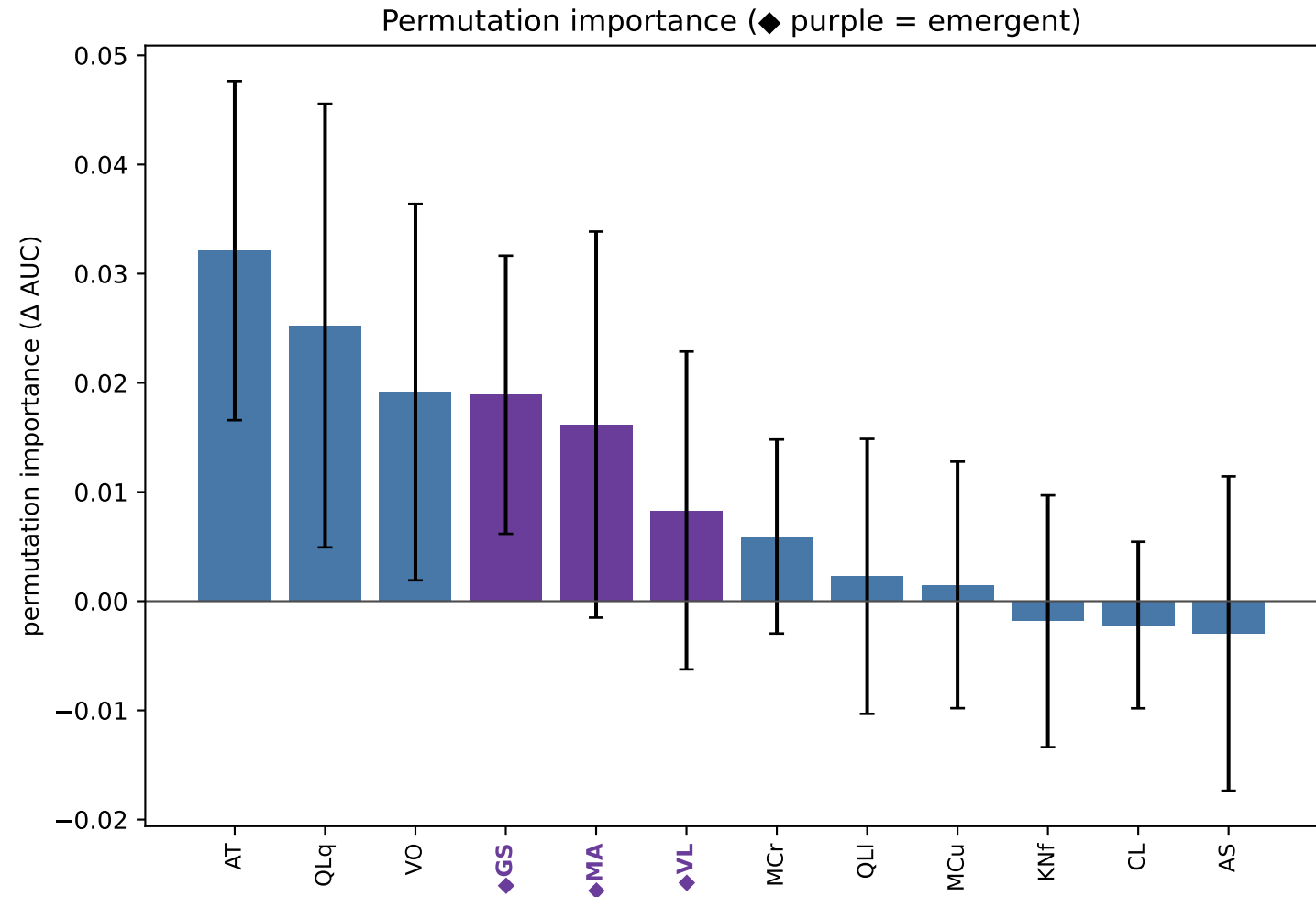
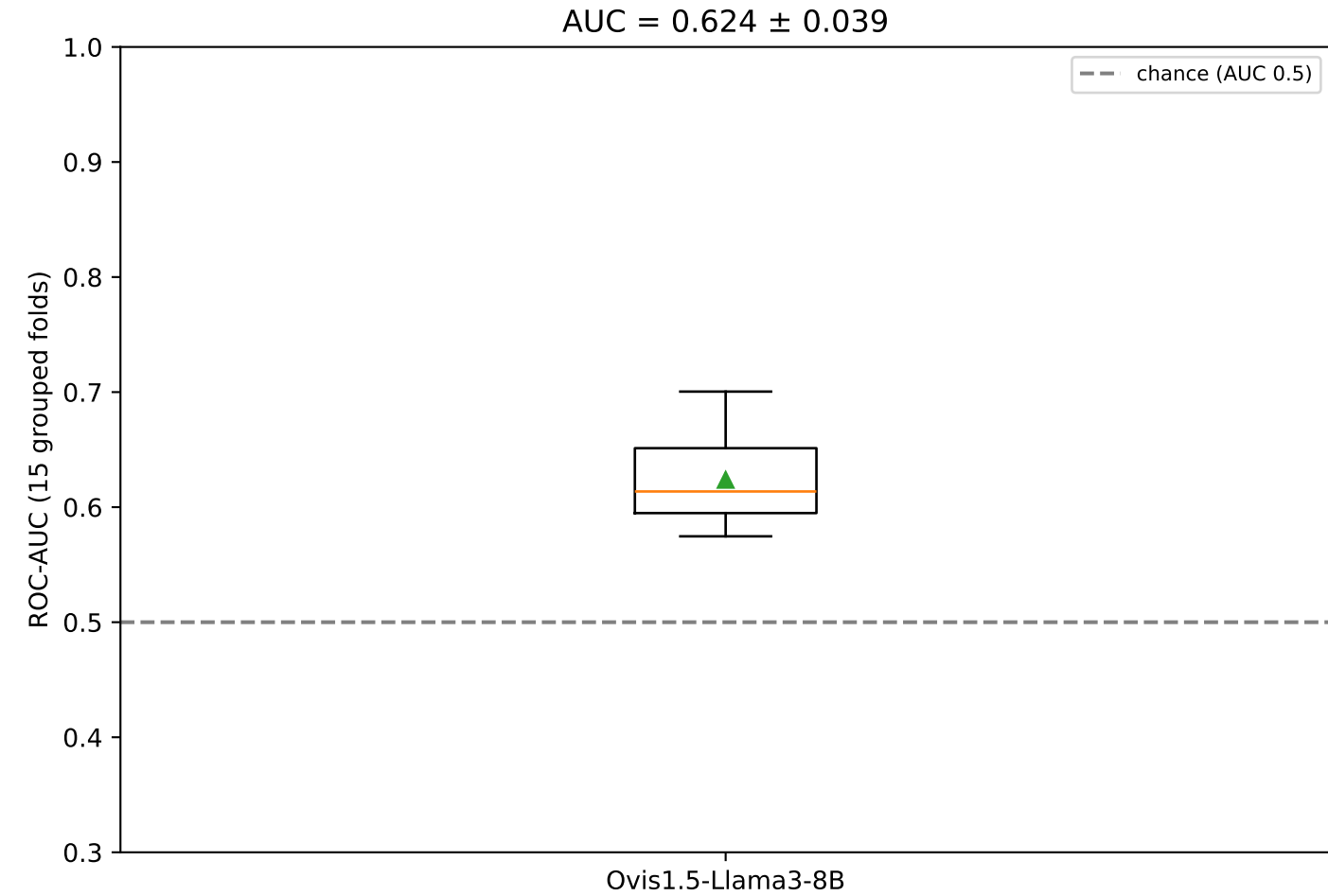


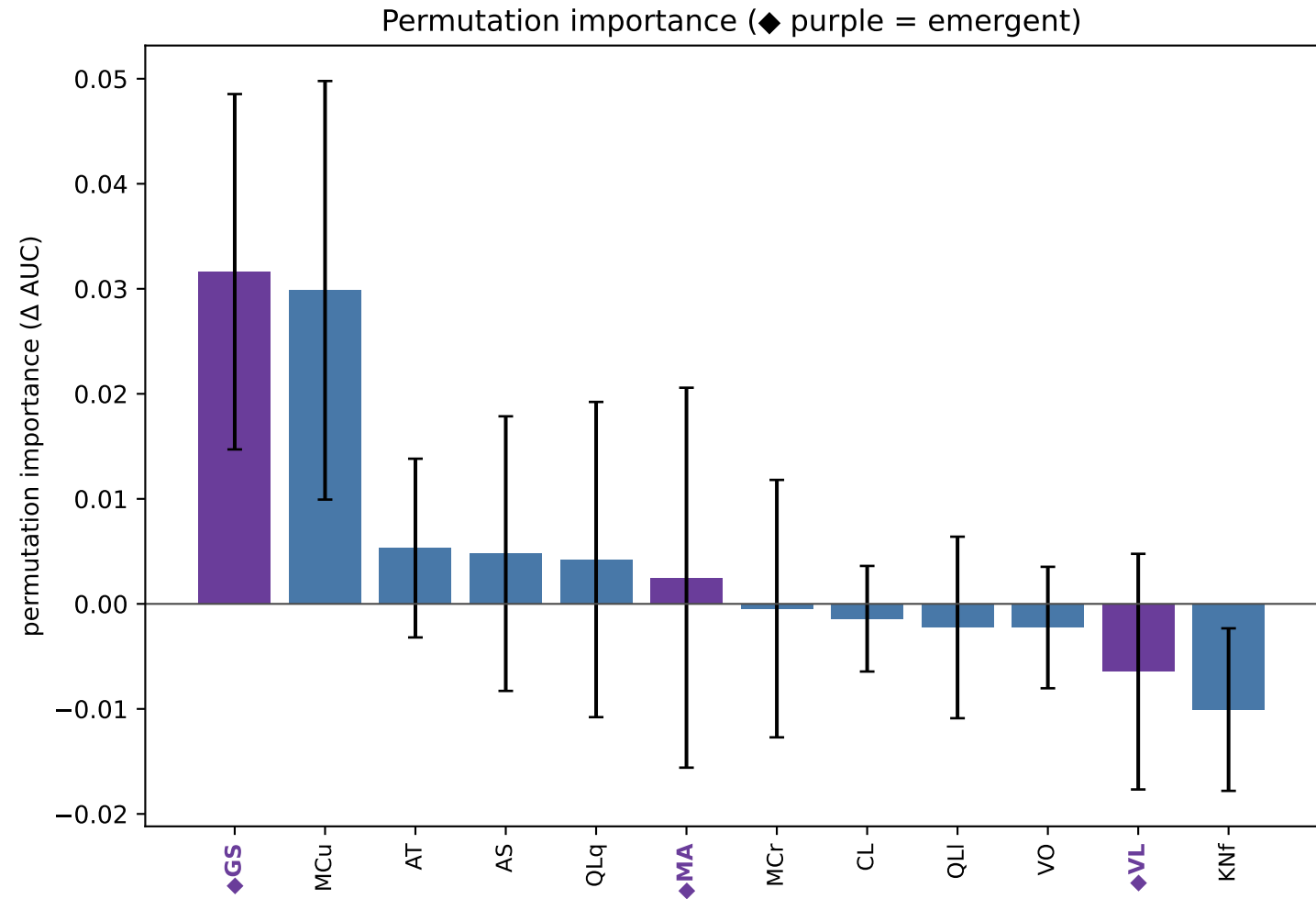
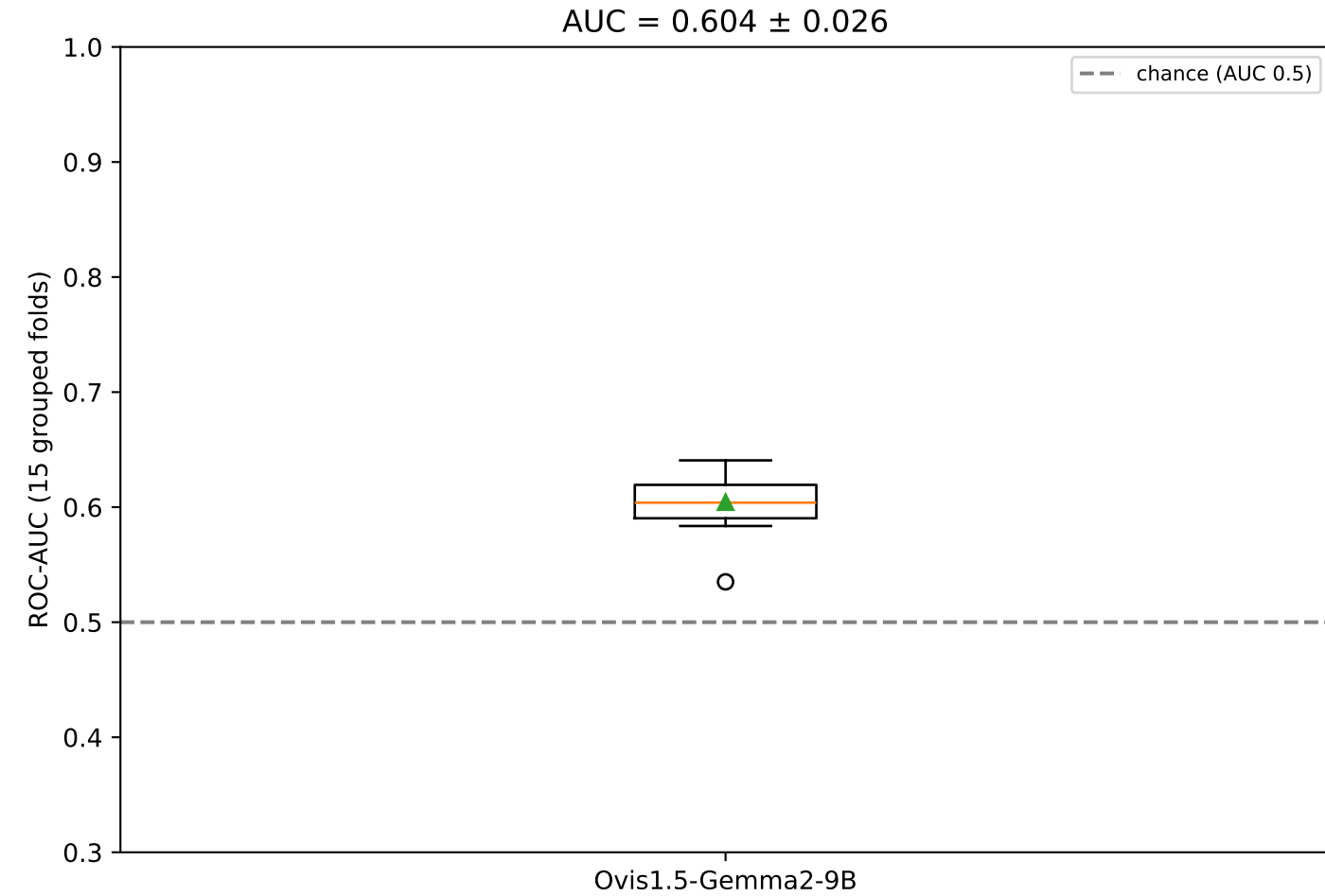
Permutation importance (◆ purple = emergent)



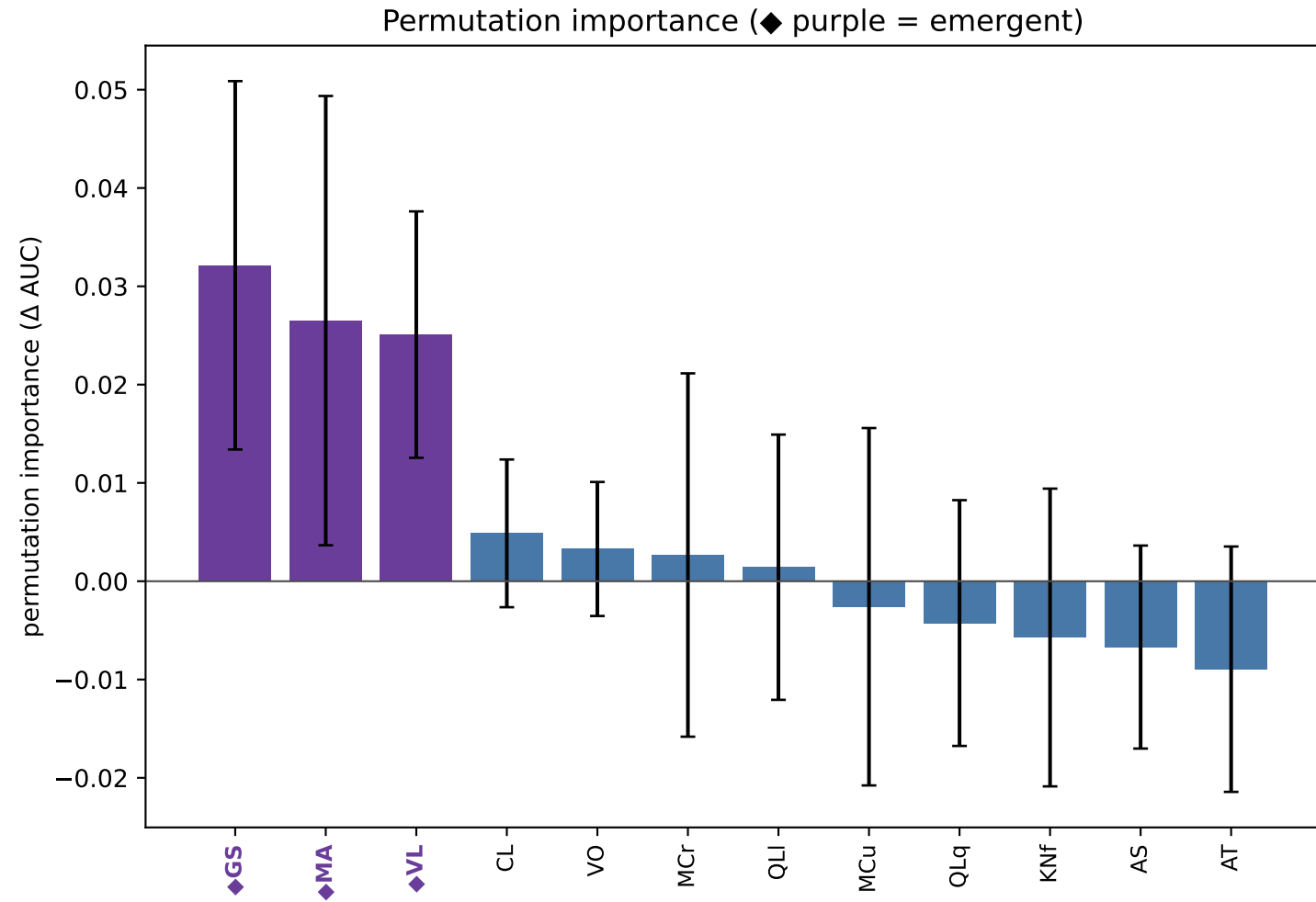
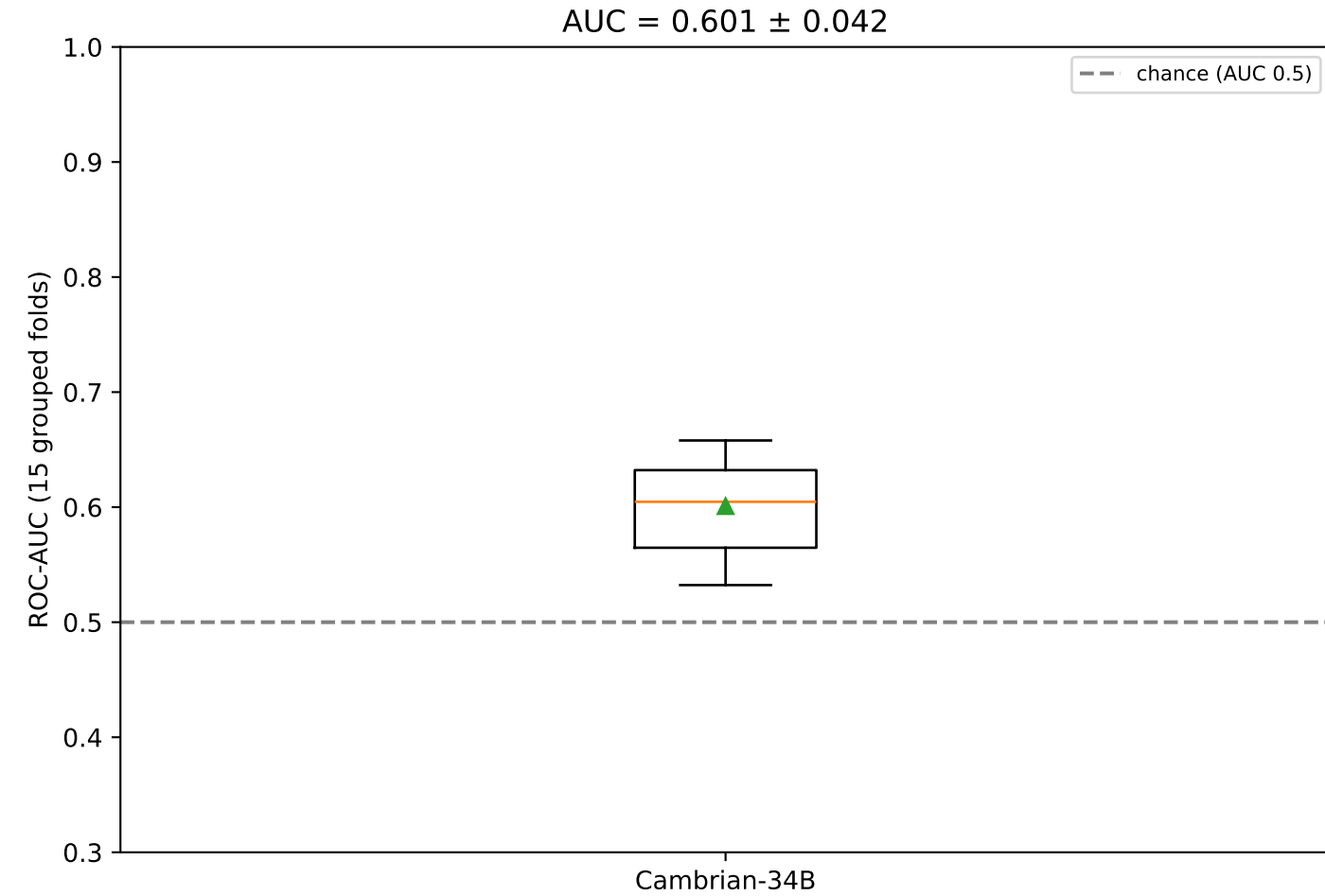


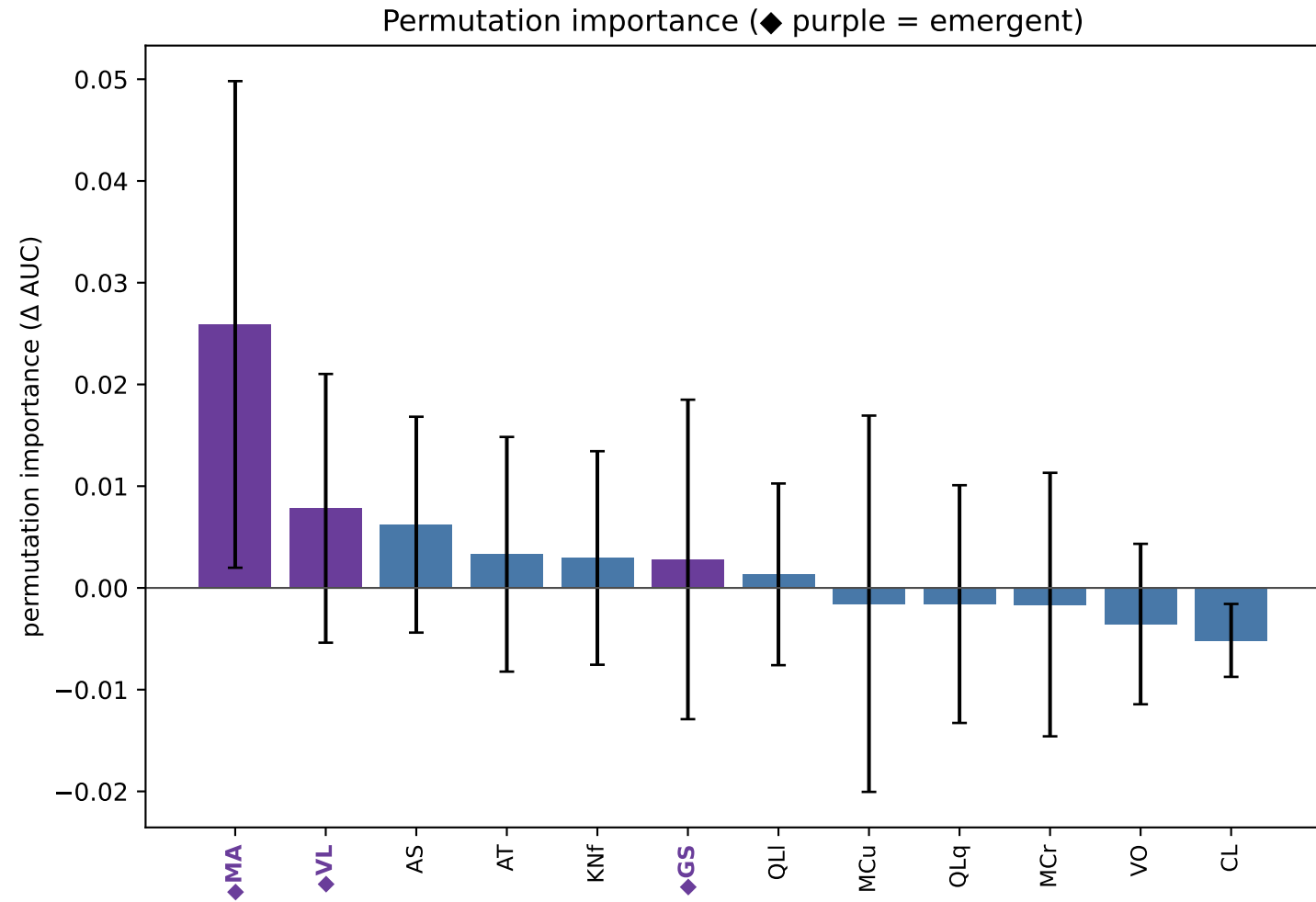
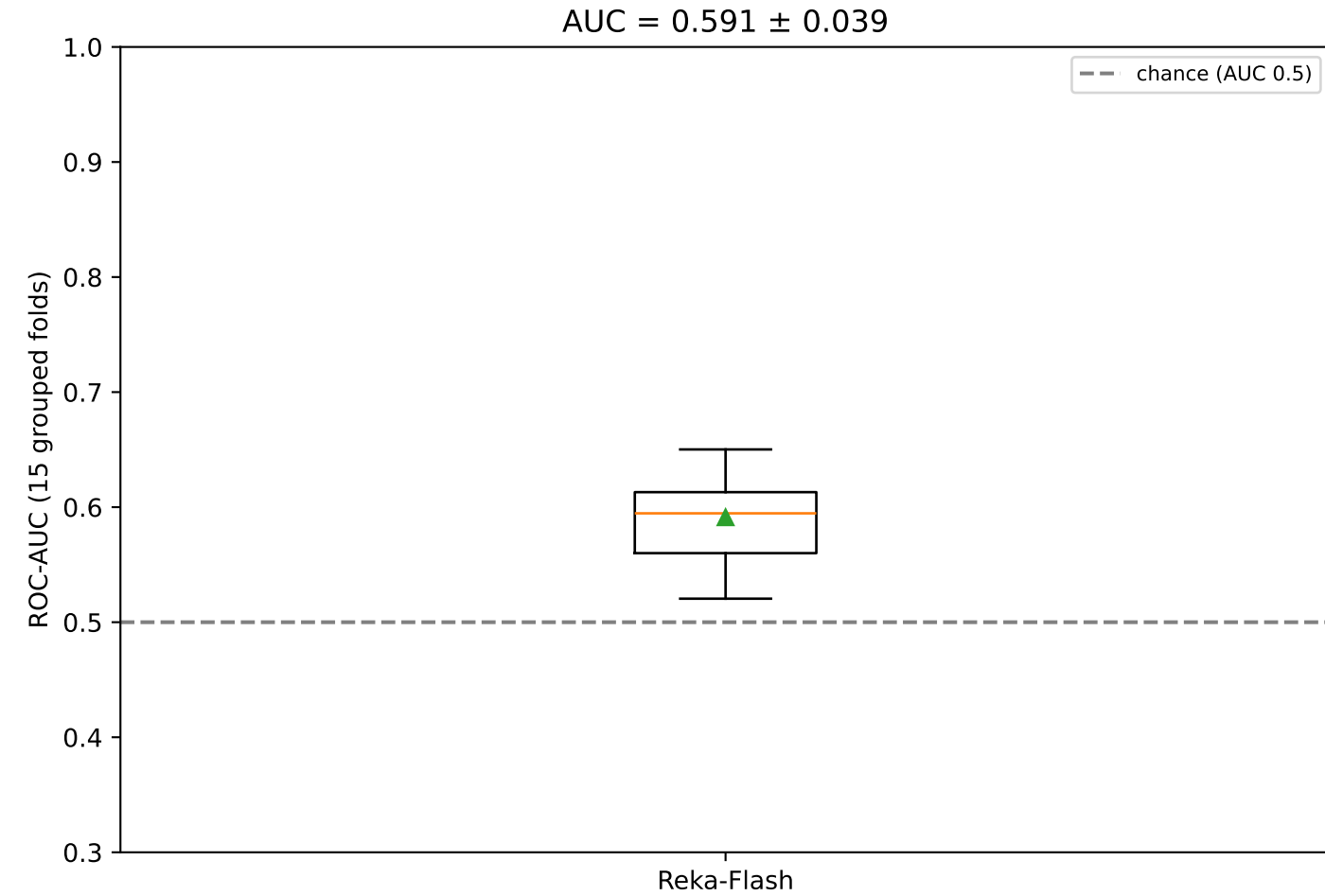


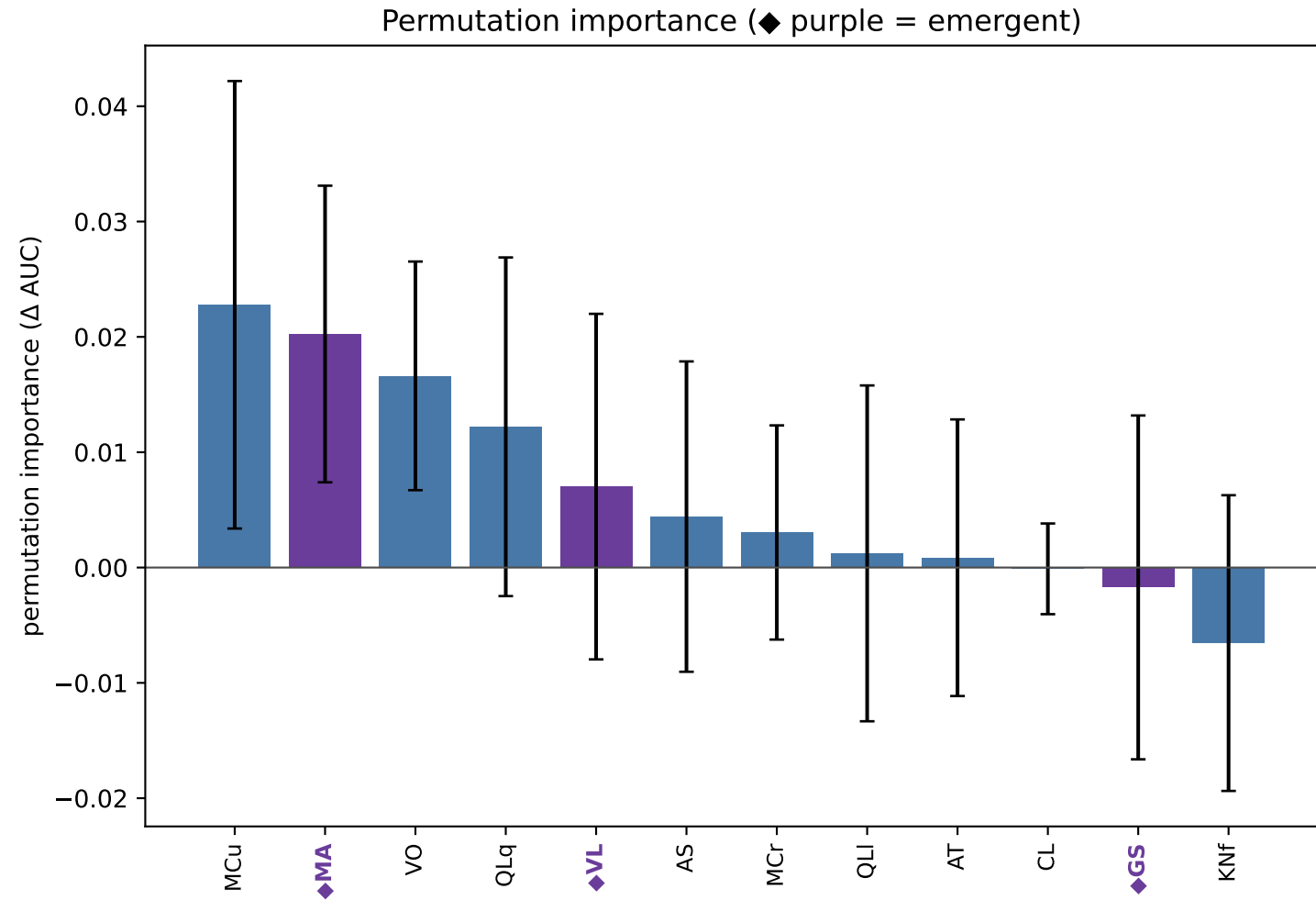
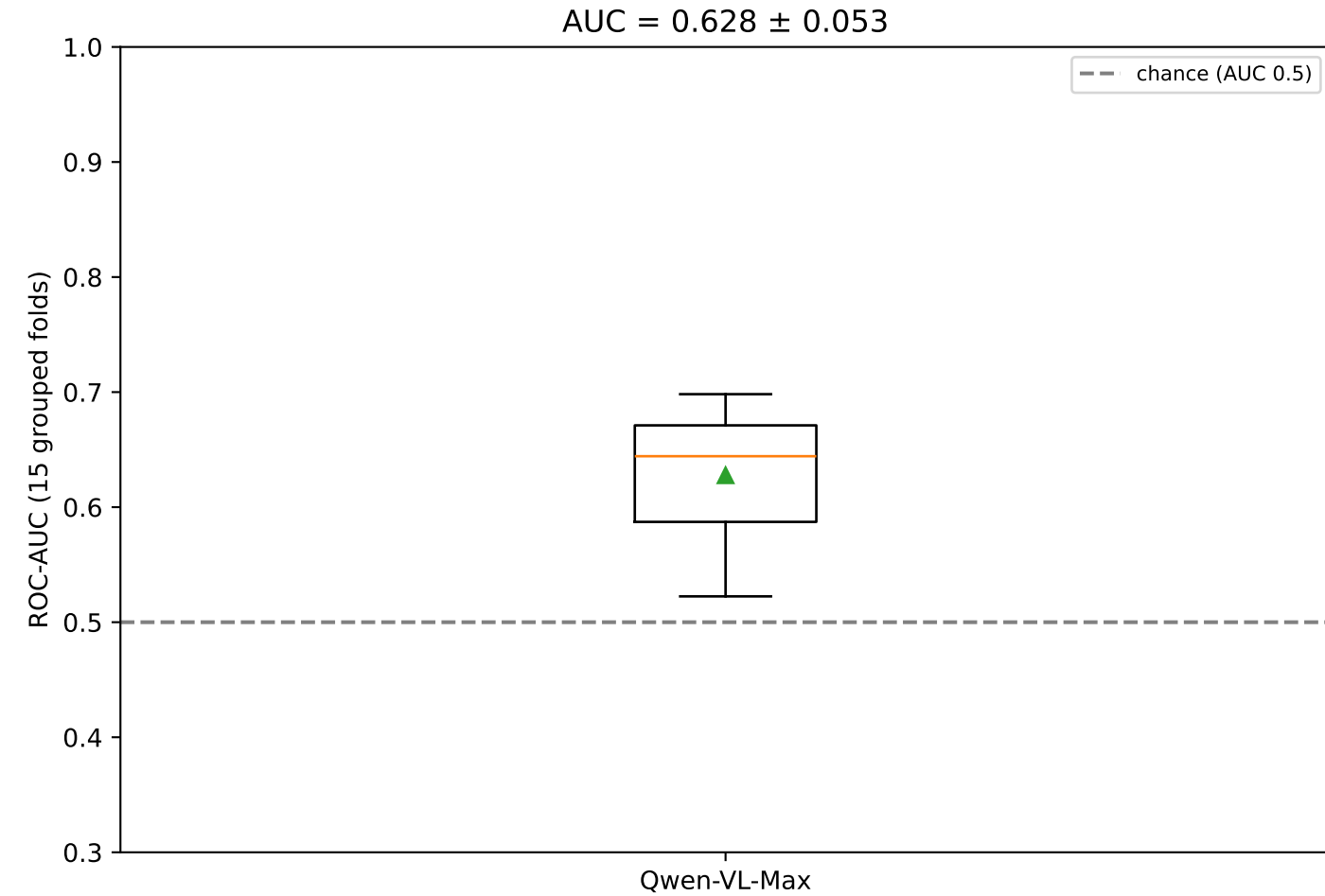


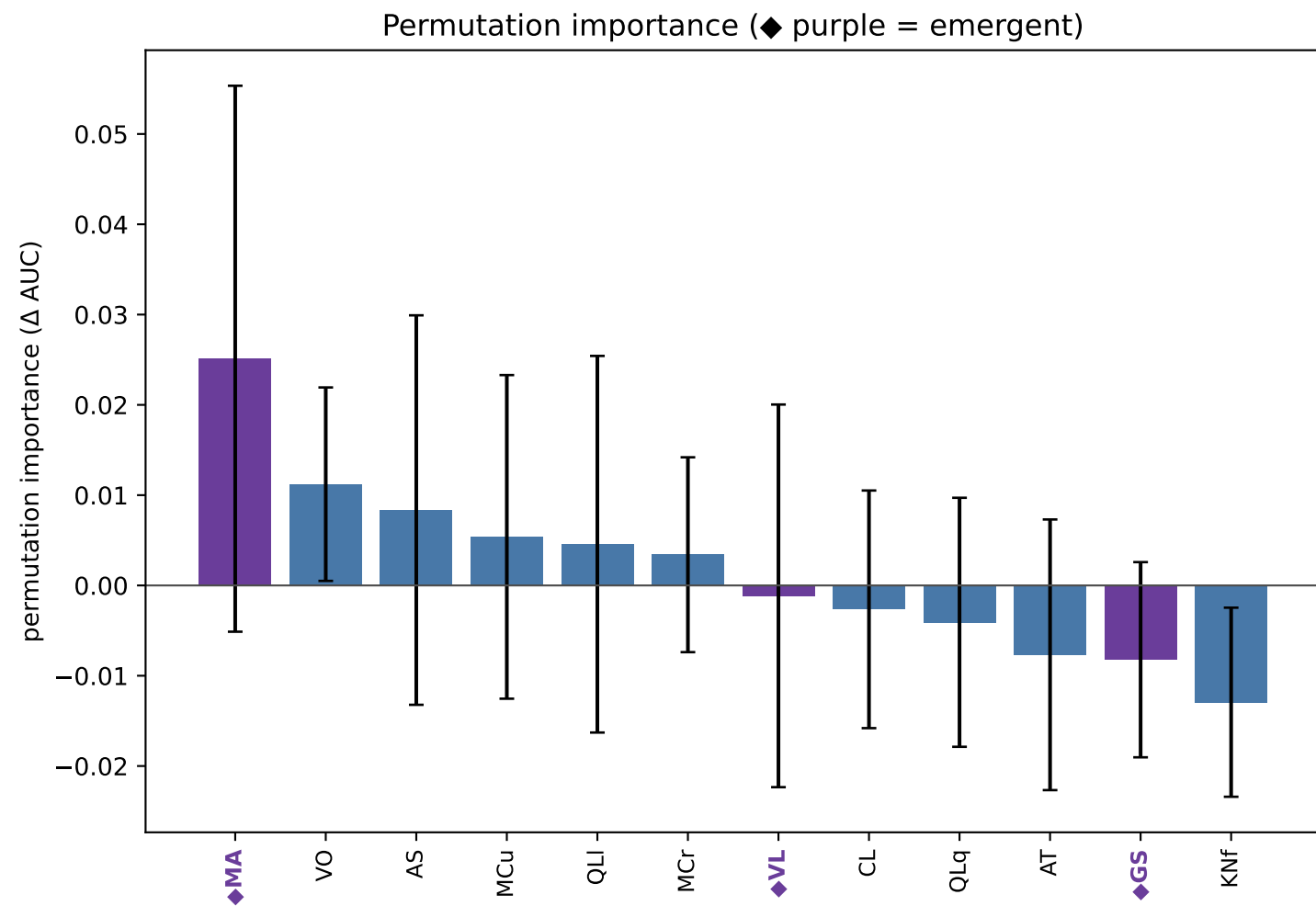
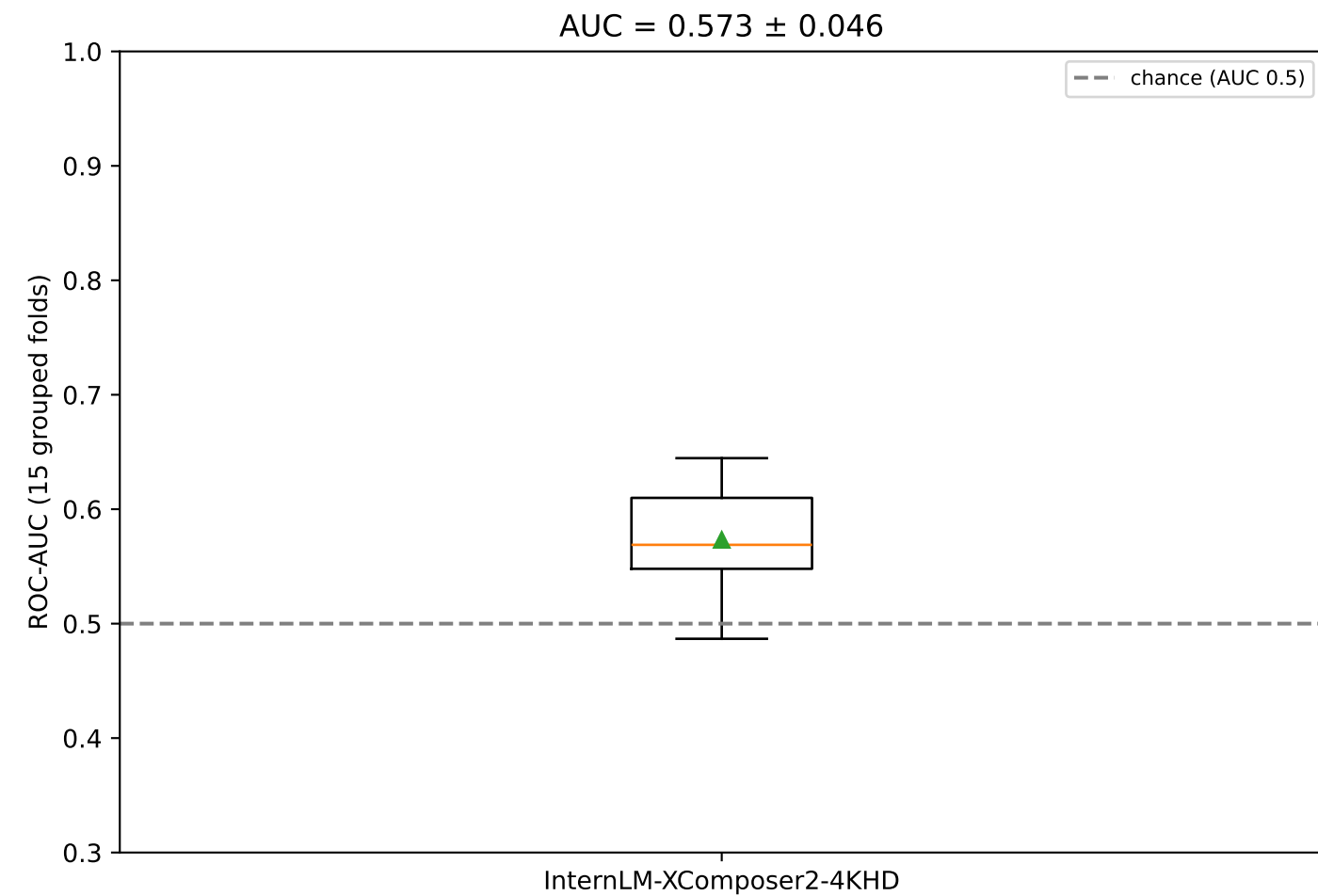


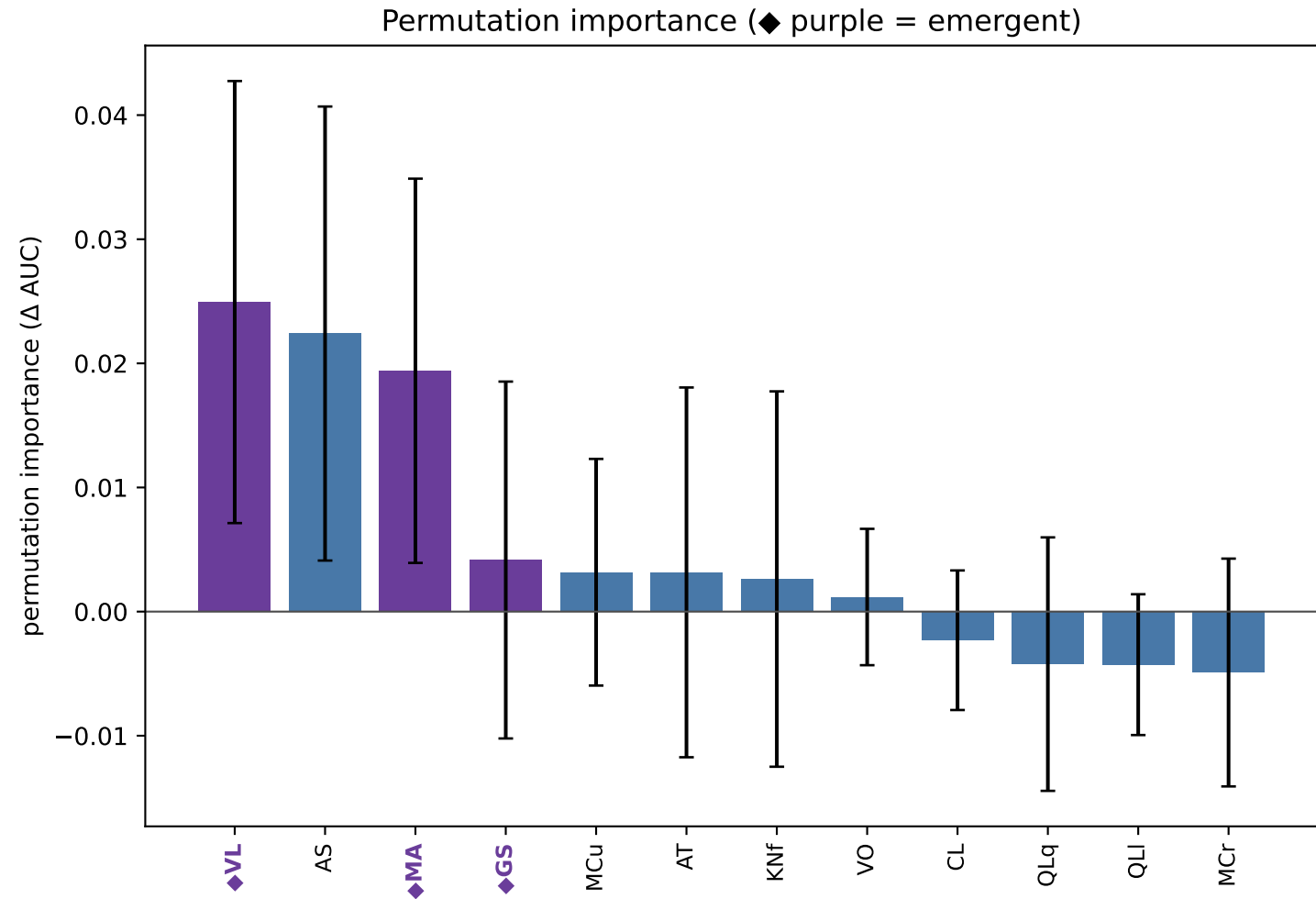
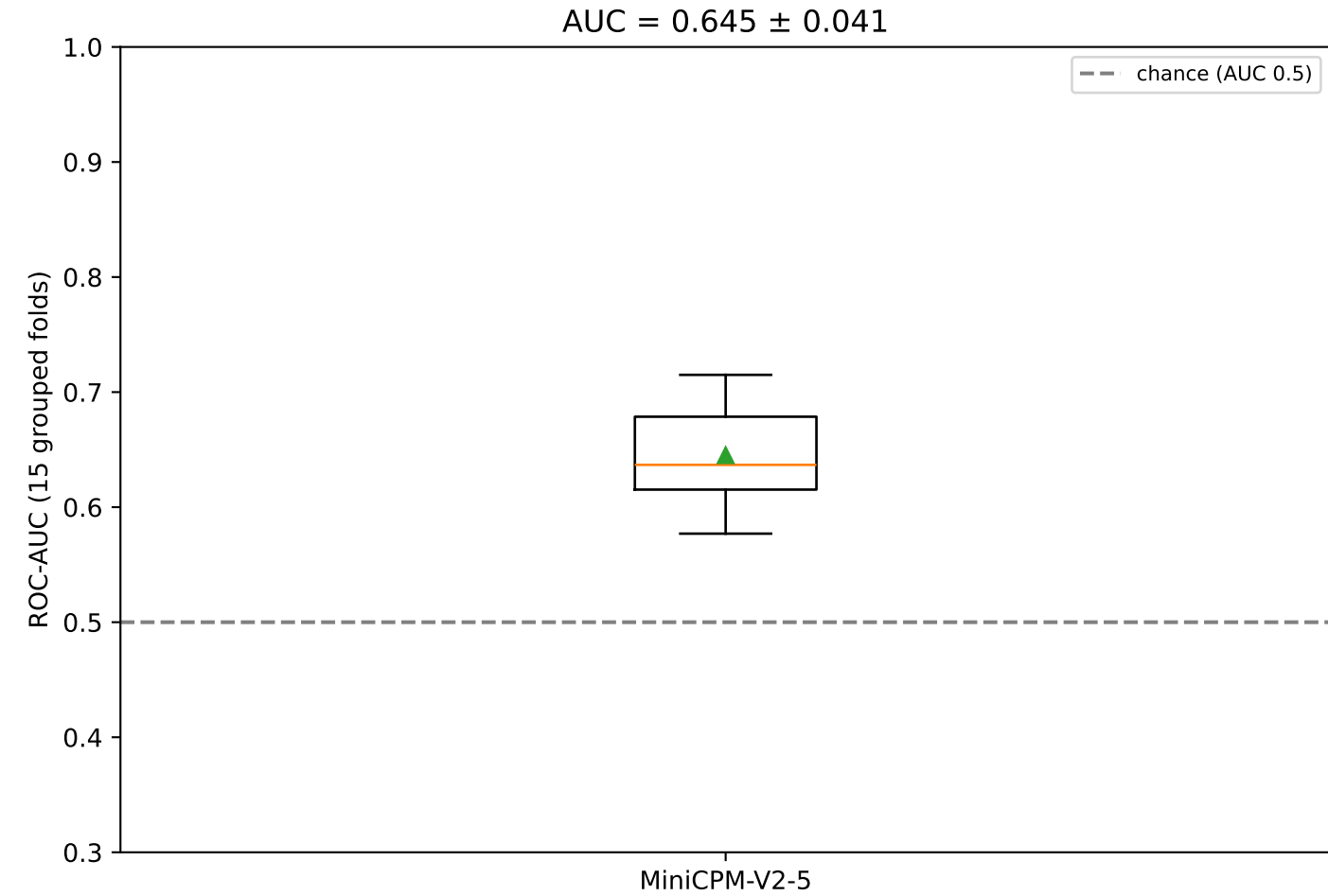
Cambrian-34B — rubric→failure RF (y=1 INCORRECT; base rate 0.738, n=800, paper-grouped 5-fold ×3)

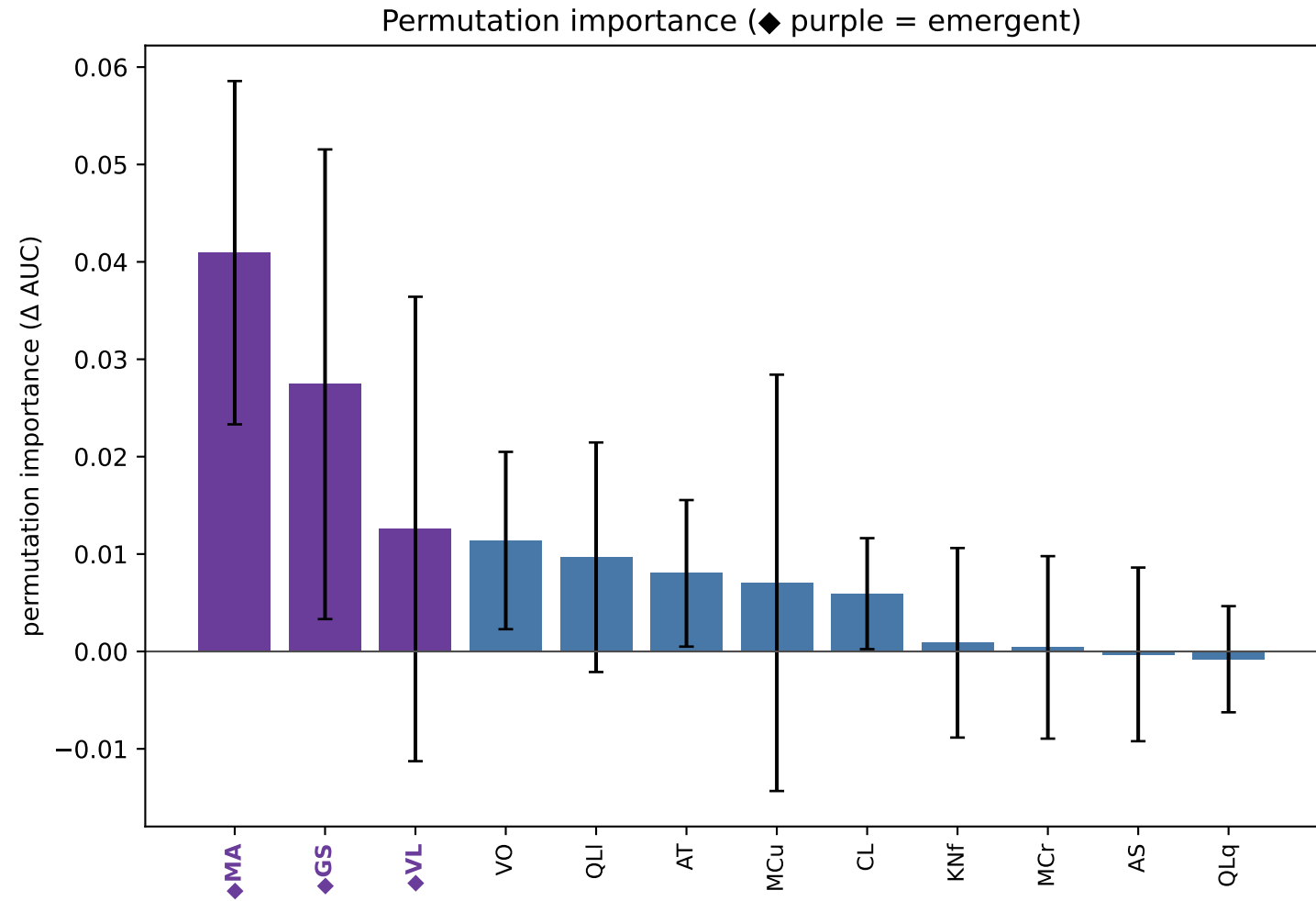
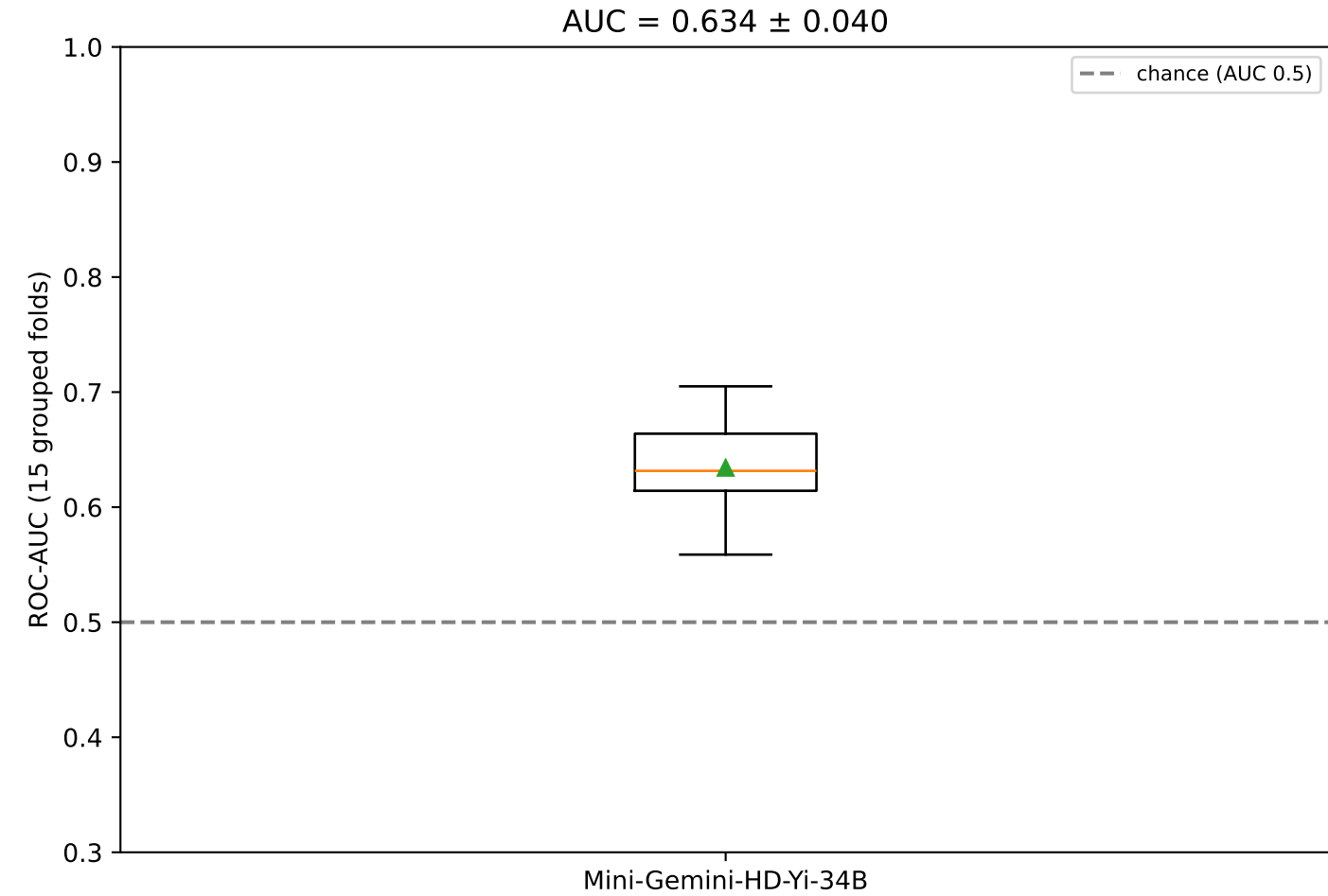




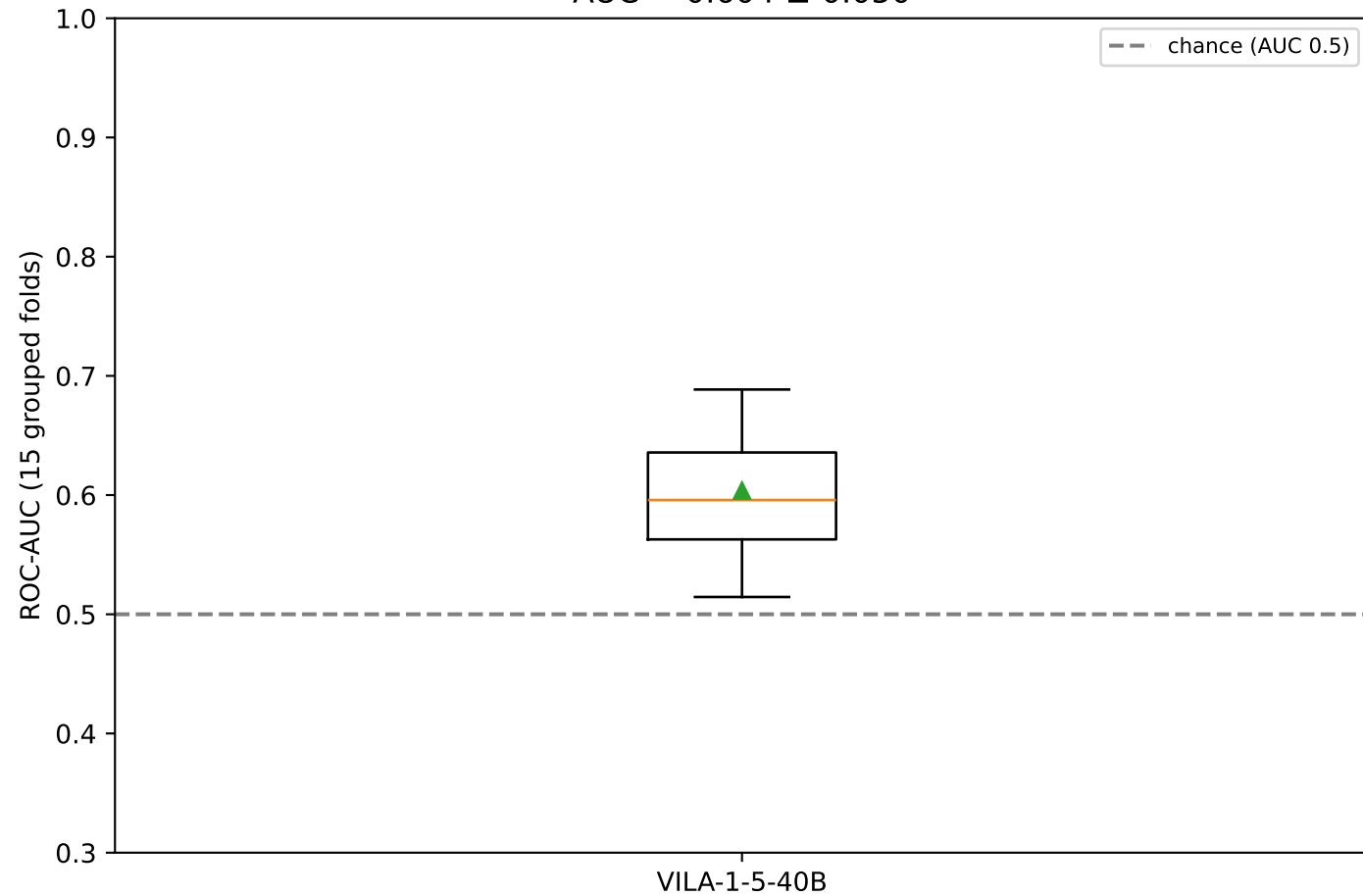




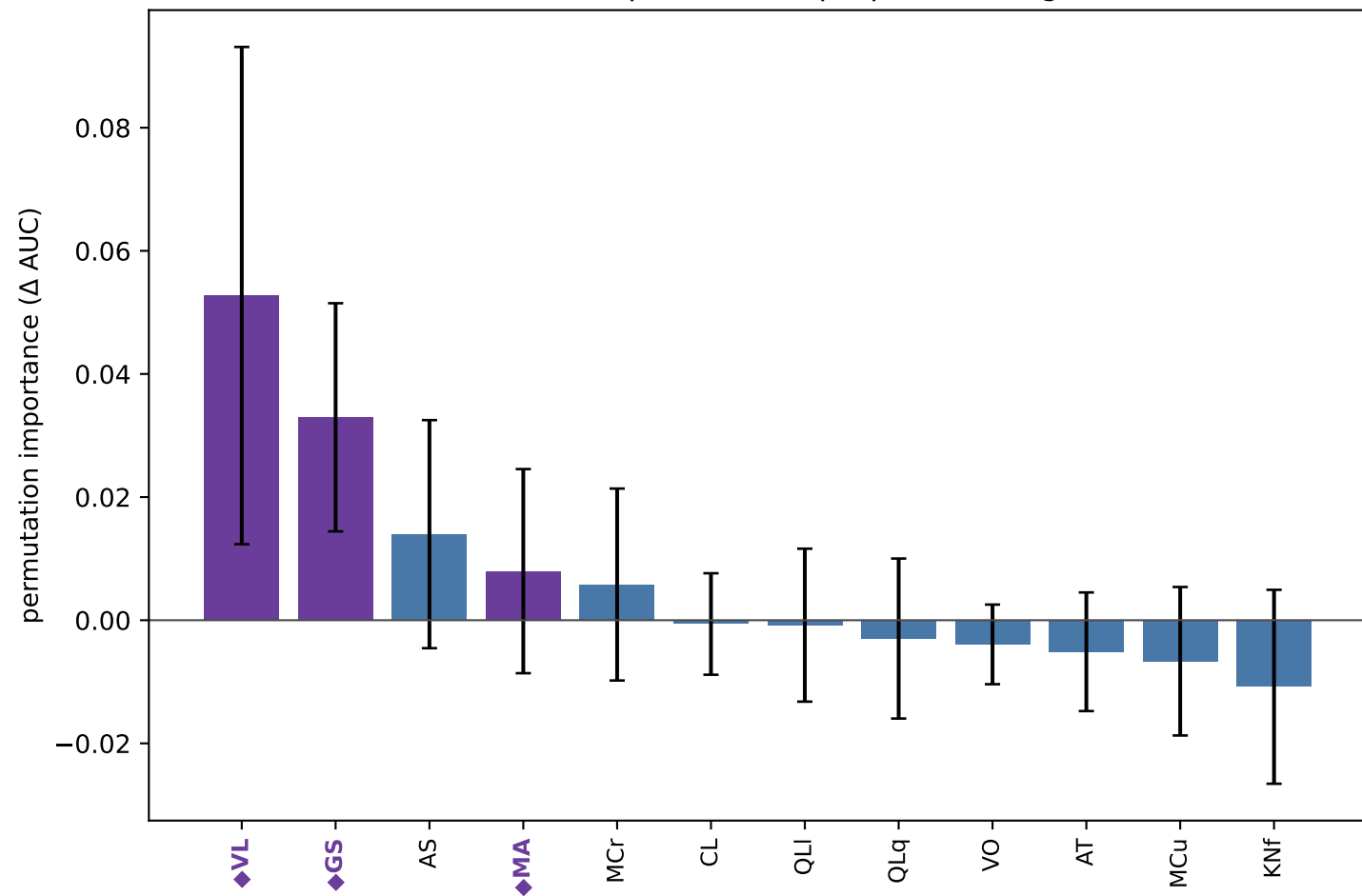


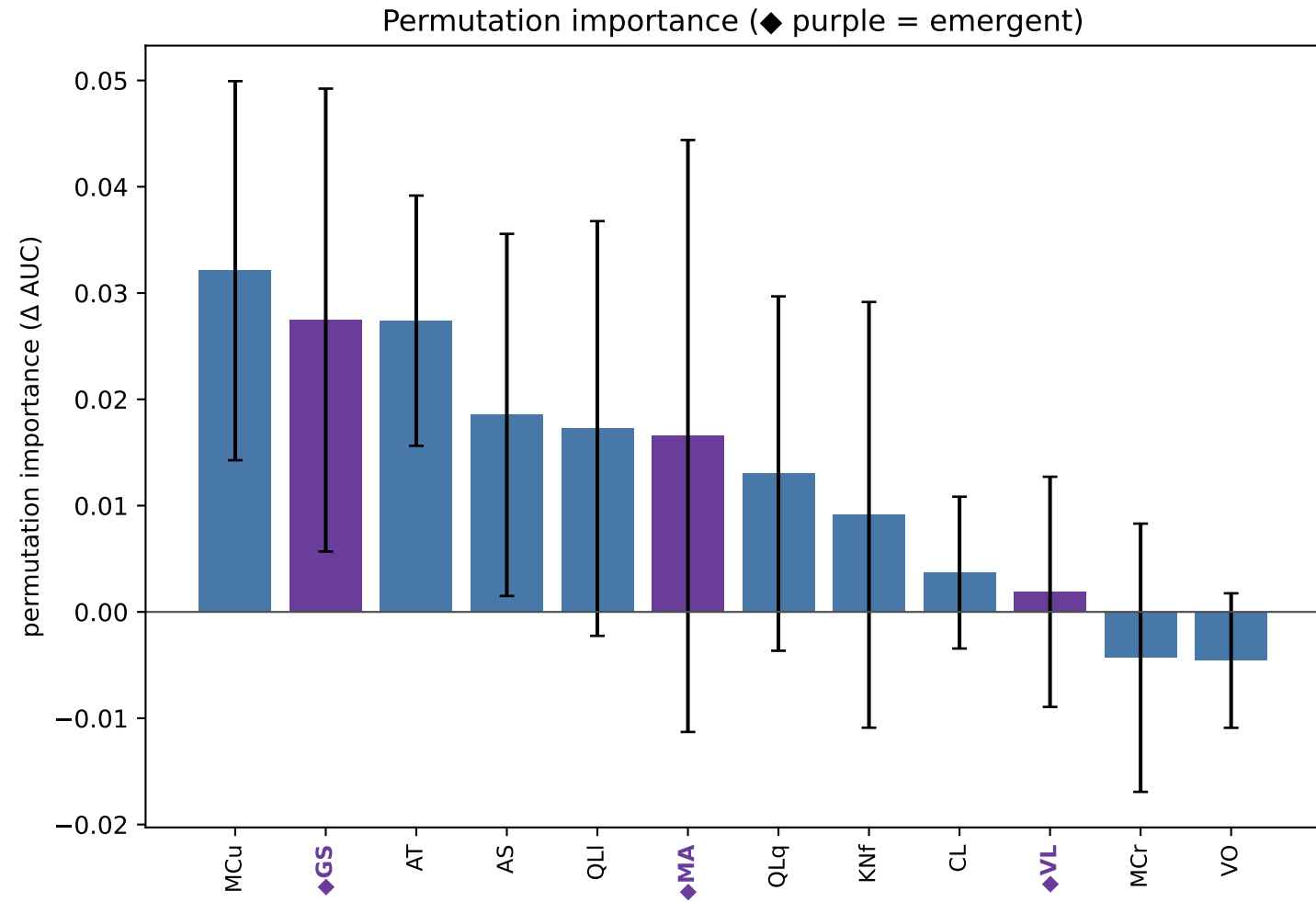
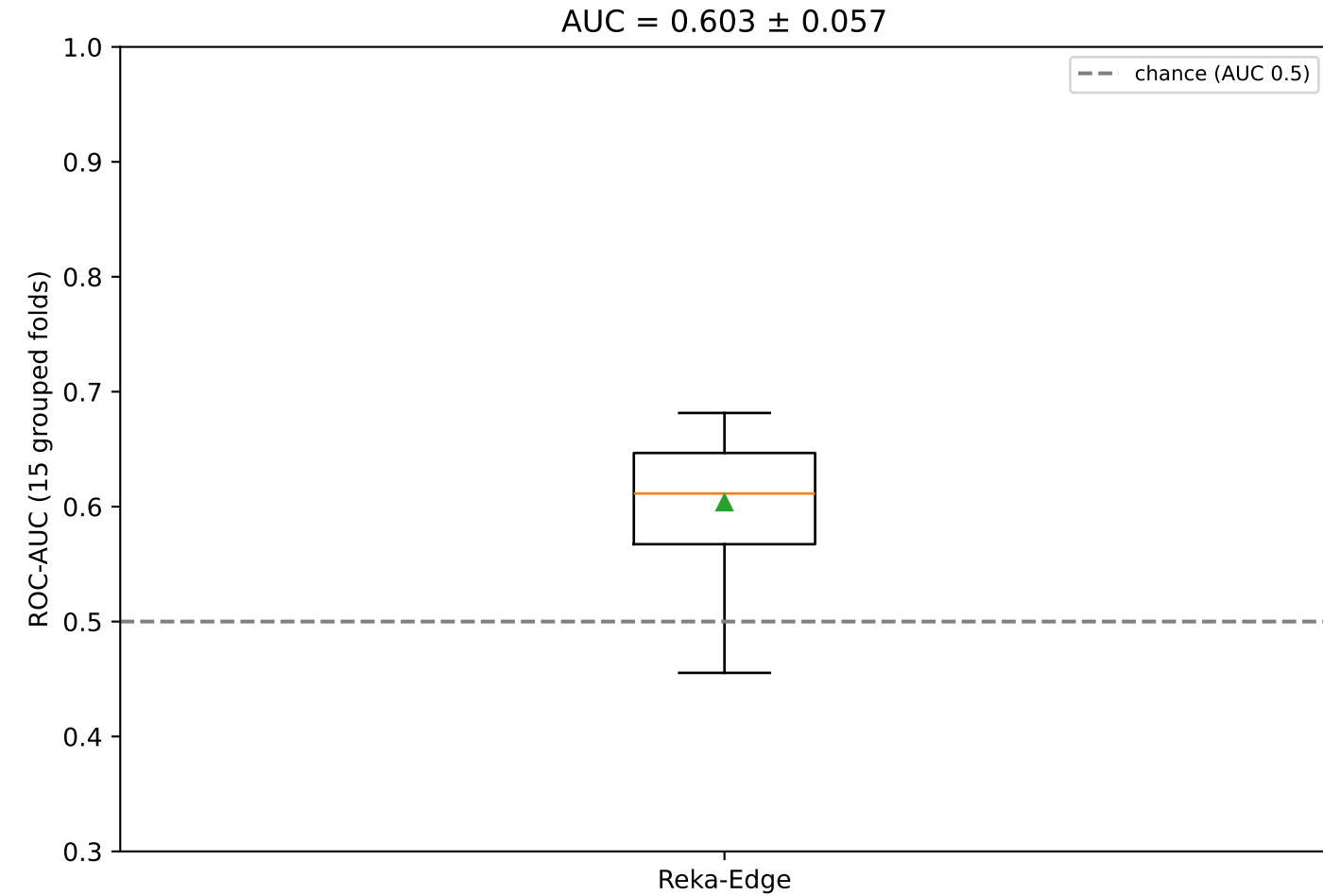


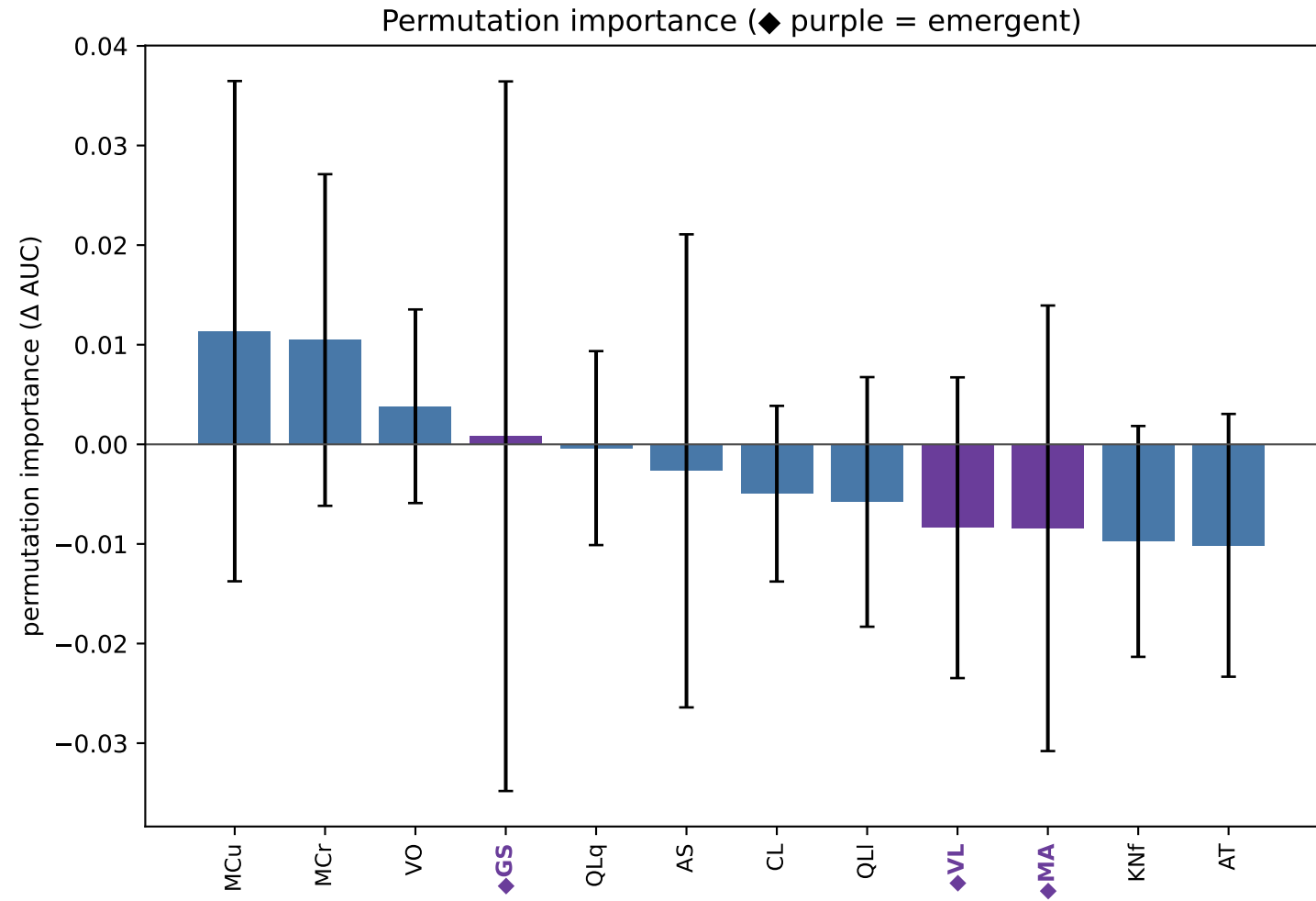
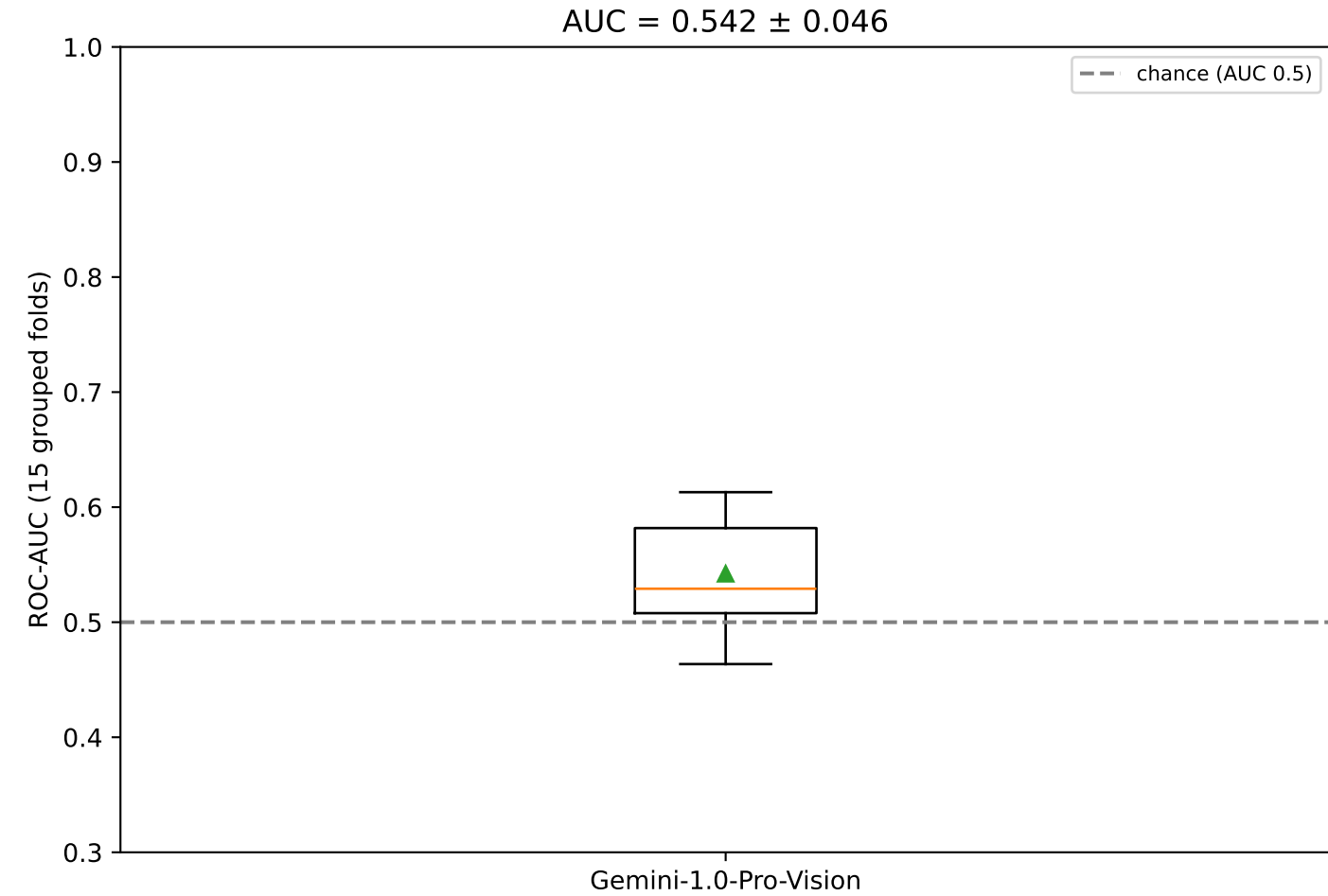
AUC = 0.604 ± 0.050

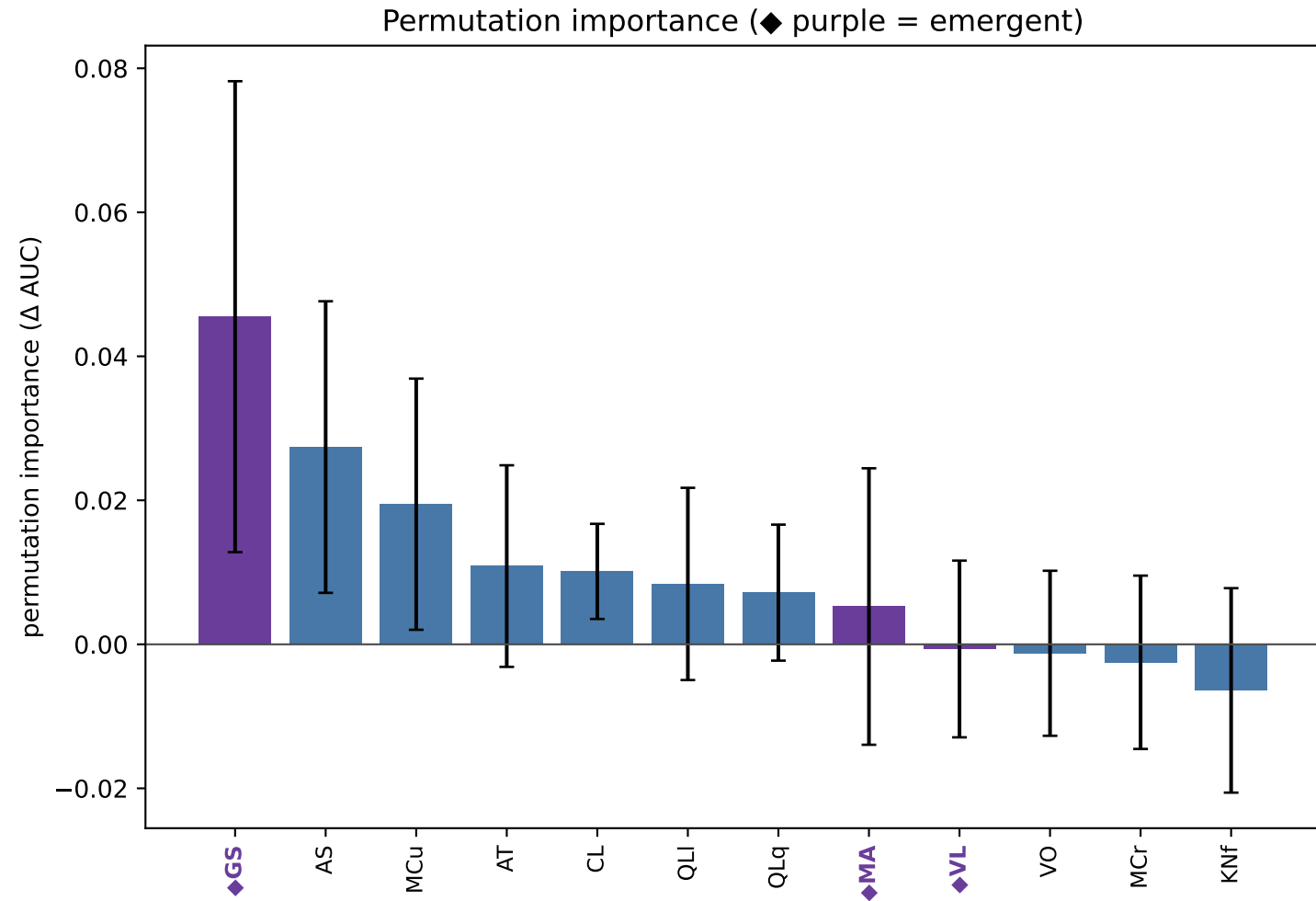
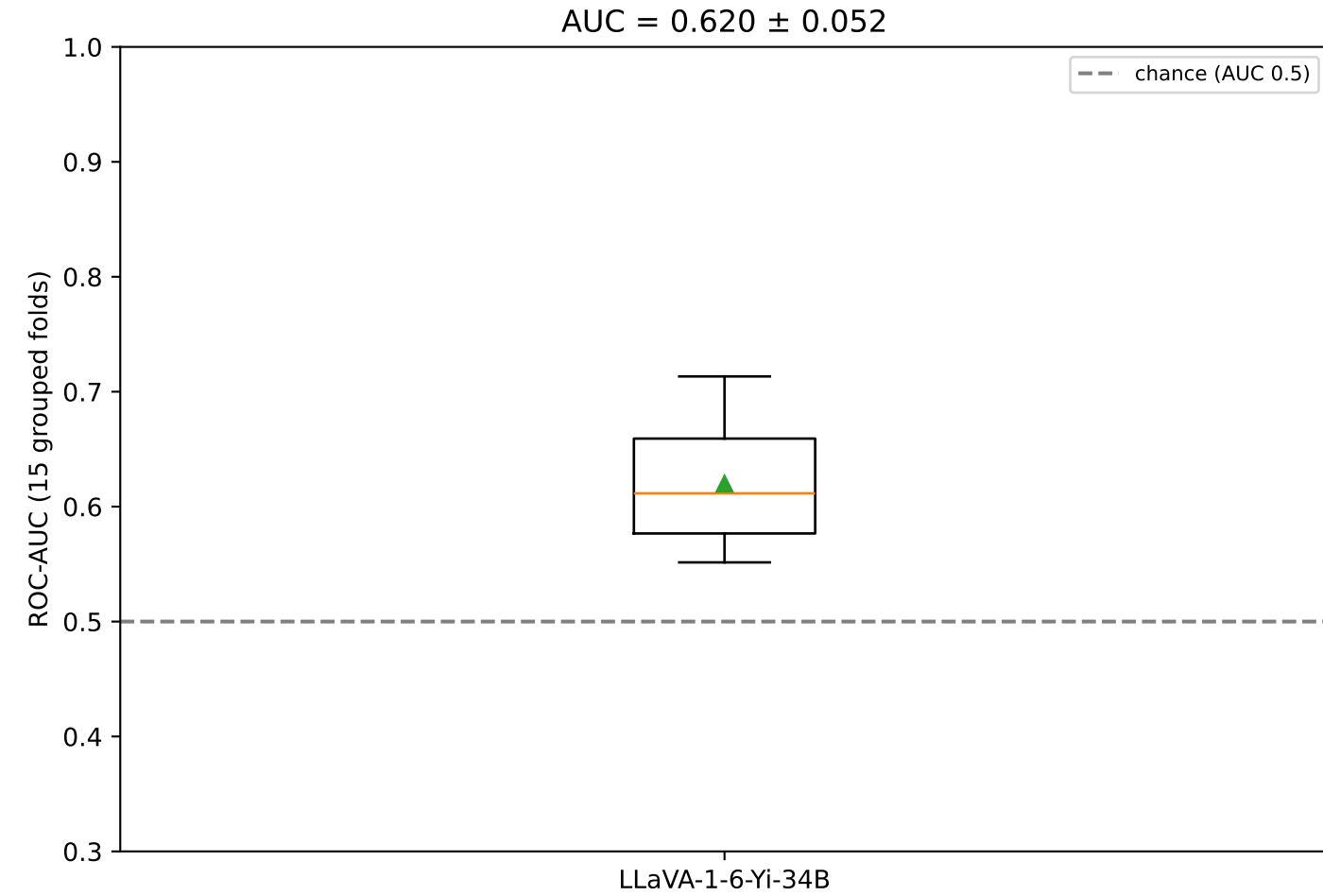


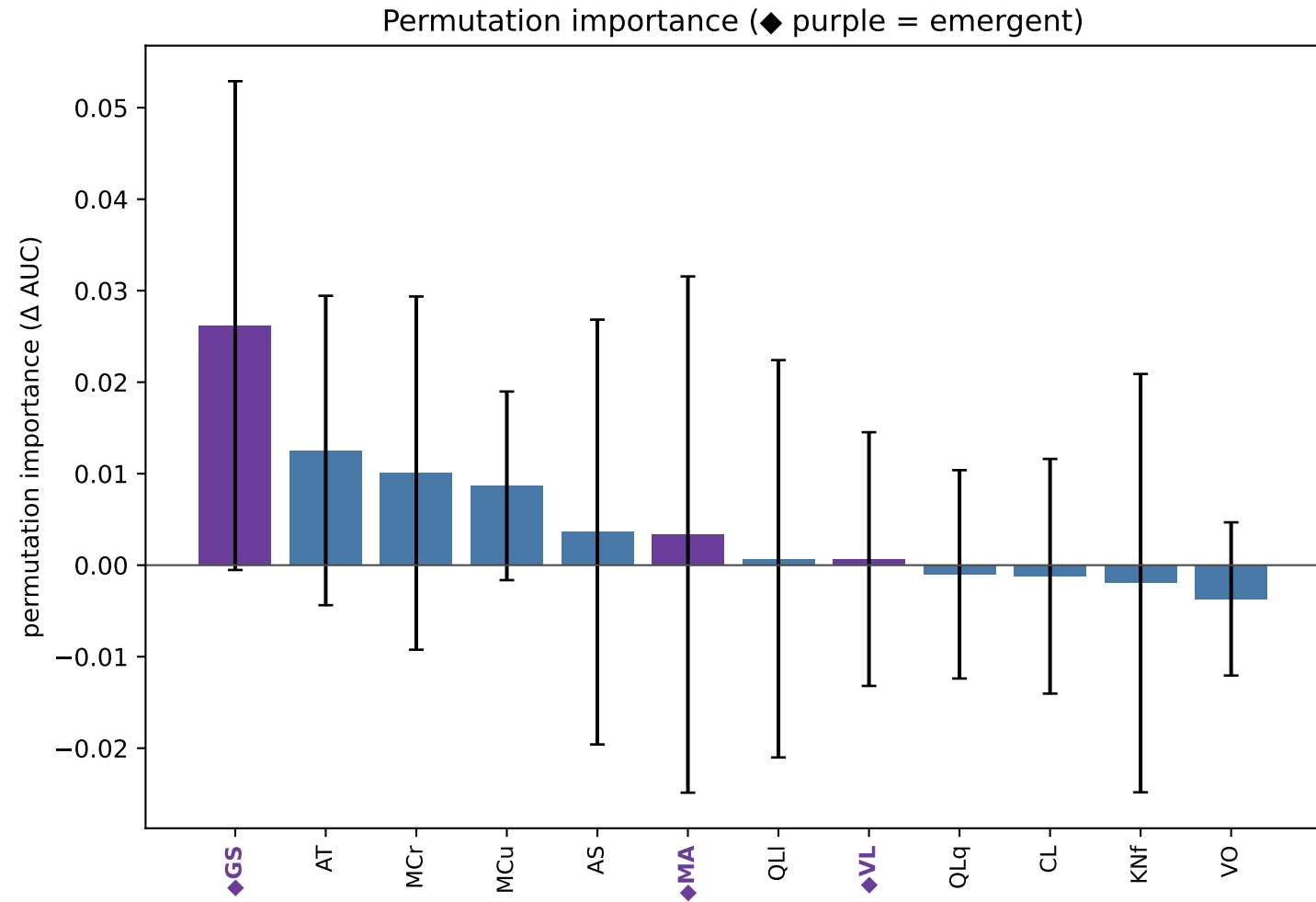
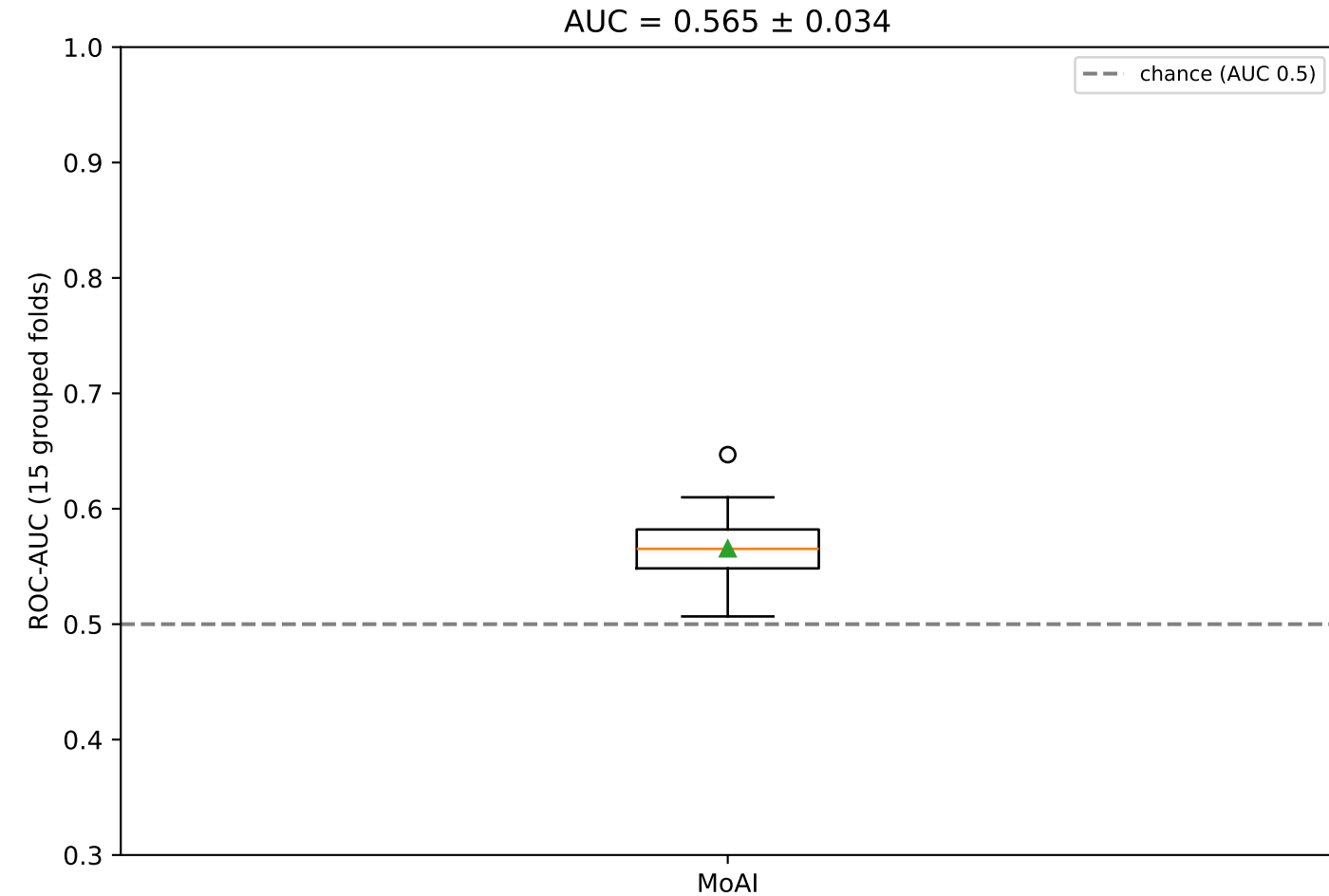
Permutation importance (◆ purple = emergent)

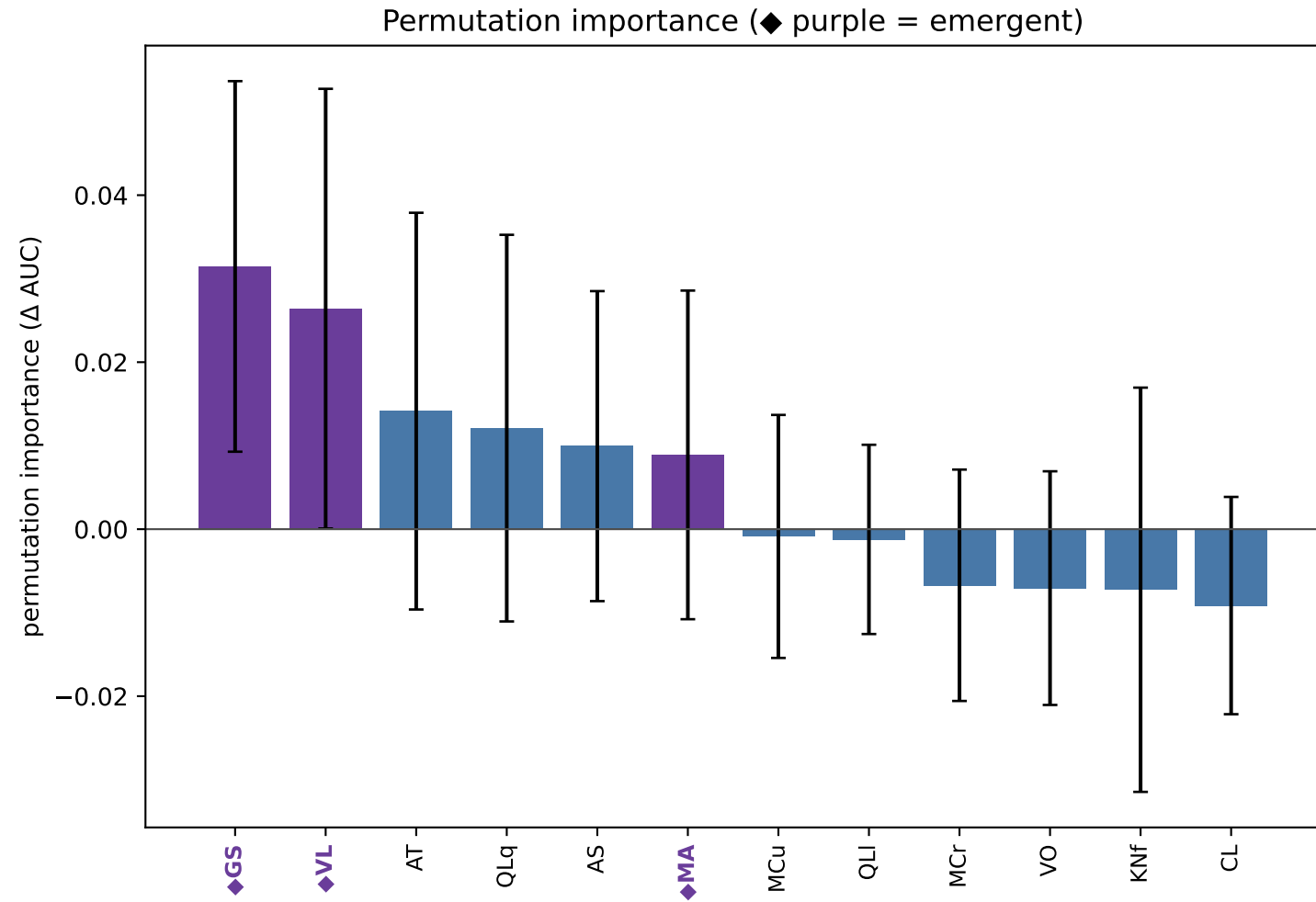
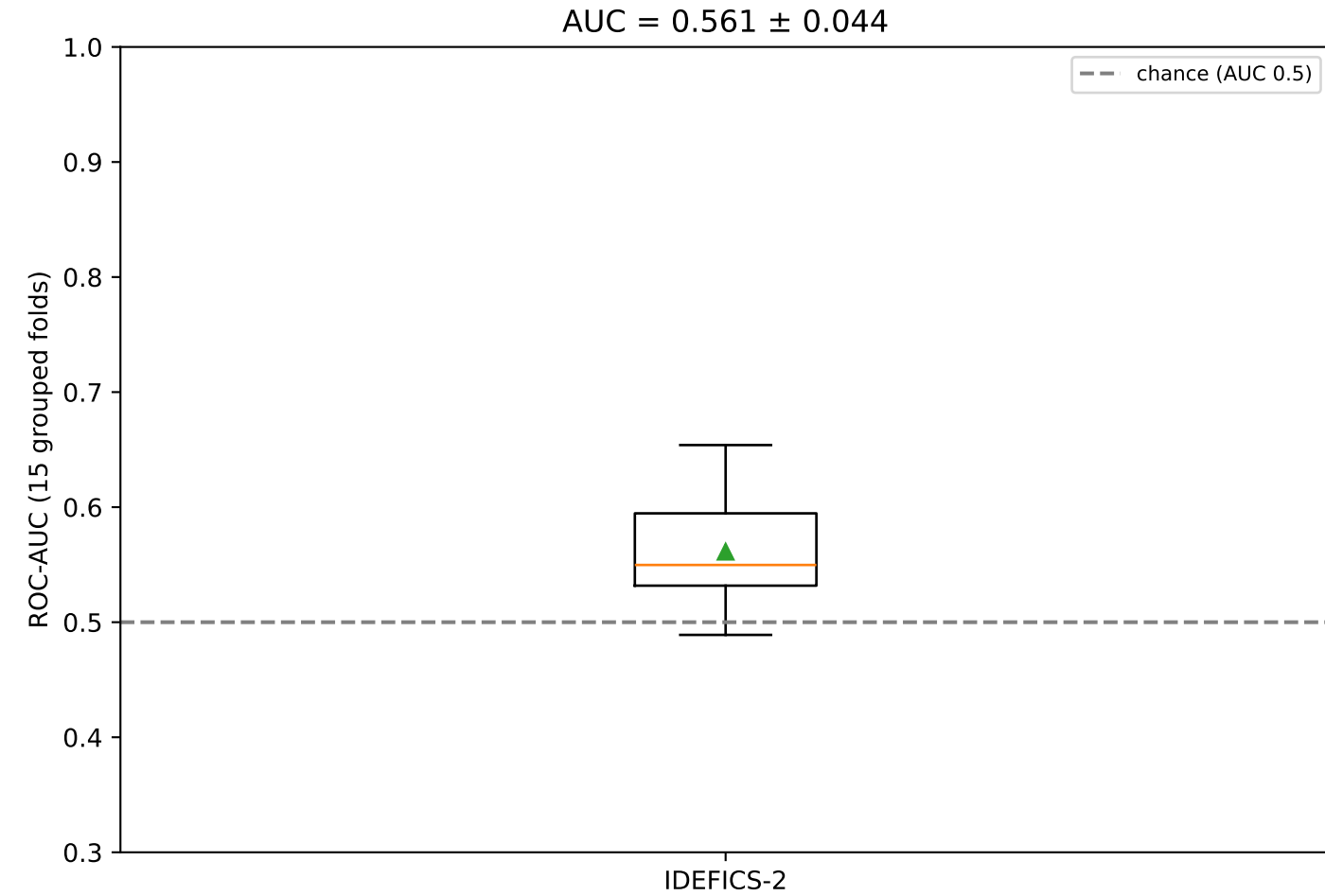


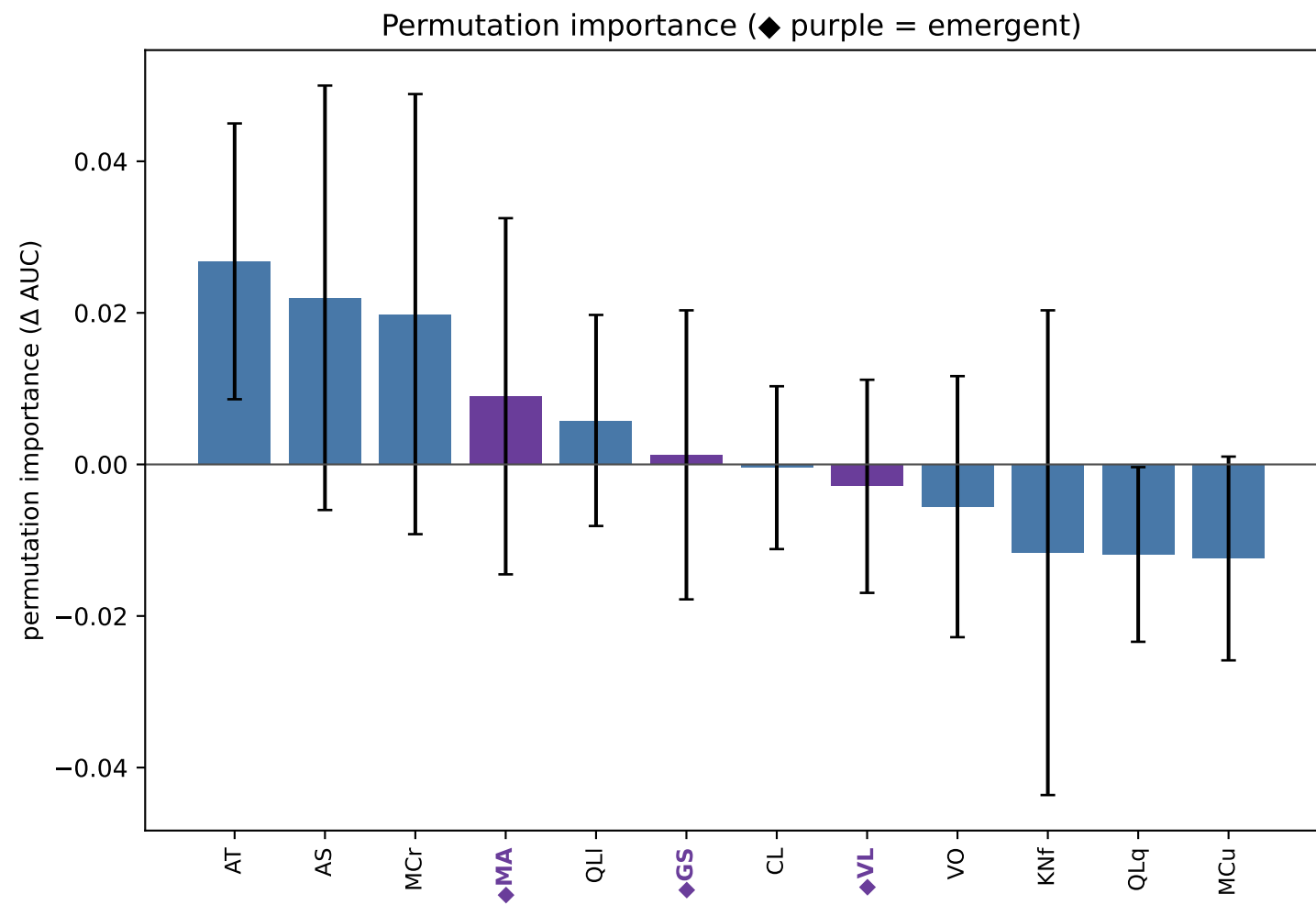
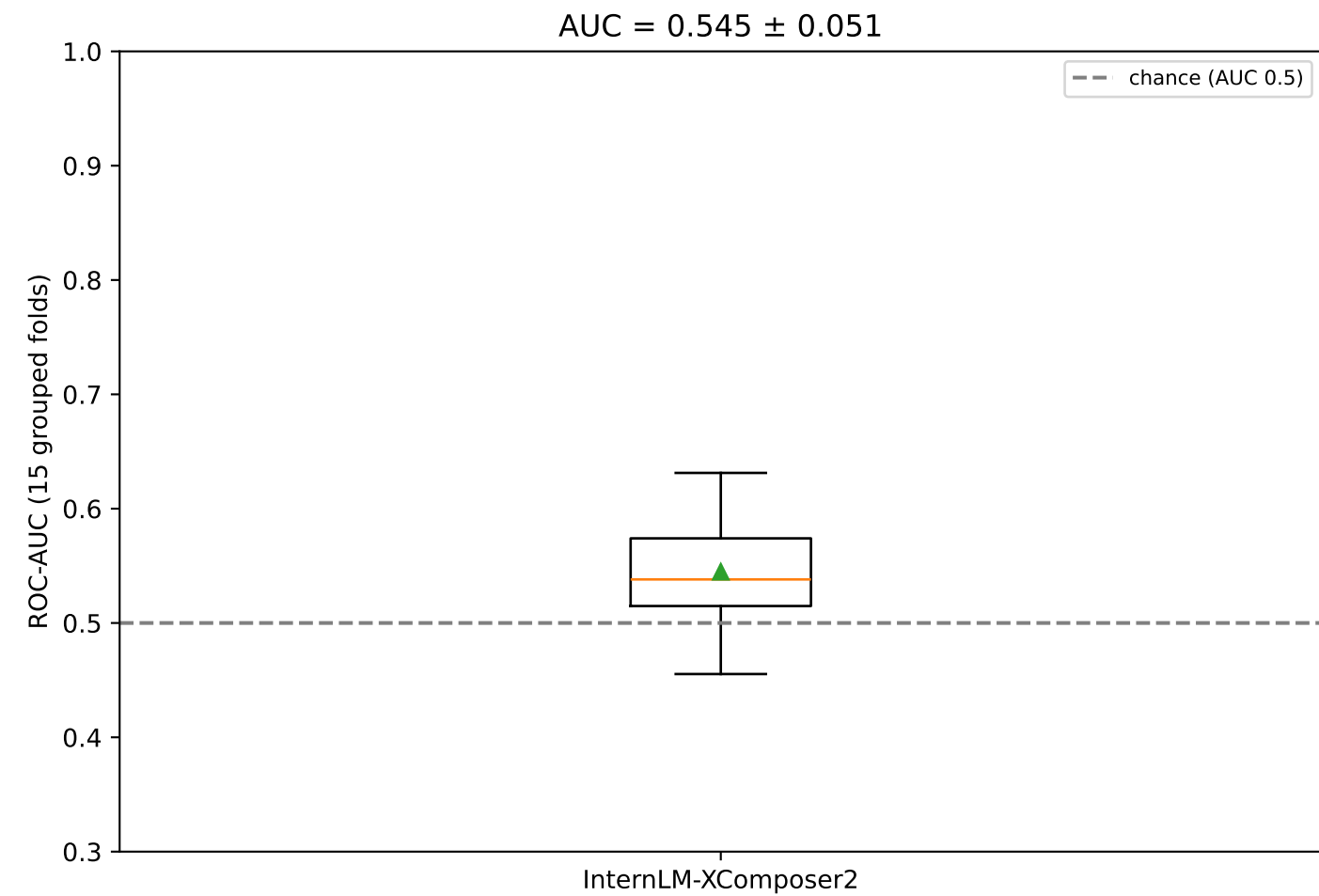


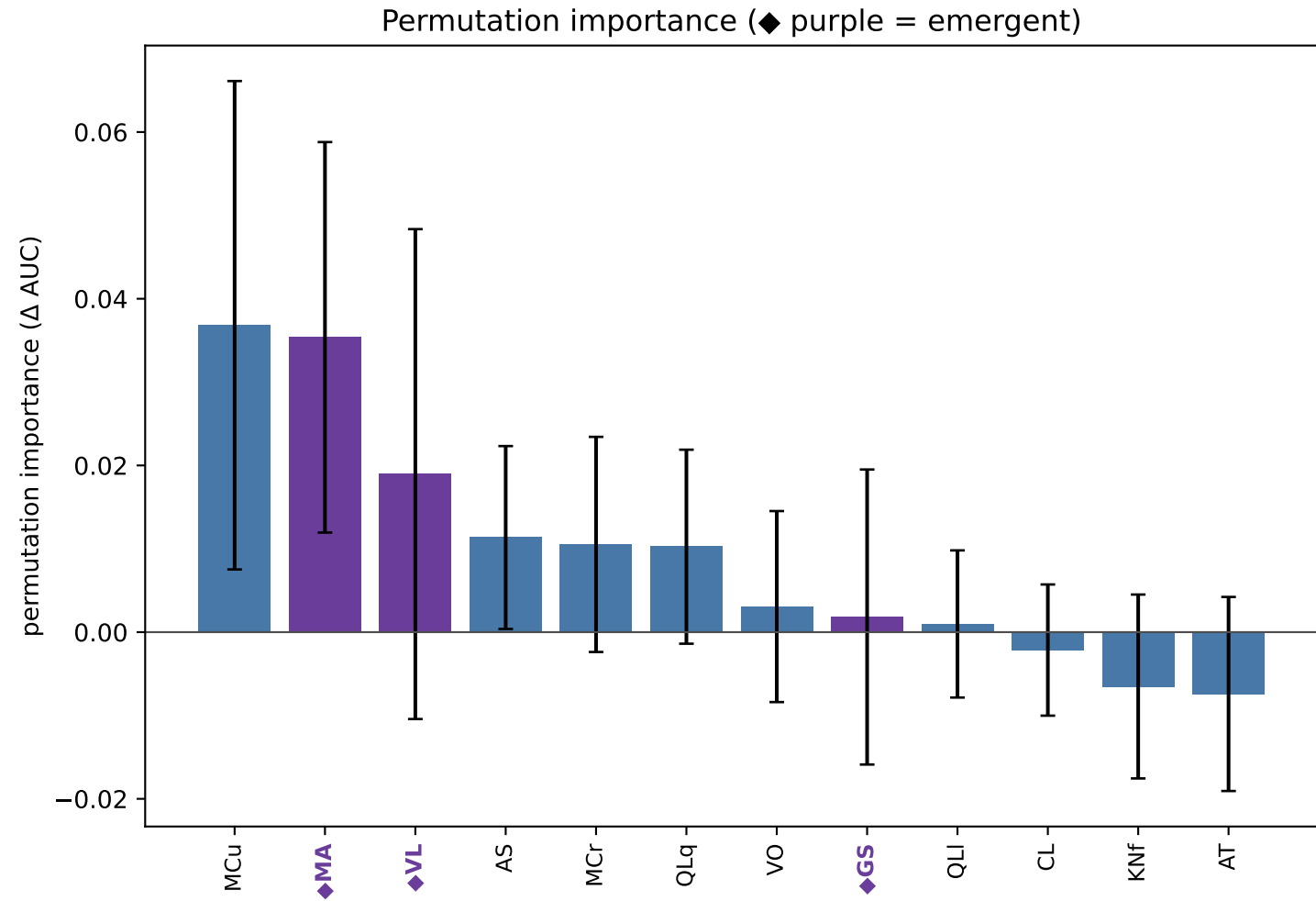
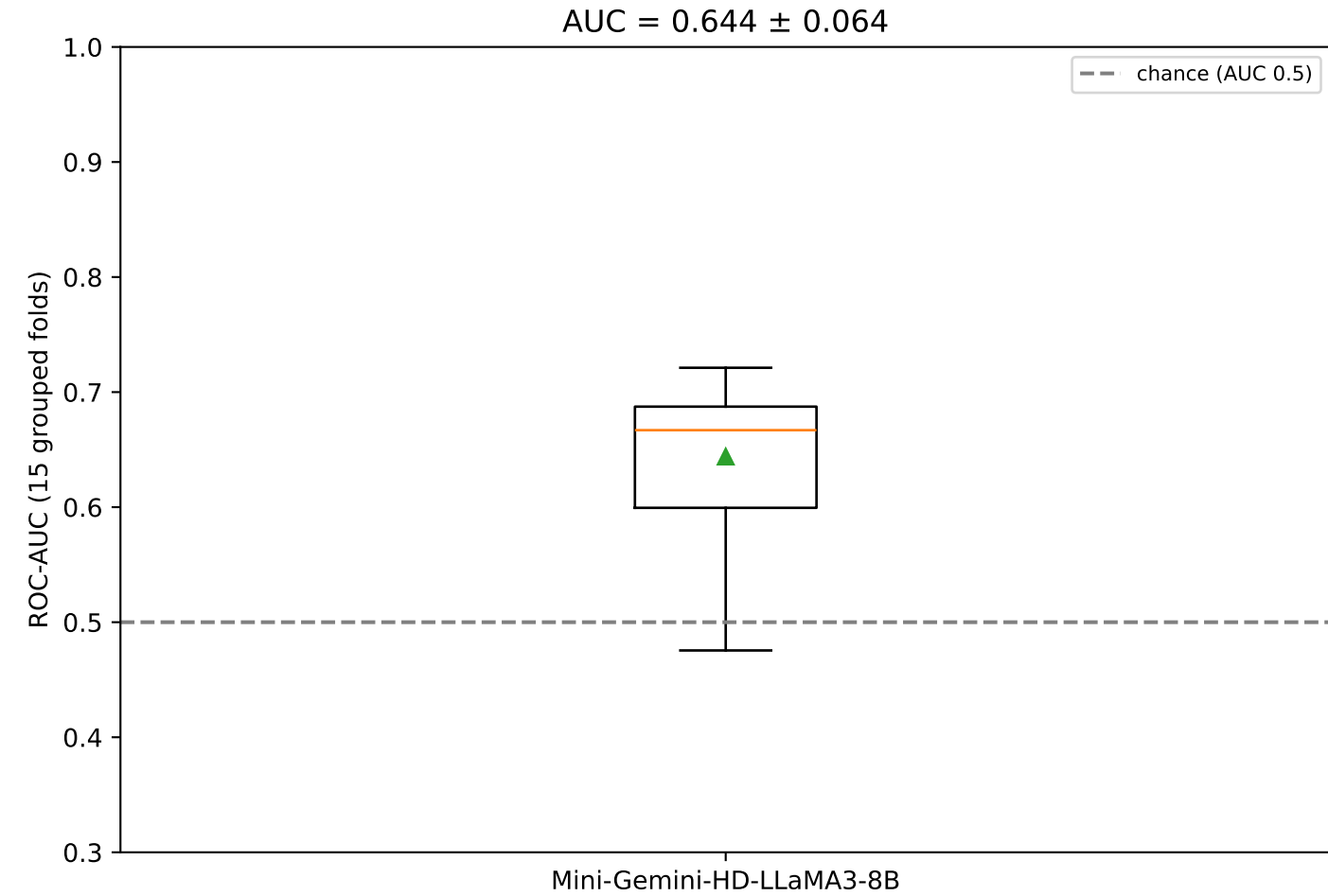


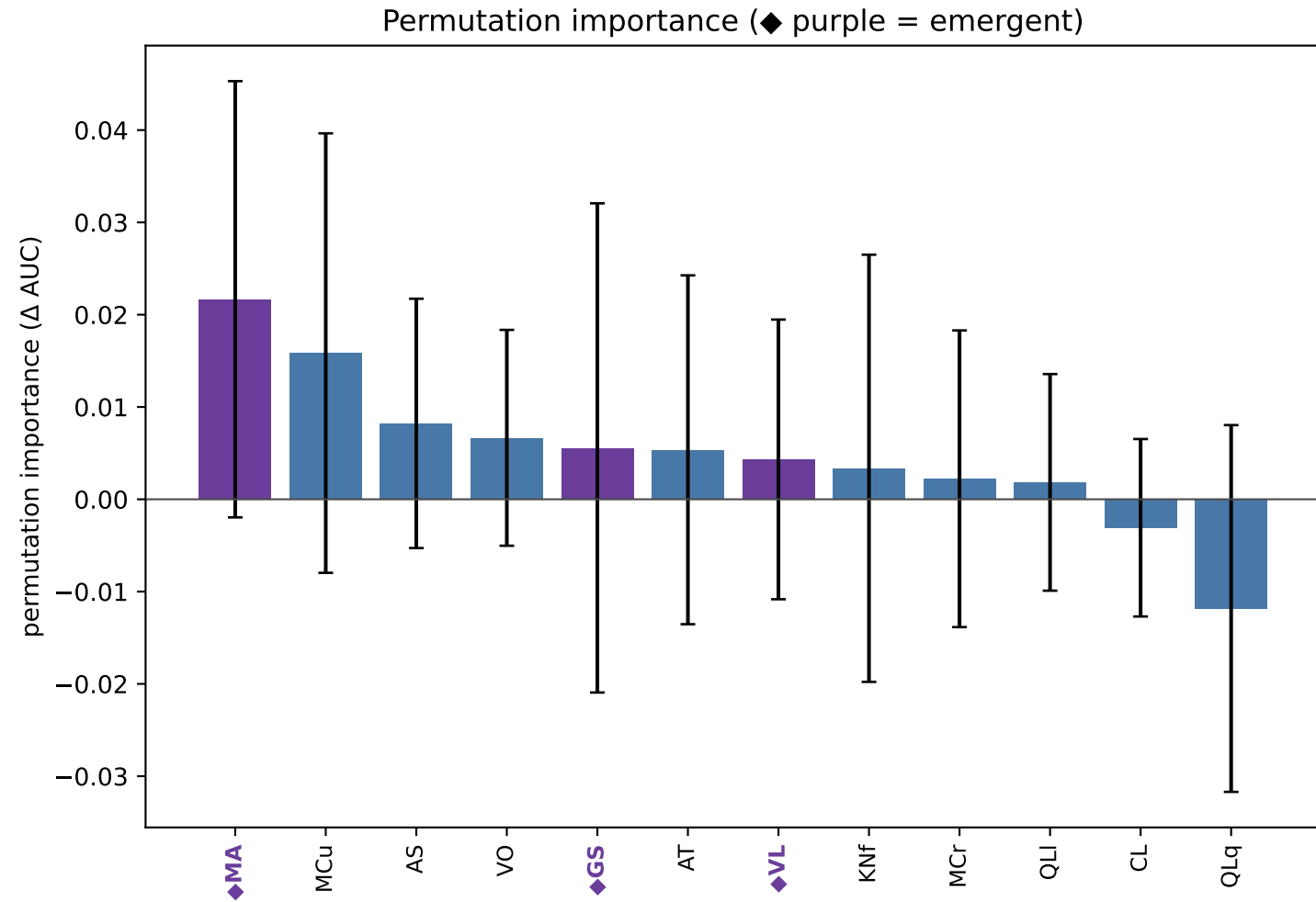
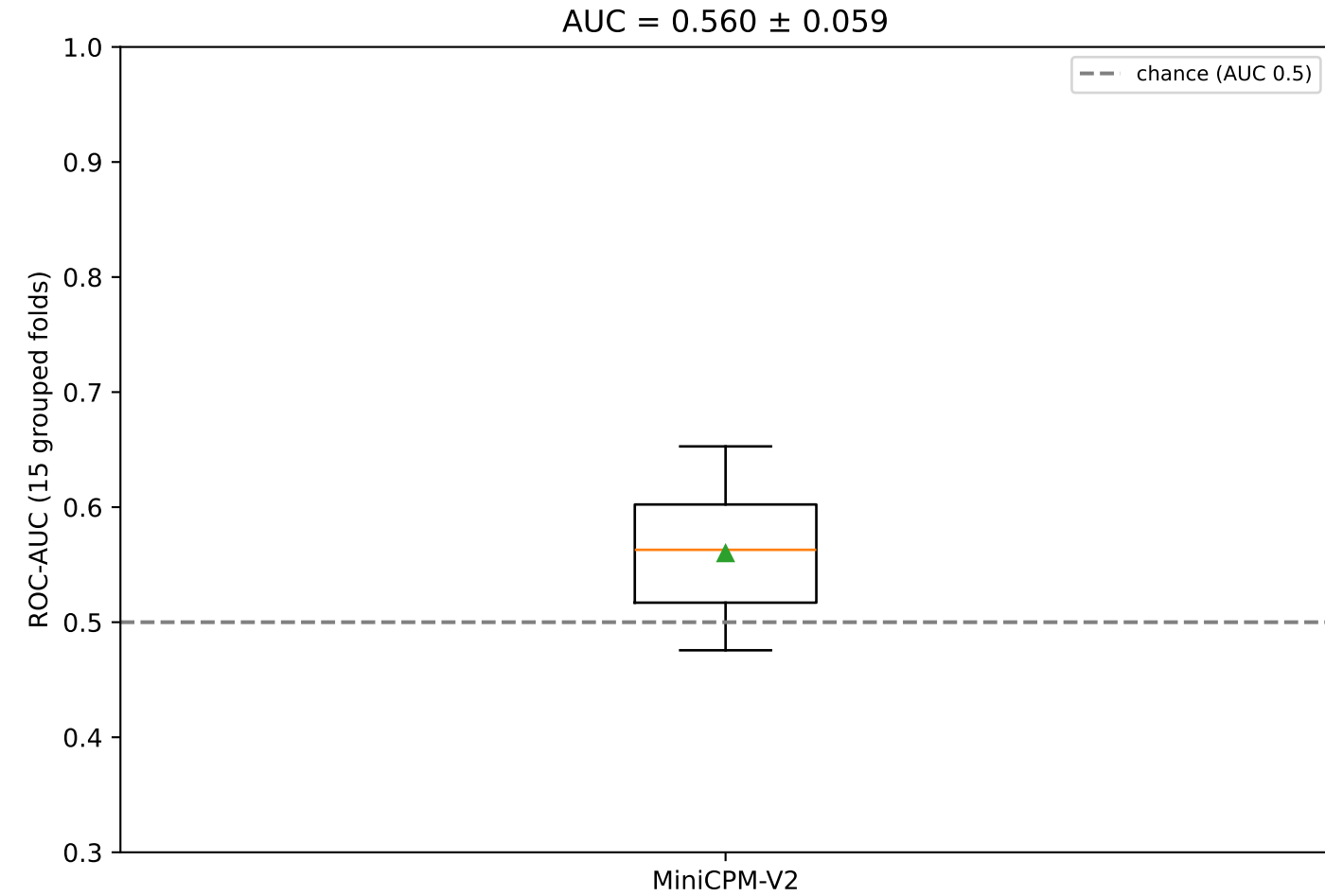




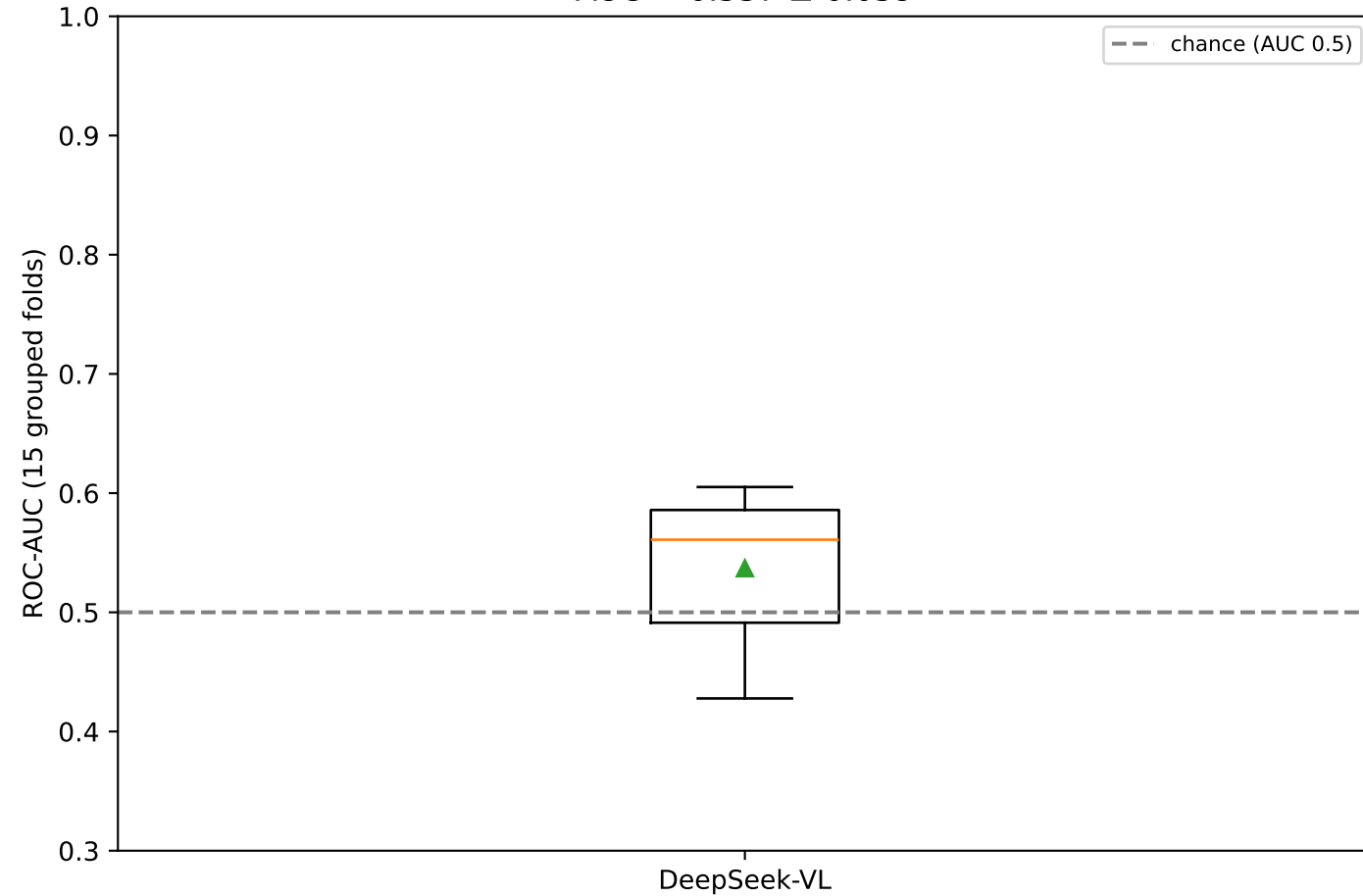




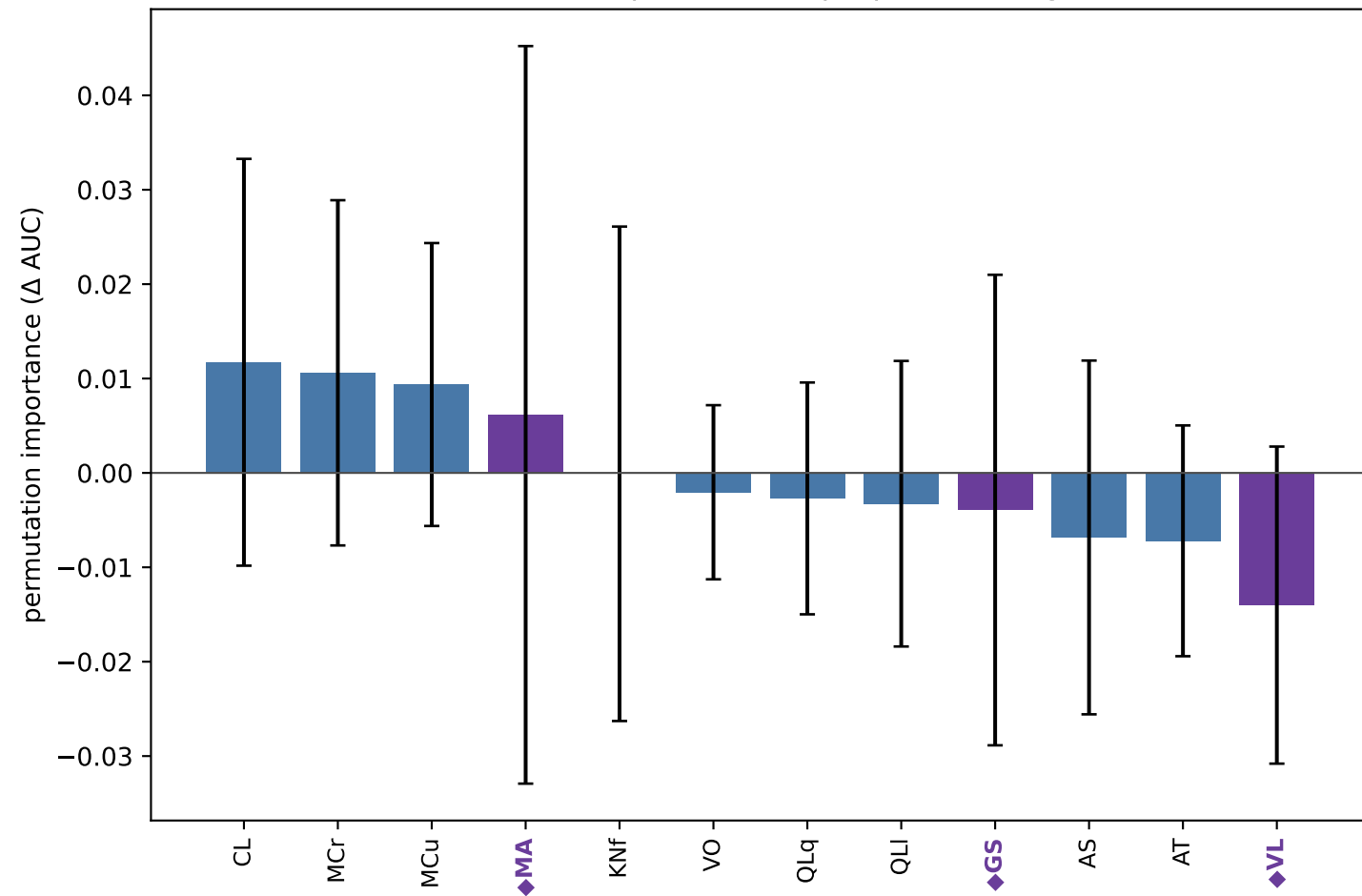


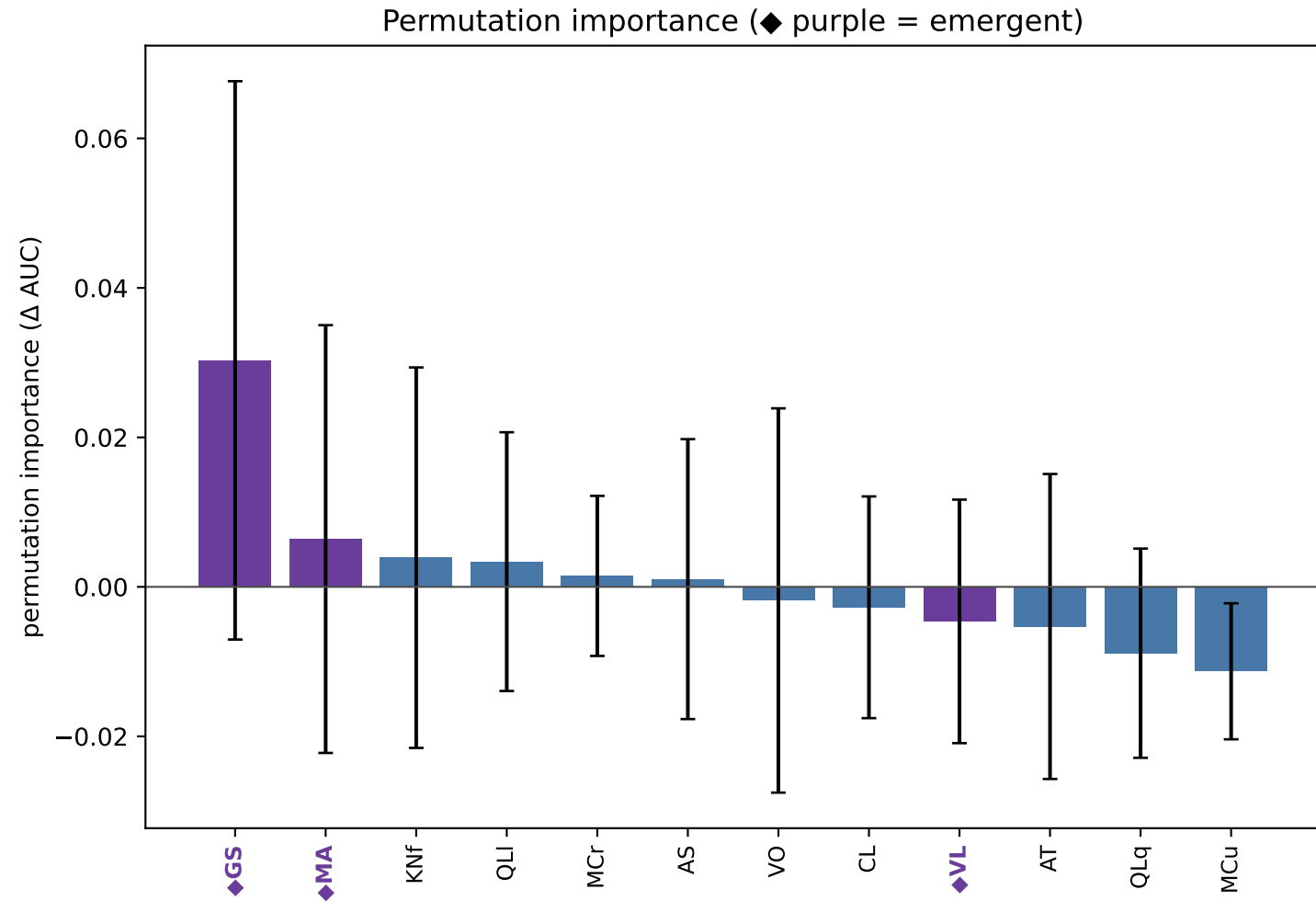
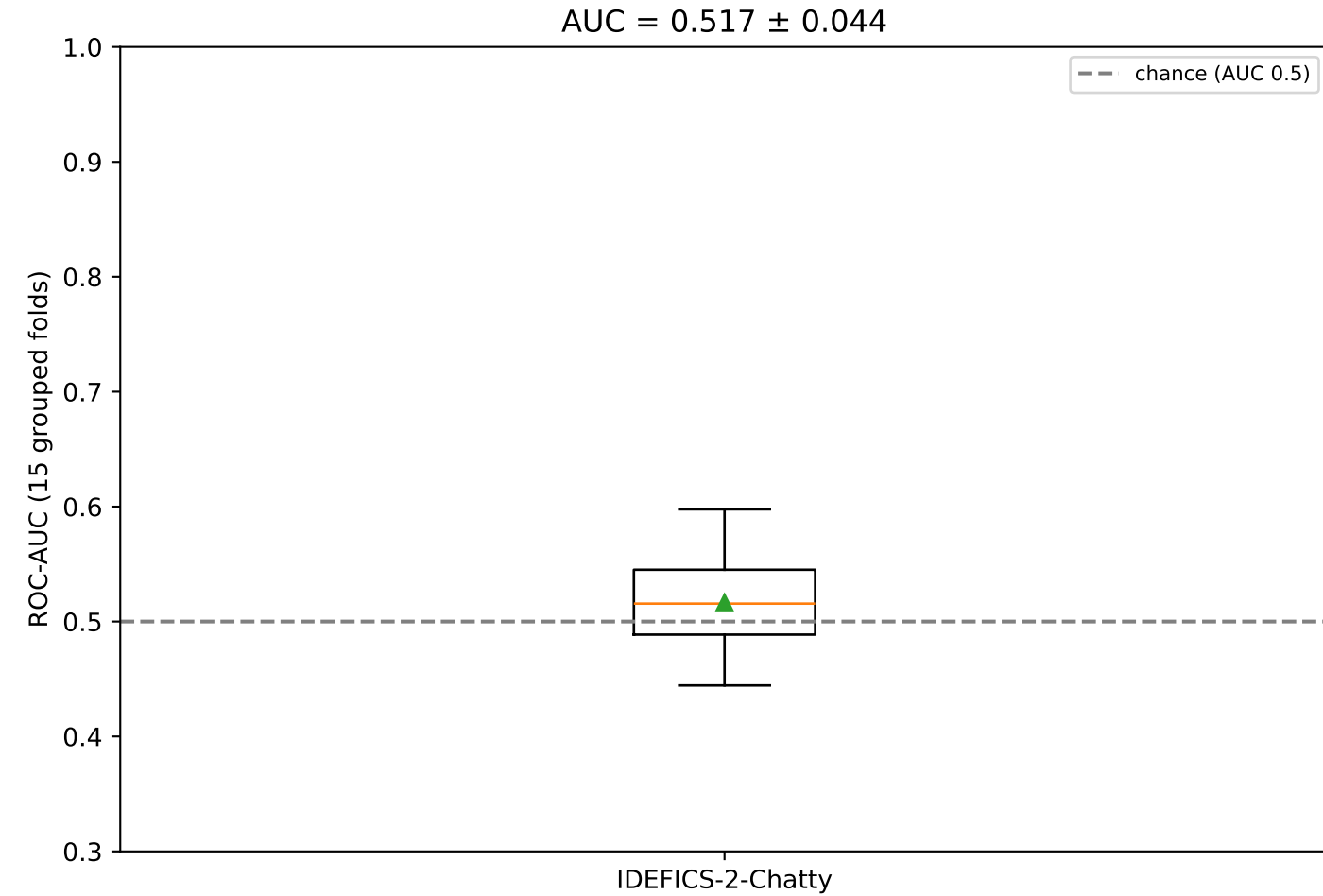


AUC = 0.537 ± 0.059

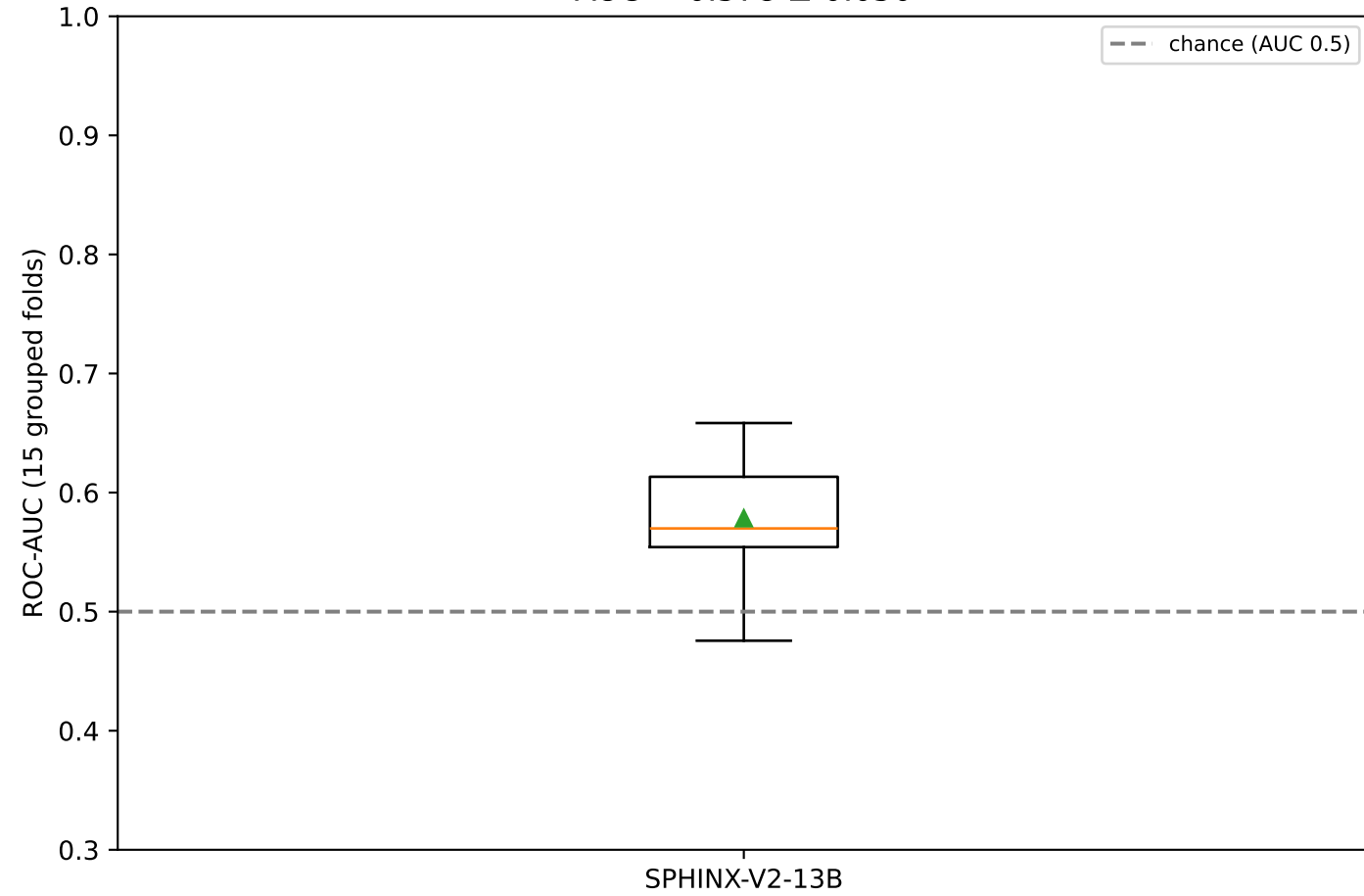


Permutation importance (◆ purple = emergent)

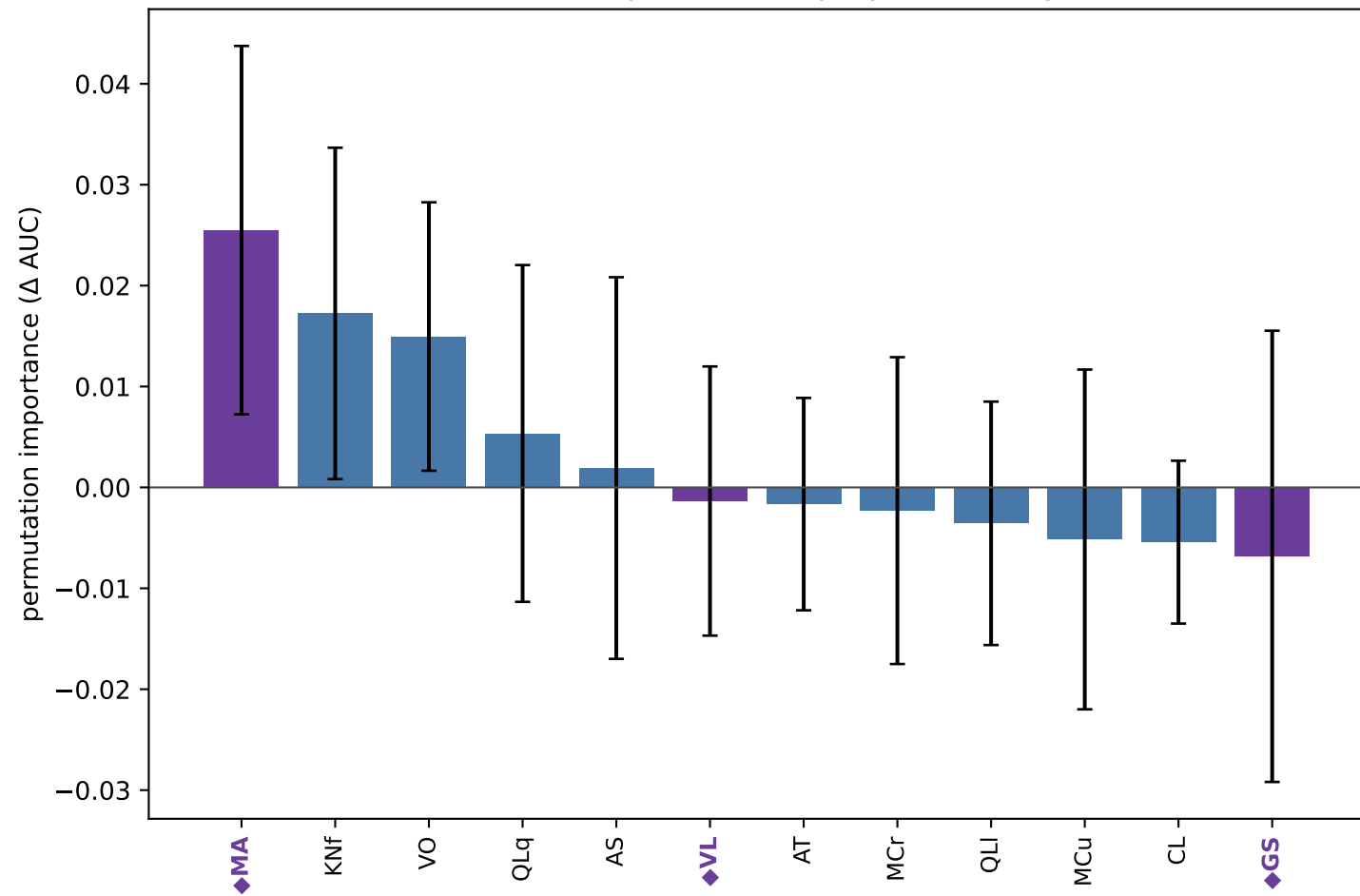


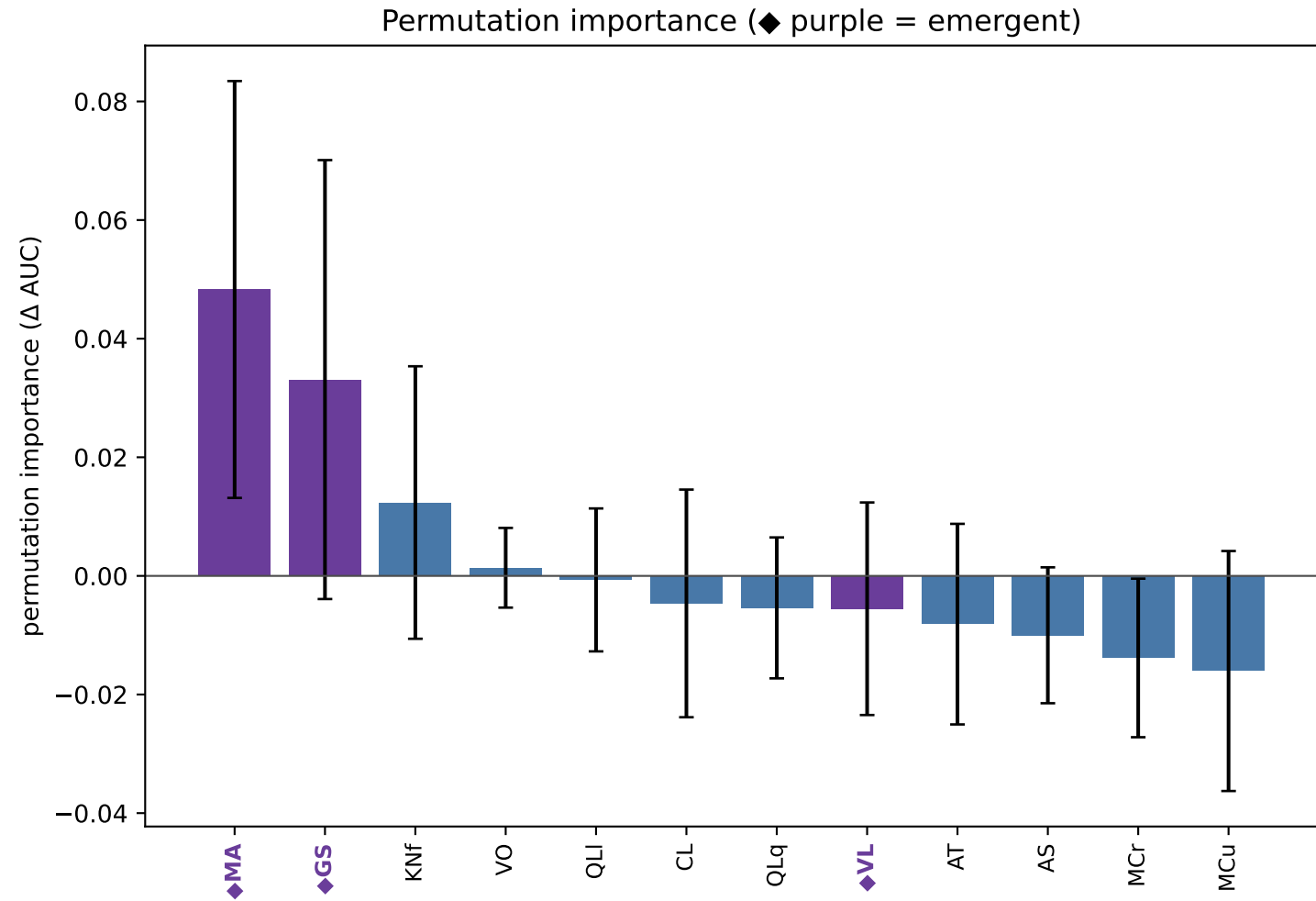
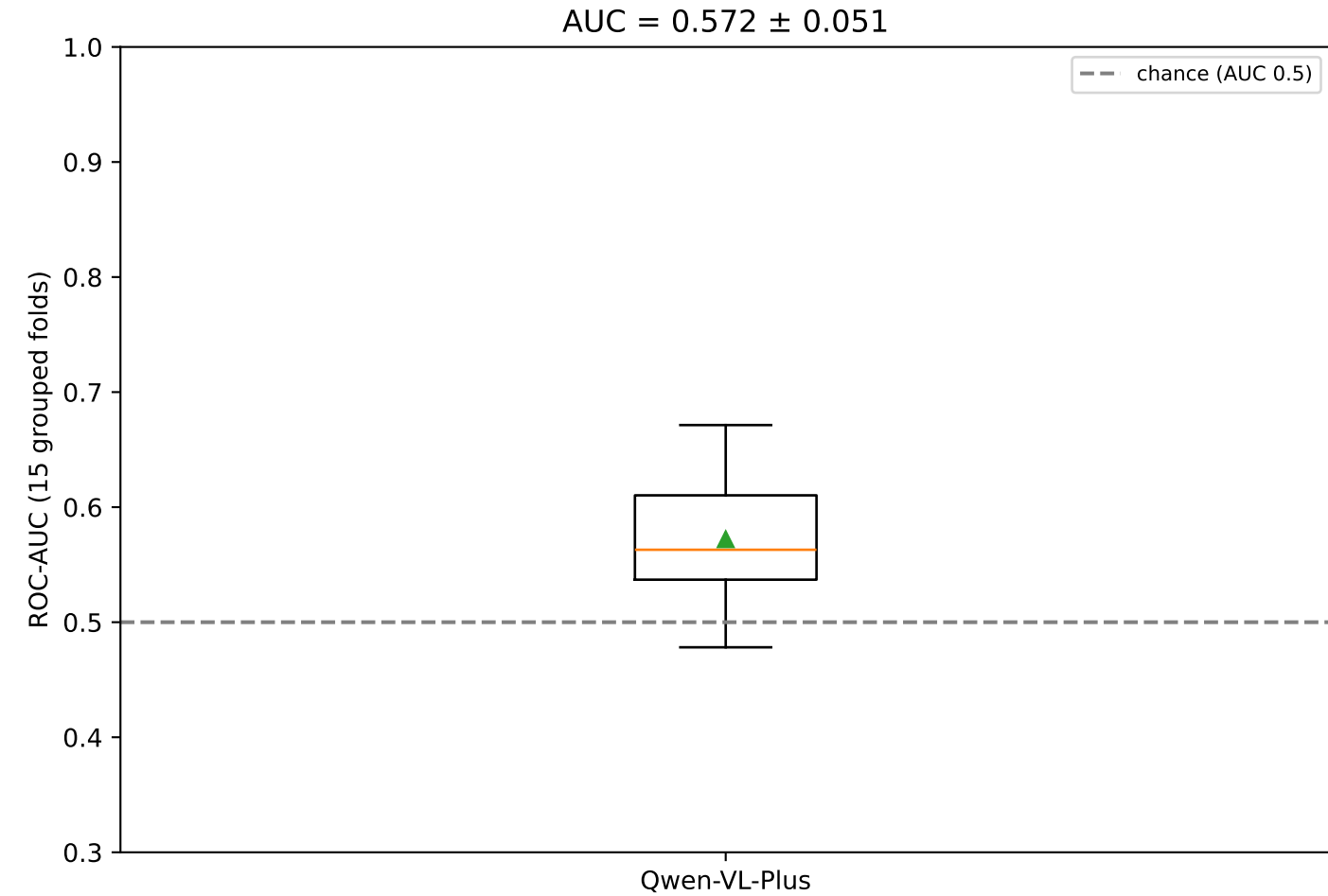


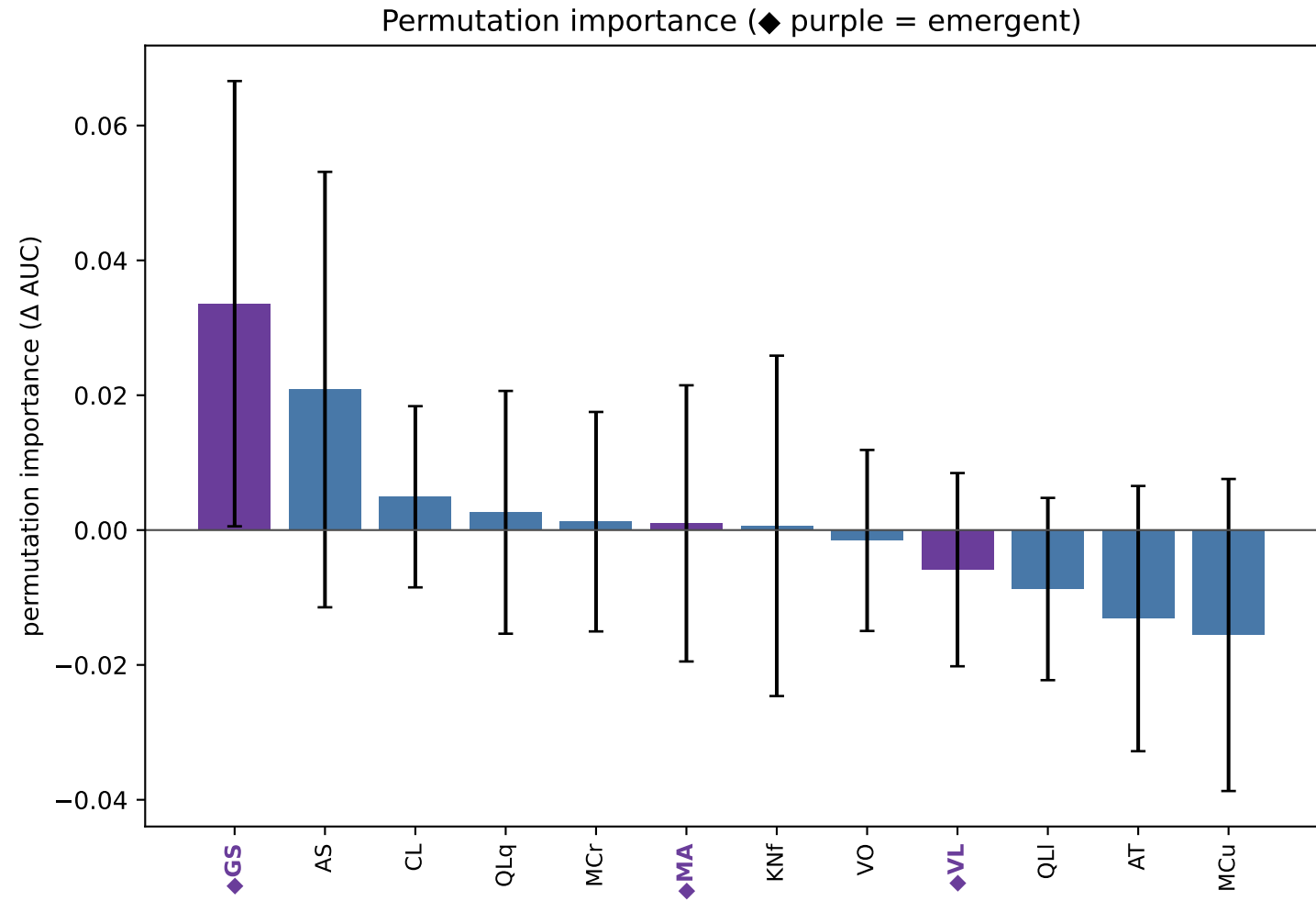
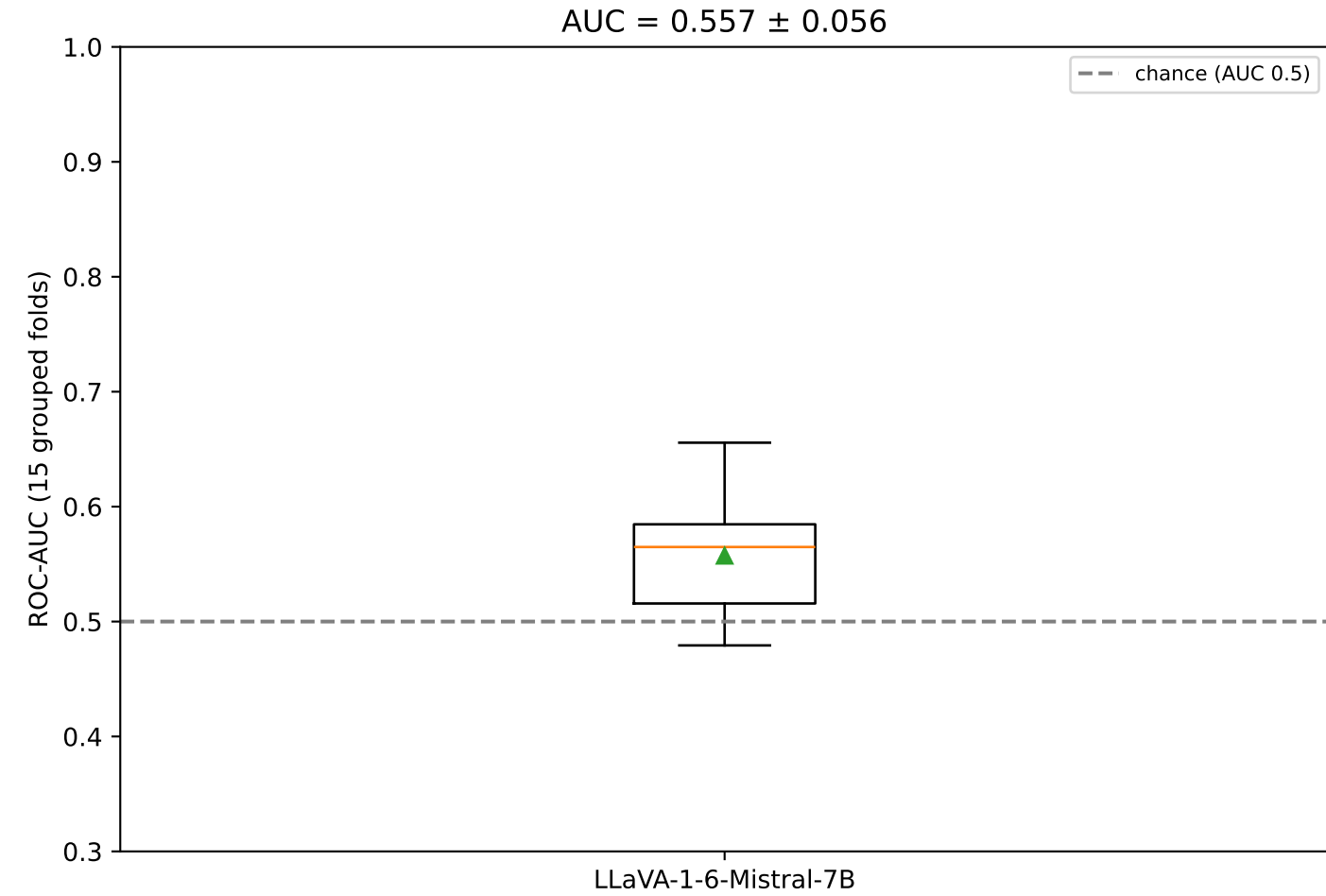
AUC = 0.578 ± 0.050



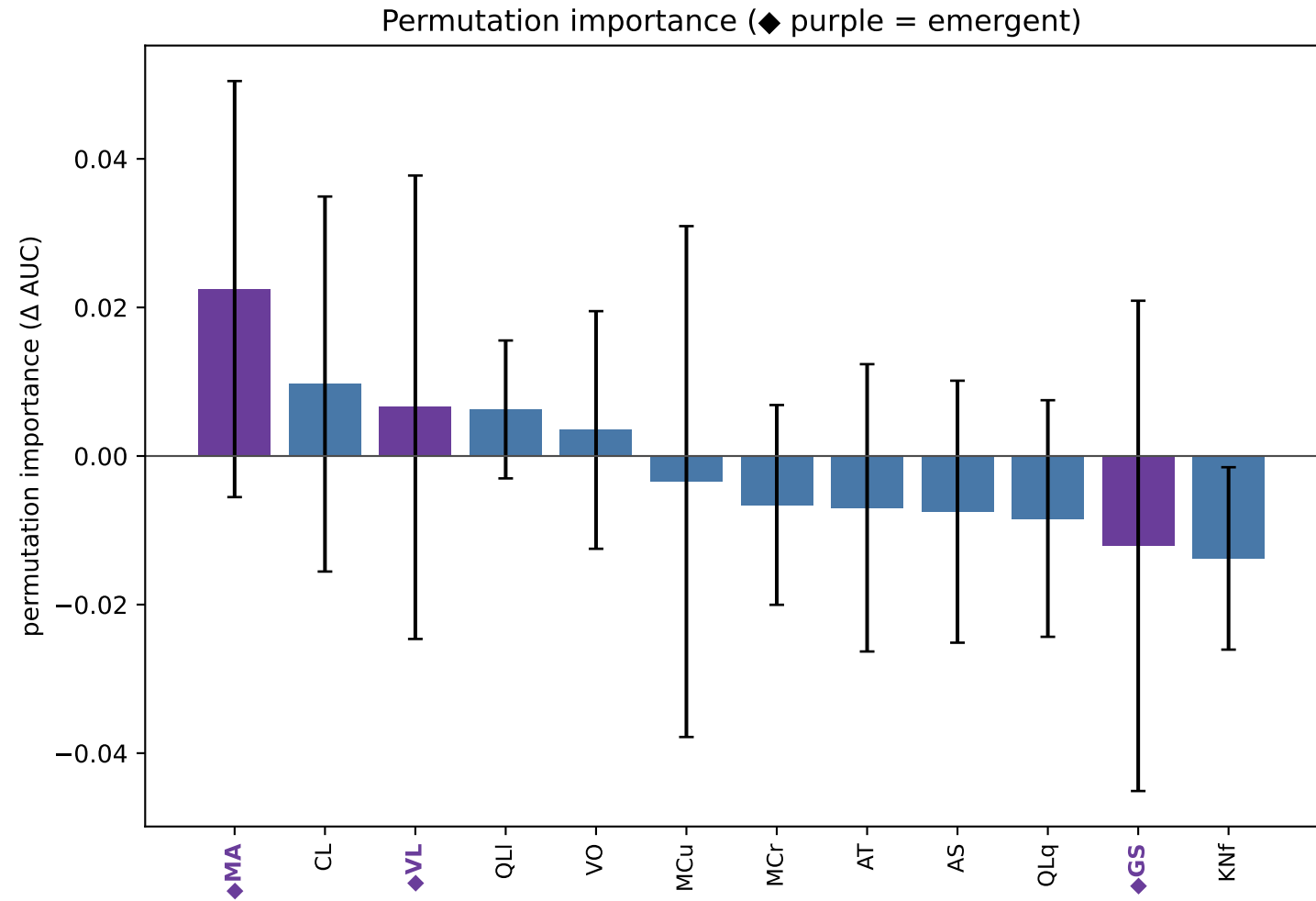
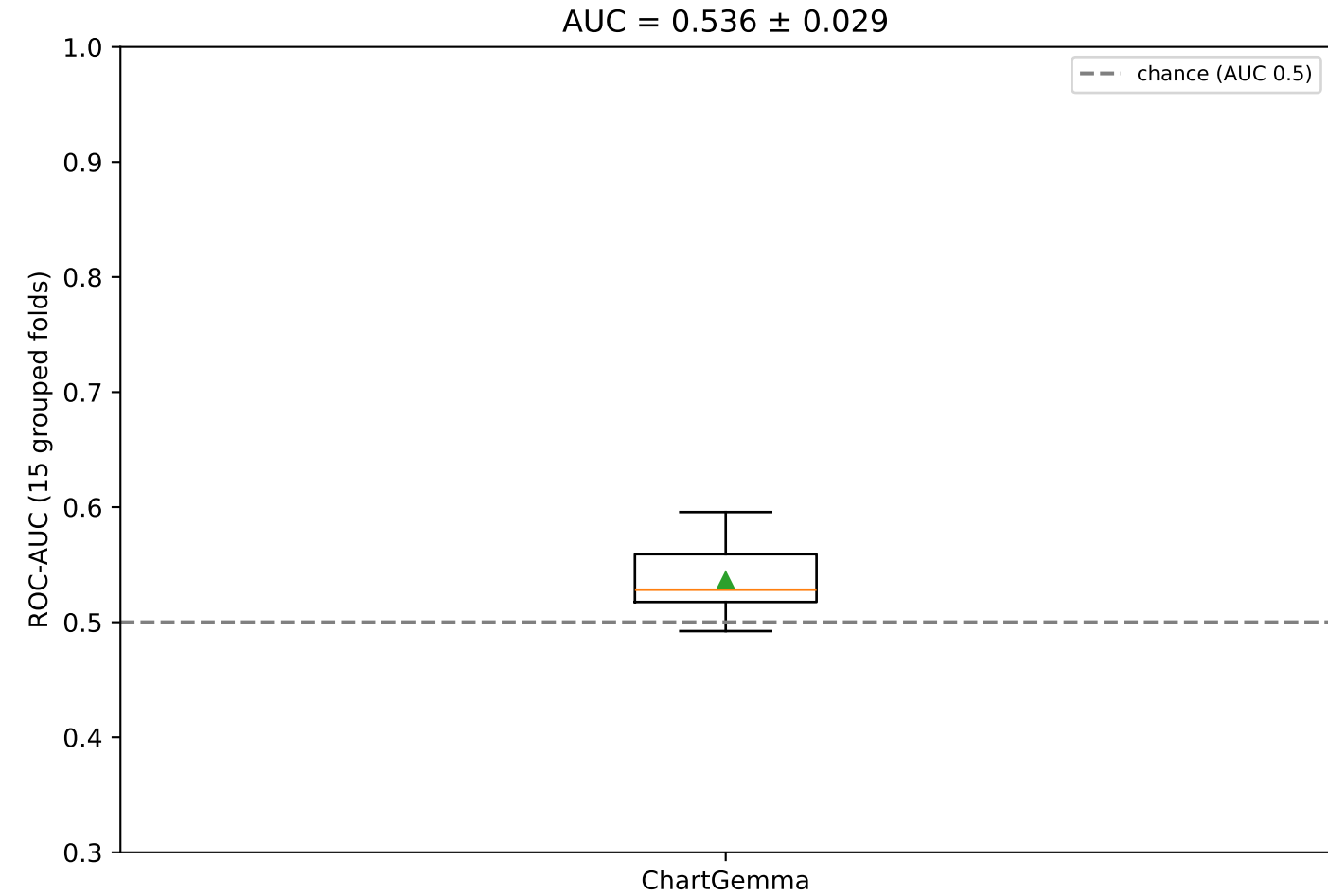
Permutation importance (◆ purple = emergent)



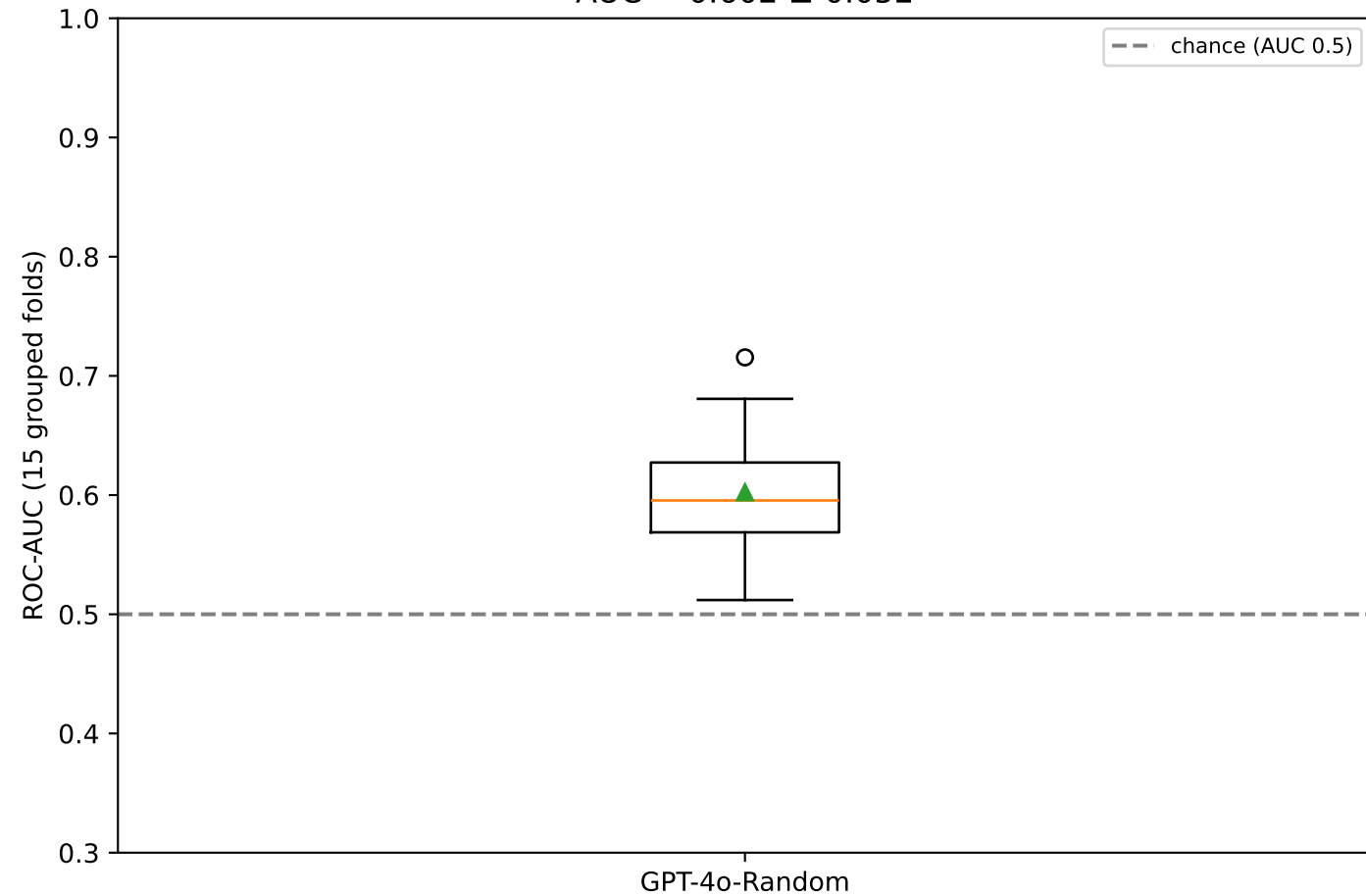




ChartGemma — rubric→failure RF (y=1 INCORRECT; base rate 0.870, n=800, paper-grouped 5-fold ×3)



AUC = 0.602 ± 0.052



Permutation importance (◆ purple = emergent)

